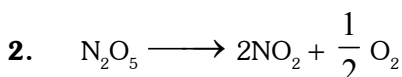
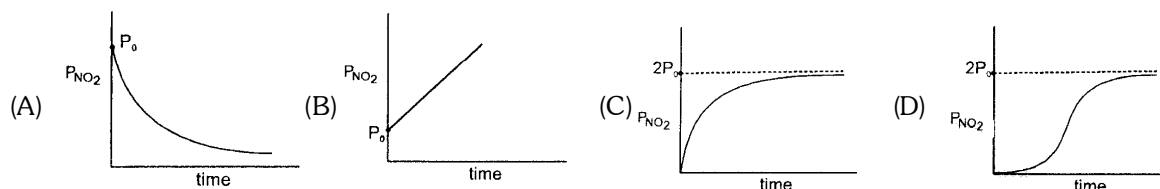


SINGLE OPTION CORRECT

1. For a process $A + B \rightarrow \text{product}$, the rate of the reaction is second order with respect to A and zero order with respect to B. When 1 mole each of A and B are taken in a one litre vessel if the initial rate is 1×10^{-2} mole/L sec. The rate of reaction when 50% of the reactants have been converted to products would be :
- (A) 1×10^{-2} mol/L sec (B) 2.5×10^{-3} mol/L sec
 (C) 5×10^{-2} mol/L sec (D) 0.5×10^{-2} mol/L sec

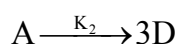
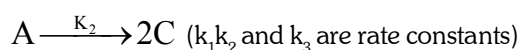
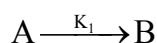


When N_2O_5 decompose, its $t_{1/2}$ does not change with its changing pressure during the reaction, so which one is the correct representation for "pressure of NO_2 " vs "time" during the reaction when initial $P_{N_2O_5}$ is equals to P_0 .



3. $A + 2B \longrightarrow C$ rate = $K[A]^1[B]^1$
 Initial concentration of A and B are 0.1 M and 100 M respectively. If concentration of A will remain 0.025 after 20 min. What is the pseudo order rate constant (assume this reaction is pseudo first order reaction)
- (A) 0.345 min^{-1} (B) 0.693 min^{-1} (C) 0.0693 min^{-1} (D) None
4. Consider the reaction $A + H_2O + H_{(aq)}^+ \xrightarrow{K} \text{product}$. A at 27°C has a time for half change of 100 min when carried out in a buffer solution of $\text{pH} = 4$, 10 min when carried out at $\text{pH} = 5$. Both times of half change are independent of the initial concentration of A. H_2O is present in excess and density of H_2O is 0.9 g/ml. If rate of reaction when $[A] = 0.2 \text{ M}$ and $\text{pH} = 5$ is 10 M/s . Find true rate constant assuming partial order of H_2O is one?
- (A) 10^{-10} (B) 10^{-5} (C) 9×10^{-6} (D) None

5. Consider the following first order parallel reactions :



Which of the following statement is correct ?

- (A) B, C and D are produced in 1 : 2 : 3 molar ratio.
 (B) B, C and D are produced in $k_1 : k_2 : k_3$ molar ratio.
 (C) B, C and D are produced in $k_1 : 2k_2 : 3k_3$ molar ratio.
 (D) None of these

6. For a first order reaction $A \rightarrow B + 2C + 3D$ (A is optically inactive and B, C and D are dextrorotatory), the optical rotation at time t and ∞ are r_t and r_∞ respectively, the expression for rate constant is:

(A) $k = \frac{1}{t} \ln \frac{r_t}{r_\infty - r_t}$ (B) $k = \frac{1}{t} \ln \frac{r_\infty}{r_\infty - r_t}$ (C) $k = \frac{1}{t} \ln \frac{r_\infty - r_t}{r_t}$ (D) $k = \frac{1}{t} \ln \frac{r_\infty - r_t}{r_\infty}$

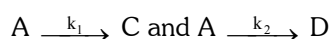
7. For the Homogeneous 1st order gaseous reaction $A \rightarrow B + C + D + E + \dots + X + Y + Z$
The initial pressure was P_0 while after time t was P . The expression for rate constant K in terms of P_0 , P and t is :

(A) $k = \frac{2.303}{t} \log \frac{P_0}{P_0 - P}$ (B) $k = \frac{2.303}{t} \log \frac{25 P_0}{26 P_0 - P}$
(C) $k = \frac{2.303}{t} \log \frac{24 P_0}{25 P_0 - P}$ (D) $k = \frac{2.303}{t} \log \frac{24 P}{P_0 - 25 P}$

Passage # (Q.8 to 10)

Competing First Order Reaction

Frequently a species can react in different ways to give a variety of products. For examples. Toluene can be nitrated at the ortho, meta, or para positions, we shall consider the simplest case, that of two competing irreversible first-order reactions :



where the stoichiometric coefficients are taken as unity for simplicity. The rate law is

$$\left(\frac{d[A]}{dt} \right) = -k_1 [A] - k_2 [A] = -(k_1 + k_2) [A]$$

$$\Rightarrow [A] = [A]_0 e^{-(k_1+k_2)t} \text{ For C, we have } \left(\frac{d[C]}{dt} \right) = k_1 [A]$$

$= k_1 [A]_0 e^{-(k_1+k_2)t}$. Multiplication by dt and integration from time 0 (where $[C]_0 = 0$) to an arbitrary time t gives

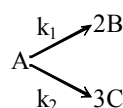
$$[C] = \frac{k_1 [A]_0}{k_1 + k_2} (1 - e^{-(k_1+k_2)t})$$

Similarly, integration of $\left(\frac{d[D]}{dt} \right)$ gives

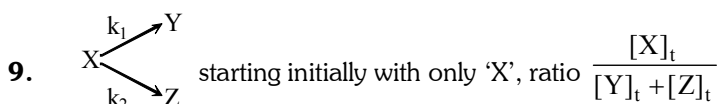
$$[D] = \frac{k_2 [A]_0}{k_1 + k_2} (1 - e^{-(k_1+k_2)t})$$

The sum of the rate constants $k_1 + k_2$ appears in the exponentials for both $[C]$ and $[D]$. Figure shows plots

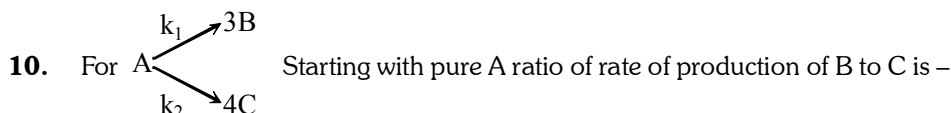
of $[A]$, $[C]$ and $[D]$ versus t for $k_1 = 2k_2$. At any time we also have $\frac{[C]}{[D]} = \frac{k_1}{k_2}$.

8.  starting initially with only A, which of the following is correct at time t –

(A) $[A]_0 = [A]_t + [B]_t + [C]_t$ (B) $[A]_0 = [A]_t + 2[B]_t + 3[C]_t$
(C) $[A]_0 = [A]_t + \frac{[B]_t}{2} + \frac{[C]_t}{3}$ (D) $[A]_0 = \frac{2}{3} [A]_t + [B]_t + [C]_t$



- (A) independent of time
(B) $\frac{1}{(e^{kt} - 1)}$
(C) Depends upon initial concentration of X
(D) $[A]_0 (e^{kt} - 1)$



- (A) Independent of time
(B) Independent of temperature
(C) depends upon initial concentration of A
(D) Independent of mechanism of reaction

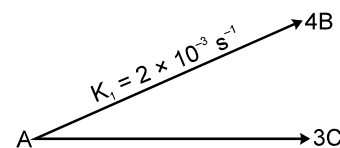
ANSWER-KEY (QUIZ # 01)

1. (B) 2. (C) 3. (C) 4. (B) 5. (C) 6. (B) 7. (C) 8. (C) 9. (B)
10. (A)

QUIZ # 02

SINGLE OPTION CORRECT

1. For the following parallel chain reaction



what will be that value of overall half-life of A in minutes? [Given that $\frac{[B]_t}{[C]_t} = \frac{16}{9}$]

- (A) 3.3 (B) 6.3 (C) 3.6 (D) None

2. A first order reaction, $A(aq) \rightarrow 2B(l) + C(g)$ takes place in a closed container at 300 K. When 0.4 litre of a solution containing 32% (w/v) of A is decomposed at 300 K, it is found that after 10 minutes 28 gms of gas C is produced. Time required to produced same amount of gas C, when 0.4 litre solution containing 64% (w/v) of A is decomposed at same temperature is

{ take $\log 2 = 0.301$, $\log 3 = 0.477$ }

- (A) 124.5 sec (B) 249 sec (C) 600 sec (D) 1200 sec

3. Two 1st order reactions have half-lives in the ratio 3 : 2. Then the ratio of time intervals $t_1 : t_2$, will be? Where t_1 is the time period for 25% completion of the first reaction and t_2 is time required for 75% completion of the second reaction. [$\log 2 = 0.3$, $\log 3 = 0.48$]

- (A) 0.2 : 1 (B) 0.42 : 1 (C) 0.28 : 1 (D) 0.3 : 1

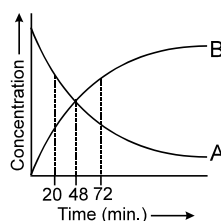
4. The rate of decomposition of N_2O_5 in CCl_4 at 318 K was studied and following results obtained

t(min) :	0	135	342	683	1693
Conc(M) :	2.08	1.91	1.67	1.35	0.57
$\log_e C$:	0.732	0.647	0.513	0.300	-0.562
1/C :	0.48	0.526	0.60	0.740	1.75

The order of the reaction is :

- (A) zero (B) 2nd (C) 1st (D) 3rd

5. For a first order reaction, $nA \rightarrow B$ whose concentration vs time curve is as shown in the figure. If half-life for this reaction is 24 minutes. Find out the value of n.

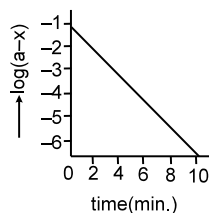


- (A) 1 (B) 2 (C) 3 (D) 4
6. Decomposition of A follows first order kinetics by the following equation.
 $4A(g) \longrightarrow B(g) + 2C(g)$
 If initially, total pressure was 800 mm of Hg and after 10 minutes it is found to be 650 mm of Hg. What is half-life of A? (Assume only A is present initially)
- (A) 10 mins (B) 5 mins (C) 7.5 mins (D) 15 mins
7. For the reaction $2NO_2 \rightarrow N_2O_2 + O_2$, rate expression is as follows $-\frac{d[NO_2]}{dt} = K[NO_2]^n$, where $K = 3 \times 10^{-3} \text{ mol}^{-1} \text{ L sec}^{-1}$. If the rate of formation of oxygen is $1.5 \times 10^{-4} \text{ mol L}^{-1} \text{ min}^{-1}$ then the molar concentrations of NO_2 in mol L^{-1} is
- (A) 1.5×10^{-4} (B) 0.0151 (C) 0.214 (D) 0.316
8. For the reaction $A_2(g) + \frac{1}{8} B_8(g) \longrightarrow A_2B(g)$ following data was obtained for a large concentration of B_8 . What is the reaction order with respect to A_2 ?

Time (min)	$[A_2](\text{moles/L}) \times 10^4$
60	292
120	198
180	104
240	010

- (A) Zero (B) One (C) Two (D) Indeterminate
9. At least how many half-lives should elapse for a 1st order reaction $A \longrightarrow$ products so that the reaction is at least 95% completed? ($\log 2 = 0.3$)
- (A) 4 (B) 5 (C) 6 (D) 7
10. For a 1st order reaction (gaseous) (cont. V, T)
- $aA \longrightarrow (b-1)B + 1C$ (with $b > a$) the pressure of the system rose by $50\left(\frac{b}{a} - 1\right)\%$ in a time of 10 min. The half life of the reaction is therefore.
- (A) 10 min (B) 20 min (C) 30 min (D) 40 min
11. A reaction $2A + B \xrightarrow{K} C + D$ is first order with respect to A and 2nd order with respect to B. Initial conc. ($t = 0$) of A is C_0 while B is $2C_0$. If at t as 30 minutes the conc. of C is $C_0/4$ then rate expression at $t = 30$ minutes is :
- (A) $R = 7 C_0^3 k / 16$ (B) $R = 27 C_0^3 k / 32$ (C) $R = 247 C_0^3 k / 64$ (D) $R = 49 k C_0^3 / 32$

12. The conversion of vinyl allyl ether to pent-4-enol follows first-order kinetics. The following plot is obtained for such a reaction. The rate constant for the reaction is



- (A) $4.6 \times 10^{-2} \text{ s}^{-1}$ (B) $1.2 \times 10^{-2} \text{ s}^{-1}$ (C) $2.3 \times 10^{-2} \text{ s}^{-1}$ (D) $8.4 \times 10^{-2} \text{ s}^{-1}$
13. For the reaction $A \longrightarrow \text{products}$, the graph of the fraction of A remaining as a function of time (x-axis) is a straight line with -ve slope. The order of the reaction is therefore
 (A) 1 (B) 2 (C) zero (D) -1
14. S_1 : The frequency factor has the same units as the rate constant, k
 S_2 : A plot $\ln \text{rate}$ vs $\ln C$ for the nth order reaction gives a straight line with slope -n and intercept k_n
 S_3 : The order of a reaction $A \rightarrow \text{product}$ in which half the reagent is reacted is half an hour, three quarters in one hour and seven - eighth in one and half hours must be 1 (unity).
 S_4 : The unit of rate constant for a second order reaction will be $\text{M}^{-1}\text{s}^{-1}$. (M is representing the molarity of solution)
 (A) T F T T (B) T T T T (C) F F T T (D) T F F T

15. For the following parallel chain reaction
- $$\begin{array}{c}
 \text{A} \begin{cases} \xrightarrow{k_1 = 2 \times 10^{-4} \text{ s}^{-1}} 2\text{B} \\ \xrightarrow{k_2 = 3 \times 10^{-4} \text{ s}^{-1}} 3\text{C} \end{cases}
 \end{array}$$
- if the sum of the concentration of B and C at any time is 2M then what will be $[B]_t$ and $[C]_t$ respectively?

- (A) $\frac{11}{12}\text{M}, \frac{13}{12}\text{M}$ (B) $\frac{3}{4}\text{M}, \frac{5}{4}\text{M}$ (C) $\frac{4}{5}\text{M}, \frac{6}{5}\text{M}$ (D) $\frac{8}{13}\text{M}, \frac{18}{13}\text{M}$

ANSWER- KEY (QUIZ # 2)

1. (A) 2. (B) 3. (D) 4. (C) 5. (C) 6. (B) 7. (D) 8. (A) 9. (B)
 10. (A) 11. (D) 12. (C) 13. (C) 14. (A) 15. (D)

QUIZ # 03

Single Option correct

1. When the concentration of 'A' is 0.1 M, it decomposes to give 'X' by a first order process with a rate constant of $6.93 \times 10^{-2} \text{ min}^{-1}$. The reactant 'A' in the presence of catalyst given 'Y' by a second order mechanism with a rate constant of $0.2 \text{ min}^{-1} \text{ M}^{-1}$. In order that half-life of both the processes is 10 min, one should start with an initial concentration of 'A' as :
 (A) 0.01 M (B) 5.0 M (C) 10.0 M (D) 0.5 M
2. Non volatile compound 'X' dimerises according to 1st order kinetics ($t_{1/2} = 10 \text{ min.}$) If 10 mol of X and 10 mol of volatile solvent ($P_0 = 400 \text{ mm of Hg}$) are mixed initially. Calculate vapour pressure of solution after 20 minutes.
 (A) 320 mm (B) 246 mm (C) 200 mm (D) can not be predicted

3. A certain reactant XO_3^- is getting converted to X_2O_7 in solution. The rate constant of this reaction is measured by titrating a volume of the solution with a reducing agent which reacts only with XO_3^- & X_2O_7 . In this process of reduction both the compounds converted to X^- . At $t = 0$, the volume of the reagent consumed is 30 mL and at $t = 9.212$ min. the volume used up is 36 mL the rate constant of the conversion of XO_3^- to X_2O_7 assuming reaction is of 1st order.
 (Given that $\ln 10 = 2.303$, $\log 2 = 0.30$)
 (A) 0.2 min^{-1} (B) 0.02 min^{-1} (C) 0.01 min^{-1} (D) 0.1 min^{-1}
4. In a hypothetical reaction $\text{A(aq)} \rightleftharpoons 2\text{B(aq)} + \text{C(aq)}$ (1st order decomposition)
 'A' is optically active (dextro-rotatory) while 'B' and 'C' are optically inactive but 'B' takes part in a titration reaction (fast reaction) with H_2O_2 . Hence the progress of reaction can be monitored by measuring rotation of plane of plane polarised light or by measuring volume of H_2O_2 consumed in titration.
 In an experiment the optical rotation was found to be $\theta = 40^\circ$ at $t = 20$ min and $\theta = 10^\circ$ at $t = 50$ min. from start of the reaction. If the progress would have been monitored by titration method, volume of H_2O_2 consumed at $t = 15$ min. (from start) is 40 ml then volume of H_2O_2 consumed at $t = 60$ min will be:
 (A) 60 ml (B) 75 ml (C) 52.5 ml (D) 90 ml
5. Select the correct statements out of I, II and III for zeroth order reaction
 I : Quantity of the product formed is directly proportional to time
 II : Larger the initial concentration of the reactant, greater the half-life period
 III : If 50% reaction takes place in 100 minutes, 75% reaction take place in 150 minutes.
 (A) I only (B) I and II only (C) II and III only (D) I, II and III
6. Following are the values of E_a and ΔH for three reactions carried out at the same temperature :
 I : $E_a = 20 \text{ kJ mol}^{-1}$, $\Delta H = -60 \text{ kJ mol}^{-1}$
 II : $E_a = 10 \text{ kJ mol}^{-1}$, $\Delta H = -20 \text{ kJ mol}^{-1}$
 III : $E_a = 40 \text{ kJ mol}^{-1}$, $\Delta H = +15 \text{ kJ mol}^{-1}$
 If all the three reaction have same frequency factor then fastest and slowest reactions are :

Fastest	Slowest
(A) I	II
(B) II	III
(C) I	III
(D) can't be predicted.	
7. Two reaction, $\text{A} \longrightarrow \text{products}$ and $\text{B} \longrightarrow \text{products}$ have rate constants K_a and K_b at temperature T and activation energies E_a and E_b respectively. If $K_a > K_b$ and $E_a < E_b$ and assuming that A for both the reactions is same then :
 (A) At higher temperatures K_a will be greater than K_b
 (B) At lower temperature K_a and K_b will differ more and $K_a > K_b$
 (C) As temperature rises K_a and K_b will be close to each other in magnitude
 (D) All of the above
8. A reaction is catalysed by H^+ ion; and in the rate law the dependence of rate is of first order with respect to the concentration of H^+ ions, in presence of HA rate constant is $2 \times 10^{-3} \text{ min}^{-1}$ and in presence of HB rate constant is $1 \times 10^{-3} \text{ min}^{-1}$. HA and HB (both strong acids) have relative strength as :
 (A) 0.5 (B) 0.002 (C) 0.001 (D) 2

Paragraph for Question Nos. 9 to 12.

The thermal decomposition of N_2O_5 occurs as : $2\text{N}_2\text{O}_5 \longrightarrow 4\text{NO}_2 + \text{O}_2$

Experimental studies suggest that rate of decomposition of N_2O_5 , rate of formation of NO_2 or rate of formation of O_2 all becomes double if concentration of N_2O_5 is doubled.

9. The correct mechanism for decomposition of N_2O_5 is :
 (A) $\text{N}_2\text{O}_5 \xrightarrow{\text{Slow}} \text{NO}_2 + \text{NO}_3$; $\text{N}_2\text{O}_5 + \text{NO}_3 \xrightarrow{\text{Fast}} 3\text{NO}_2 + \text{O}_2$
 (B) $\text{N}_2\text{O}_5 \xrightarrow{\text{Fast}} \text{NO}_2 + \text{NO}_3$; $\text{N}_2\text{O}_5 + \text{NO}_3 \xrightarrow{\text{Slow}} 3\text{NO}_2 + \text{O}_2$
 (C) $\text{N}_2\text{O}_5 \xrightarrow{\text{Fast}} 2\text{NO}_2 + \frac{1}{2}\text{O}_2$
 (D) $\text{N}_2\text{O}_5 \xrightarrow{\text{Slow}} \text{NO} + \text{NO}_2 + 2\text{O}_2$; $\text{N}_2\text{O}_5 + \text{NO}_2 \xrightarrow{\text{Fast}} 3\text{NO}_2 + \frac{1}{2}\text{O}_2$

10. If rate constants for decomposition of N_2O_5 , formation of NO_2 and formation of O_2 are K_1 , K_2 and K_3 respectively, then :
- (A) $K_1 = K_2 = K_3$ (B) $2K_1 = K_2 = 4K_3$ (C) $K_1 = 2K_2 = K_3$ (D) $K_1 = K_2 = 2K_3$
11. If rate of formation of O_2 is 16 g/hr, then rate of decomposition of N_2O_5 and rate of formation of NO_2 respectively is :
- (A) cannot be calculated without knowing rate constants
 (B) 108 g/hr, 92 g/hr
 (C) 32 g/hr, 64 g/hr
 (D) 54 g/hr, 46 g/hr
12. The container of 2 litre contains 4 mole of N_2O_5 . On heating to $100^\circ C$, N_2O_5 undergoes complete dissociation to NO_2 and O_2 . Select the correct answers if rate constant for decomposition of N_2O_5 is $6.2 \times 10^{-4} \text{ sec}^{-1}$.
- The mole ratio before and after dissociation is 4 : 2.
 - Half life of N_2O_5 is 1117 sec and it is independent of temperature.
 - Time required to complete 40% of reaction is 824 sec.
 - If volume of container is doubled, the rate of decomposition becomes half of the initial rate :
- (A) 1, 3, 4 (B) 1, 2, 3, 4 (C) 3, 4 (D) 2, 3, 4

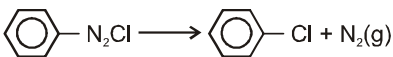
Paragraph for Question Nos. 13 to 14.

A 0.5 L reaction vessel which is equipped with a movable piston is filled completely with a 1M aqueous solution of H_2O_2 . The H_2O_2 decomposes to H_2O and $O_2(g)$ in a first order process with half life 10 hrs at 300 K. As gas decomposes, the piston moves up against constant external pressure of 750 mm Hg.

13. What is the net work done by the gas from the start of sixth hour till the end of 10 hrs ?
- (A) 100J (B) 120 J (C) 130 J (D) 150 J
14. If ΔH for decomposition of $H_2O_2(aq.)$ is X kJ/mole, how much heat has been exchanged with surrounding to maintain temperature at 300 K (in KJ) from the start of 6th hour till the end of 10th hour.
- (A) $-0.1035 X$ (B) $-0.2070 X$ (C) $-0.052 X$ (D) $0.026 X$

Multiple Correct Choice Type

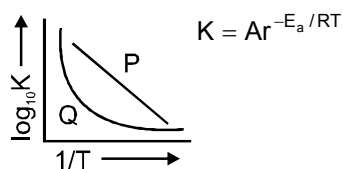
15. Table I (reactions) and table II (methods) have been matched

Table I (reactions)		Table II (method)	
P :	$A(g) \longrightarrow B(g) + C(g)$	X :	Titration method
Q :	$RCOOR' + H_2O \xrightarrow{H^+} RCOOH + R'OH$	Y :	Pressure measurement
R :		Z :	Volume measurement

Correct matching is :

- | | | | | | |
|-------|---|---|-------|---|---|
| P | Q | R | P | Q | R |
| (A) X | Y | Z | (B) Y | X | Z |
| (C) Z | X | Y | (D) Z | Y | X |
16. Which of the following statements are correct about half-period.
- (A) It is proportional to initial concentration for zeroth order
 (B) Average life = 1.44 (half-life for first order reaction)
 (C) Time of 75% completion of reaction is thrice of half-life (initial half life) period in second order reaction
 (D) 99.9% reaction takes place in 100 minutes for the case when rate constant is 0.0693 min^{-1}

17. Activation energy of a chemical reaction can be determined by :
- evaluating rate constant at standard temperature
 - evaluating velocities of reaction at two different temperatures
 - evaluating rate constant at two different temperatures
 - changing concentration of reactants
18. Which of the following statements are correct ?
- Law of mass action and rate law expressions are same for single step reactions.
 - Order of the slowest elementary reaction of a complex reaction gives the order of the complex reaction
 - Both order and molecularity have normally a maximum value of 3
 - Molecularity of a complex reaction,
 $A + 2B \longrightarrow C$ is 3.
19. Which of the following statement are true regarding the $\log K$ vs. $1/T$ plot shown in the given diagram ?



- Plot P shows that the energy of activation is independent of temperature.
 - Plot Q describes the behaviour of temperature dependence of energy of activation
 - Arrhenius behaviour is described by P
 - The slope of curve P gives the value $-\frac{E_a}{R}$
20. Select the correct statements :
- the molecularity of an elementary reaction indicates how many reactant species take part in the step
 - the rate law of an elementary reaction can be predicted by simply seeing the stoichiometry of reaction
 - the slowest elementary step in sequence of the reactions governs the overall rate of formation of product
 - a rate law is often derived from a proposed mechanism by imposing the steady state approximation or assuming that there is a pre-equilibrium
21. In the following gaseous phase first order reaction
- $$A(g) \longrightarrow 2B(g) + C(g)$$
- initial pressure was found to be 400 mm and it changed to 1000 mm after 20 min. Then :
- Half life for A is 10 min
 - Rate constant is 0.0693 min^{-1}
 - Partial pressure of C at 30 min is 350 mm
 - Total pressure after 30 min is 1150 mm

Matrix-Match

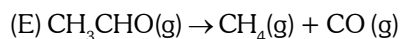
22. Match the given reaction of various order (in List I) with method of study of kinetics (in List II) :

List I (reaction)

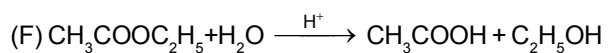
- $2N_2O_5(g) \rightarrow 2N_2O_4(g) + O_2(g)$
- $S_2O_8^{2-} + 2I^- \rightarrow 2SO_4^{2-} + I_2$
- $2H_2O_2 \rightarrow 2H_2O + O_2$
- $C_{12}H_{22}O_{11} + H_2O \rightarrow C_6H_{12}O_6 + C_6H_{12}O_6$

List II (method)

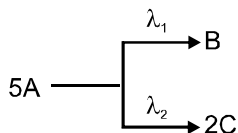
- Titration by $Na_2S_2O_3$
- Titration by $KMnO_4$
- Titration by $NaOH$
- Measuring pressure of the mixture in inert medium.



5. Measuring angle of rotation.

6. Measuring pressure of the gas in CCl_4 medium(use the fact that N_2O_4 dissolves in CCl_4 medium)**Subjective type question**

- 23.**
- A follows parallel path I order reactions giving B and C as :



If initial concentration of A is 0.25 M calculate the concentration of C after 5 Hrs. of reaction

Given $\lambda_1 = 1.5 \times 10^{-5} \text{ sec}^{-1}$; $\lambda_2 = 5 \times 10^{-6} \text{ sec}^{-1}$

- 24.**
- The reaction,
- $\text{A} + \text{OH}^- \longrightarrow \text{Products}$
- , obeys rate law expression as,
- $\frac{-d[\text{A}]}{dt} = K [\text{A}] [\text{OH}^-]$
- . If initial concentrations of [A] and
- $[\text{OH}^-]$
- are 0.002 M and 0.3 M respectively and if it takes 30 sec for 1% A to react at
- 25°C
- , calculate the rate constant for the reaction.

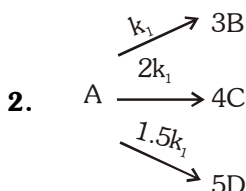
ANSWERKEY (QUIZ # 3)

- | | | | | | | | |
|---|-------------------------------------|------------|-------------|---------------|---------------|-----------|--------|
| 1. (D) | 2. (B) | 3. (D) | 4. (B) | 5. (D) | 6. (B) | 7. (D) | 8. (D) |
| 9. (A) | 10. (B) | 11. (B) | 12. (C) | 13. (C) | 14. (A) | 15. (B,C) | |
| 16. (A,B,C,D) | 17. (B,C) | 18. (A, C) | 19. (A,B,C) | 20. (A,B,C,D) | 21. (A,B,C,D) | | |
| 22. (A)- 6; (B) - 1; (C) - 2; (D) - 5; (E) - 4; (F) - 3 | 23. $7.56 \times 10^{-3} \text{ M}$ | | | | | | |
| 24. $1.2 \times 10^{-3} \text{ Lmol}^{-1} \text{ sec}^{-1}$ | | | | | | | |

CHEMICAL KINETICS

SINGLE CHOICE QUESTIONS

1. For the system $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$, $\Delta H = -80 \text{ kJ/mole}$
 If the activation energy for the forward step is 100 kcal/mole. What is the ratio of temperature at which the forward and backward reaction shows the same % change of rate constant per degree rise of temperature? (1 cal = 4.2 J)
- (A) $\sqrt{0.72}$ (B) $\sqrt{0.84}$ (C) $\sqrt{0.42}$ (D) $\sqrt{1}$



All reactions are of 1st order

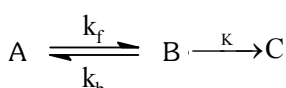
At time $t = t_1$ ($t_1 > 0$) $\frac{[B]}{[C]} = \alpha$. Therefore at

time $t = t_2$ (where $t_2 \geq t_1$) $\frac{[C]}{[D]} = \beta$ which of the following is correct.

- (A) $\alpha > \beta$ (B) $\alpha < \beta$ (C) $\alpha\beta = 0.4$ (D) both B and C
3. In the formation of HBr from H_2 & Br_2 , following mechanism is observed.
- (A) $Br_2 \rightleftharpoons 2Br\cdot$ (Equilibrium step)
 (B) $H_2 + Br\cdot \longrightarrow HBr + H\cdot$ (Slow step)
 (C) $H\cdot + Br_2 \longrightarrow HBr + Br\cdot$ (Fast step)

Calculate the rate of reaction, if concentration of hydrogen is twice that of bromine ($[Br_2] = 1 \text{ M}$) and the rate constant is equal to one rutherford.

- (A) $2 \times 10^6 \text{ dps}$ (B) $10 \times 10^9 \text{ dps}$ (C) $20.2 \times 10^{10} \text{ dps}$ (D) $4 \times 10^2 \text{ dps}$
4. In the following reaction

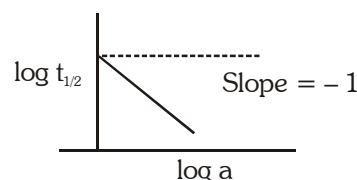


The equilibrium concentration of 'A' is 0.4 M. And the K_f, K_b & K values are $2 \times 10^6, 4 \times 10^3$ and $4 \times 10^{-1} \text{ sec}^{-1}$ respectively. Calculate rate of reaction

- (A) $2 \times 10^2 \text{ M sec}^{-1}$ (B) $4 \times 10^{-1} \text{ M sec}^{-1}$ (C) $8 \times 10^3 \text{ M sec}^{-1}$ (D) $8 \times 10^1 \text{ M sec}^{-1}$
5. A graph between $\log t_{1/2}$ and $\log a$ (abscissa) a being the initial concentration of A in the reaction

$A \rightarrow \text{Product}$, is depicted in the figure the rate law is

- (A) $\frac{-d[A]}{dt} = K$ (B) $\frac{-d[A]}{dt} = K[A]$
 (C) $\frac{-d[A]}{dt} = K[A]^2$ (D) $\frac{-d[A]}{dt} = K[A]^3$

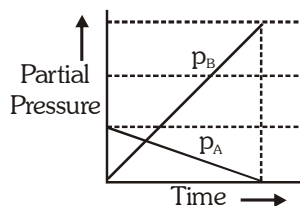


6. For the reaction $R-X + OH^- \rightarrow ROH + X^-$ The Rate is given of

Rate = $6.0 \times 10^{-5} [R-X][OH^-] + 0.20 \times 10^{-5} [R-X]$ what percentage of R-X Reaction by SN^2 mechanism when $[OH^-] = 1.0 \times 10^{-2} \text{ M}$

- (A) 96.1% (B) 3.9% (C) 80% (D) 23.07%

7. If for a reaction in which $A(g)$ converts to $B(g)$ the reaction carried out at const. V & T results into the following graph

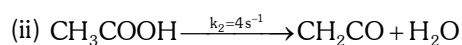
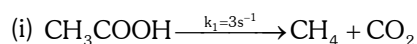


- (A) then the reaction must be $A(g) \rightarrow 3B(g)$ and is a first order reaction
 (B) then the reaction must be $A(g) \rightarrow 3B(g)$ and is a second order reaction
 (C) then the reaction must be $A(g) \rightarrow 3B(g)$ and is a zero order reaction
 (D) then the reaction must be $A(g) \rightarrow B(g)$ and is a first order reaction
8. For a certain reaction the variation of the rate constant with temperature is given by the equation

$$\ln k_t = \ln k_0 + 0.0693 t \quad (t \geq 0^\circ\text{C})$$

The value of the temperature coefficient of the reaction rate is therefore

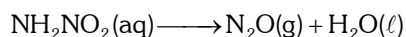
- (A) 0.1 (B) 1.0 (C) 10 (D) 2
9. At high temperature acetic acid decomposes into CO_2 & CH_4 and simultaneously into CH_2CO (ketene) and H_2O



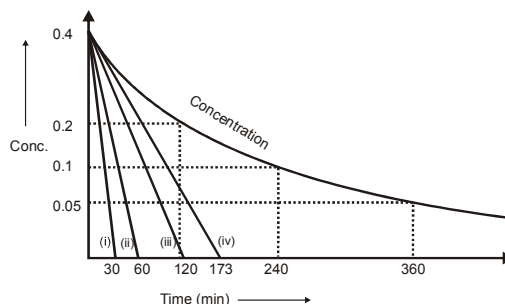
What is the fraction of acetic acid reacting as per reaction (i)

- (A) $\frac{3}{4}$ (B) $\frac{3}{7}$ (C) $\frac{4}{7}$ (D) None of these

10. Consider the reaction



The concentration of nitramide as a function of time is shown below for a particular run.



Which the represents the correct tangent to the graph at the origin ($t = 0$) ?

- (A) (i) (B) (ii) (C) (iii) (D) (iv)
11. For the decomposition of $\text{H}_2\text{O}_2(\text{aq})$ it was found that V_{O_2} ($t = 15$ min.) was 100 mL (at 0°C and 1 atm) while V_{O_2} (maximum) was 200 mL (at 0°C and 2 atm). If the same reaction had been followed by the titration method and if $V_{\text{KMnO}_4}^{(\text{cM})}$ ($t = 0$) had been 40 mL, what would $V_{\text{KMnO}_4}^{(\text{cM})}$ ($t = 15$ min) hve been ?
- (A) 30 mL (B) 25 mL (C) 20 mL (D) 15 mL

12. For the system $A \xrightleftharpoons[k_{-1}]{k_1} B$, $A \xrightleftharpoons[k_{-2}]{k_2} C$ which was started with only A the equilibrium concentration $[A]_{eq}$ is correctly related to $[B]_{eq}$ and $[C]_{eq}$ as :

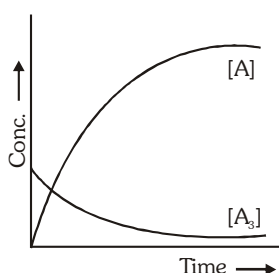
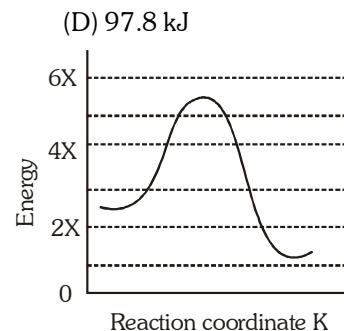
(A) $\frac{k_{-1}[B]_{eq} + k_{-2}[C]_{eq}}{(k_1 + k_2)}$ (B) $\frac{k_{-1}[B]_{eq} - k_{-2}[C]_{eq}}{(k_2 - k_1)}$ (C) $\frac{k_{-1}[B]_{eq} + k_{-2}[C]_{eq}}{(k_1 - k_2)}$ (D) $\frac{k_{-1}[B]_{eq} - k_{-2}[C]_{eq}}{(k_1 + k_2)}$

13. For a certain reaction that follows second order kinetics, $A + B \longrightarrow C + D + E$ the value of the rate constant, k , was measured at several different temperatures and the data are shown below :

Temperature ($^{\circ}\text{C}$)	k , ($\text{M}^{-1}\text{s}^{-1}$)
100	6.264
150	45.464
200	217.008
250	768.232

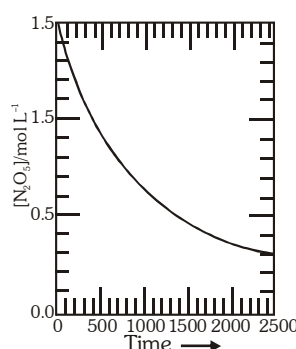
Calculate the value of the activation energy, E_a for this reaction.

- (A) 143 kJ (B) 27.4 kJ (C) 52.0 kJ (D) 97.8 kJ
14. $X = 50\text{J}$ in the diagram. Choose the correct statements regarding the reaction represented by the diagram. Answer format : Mark (A) if I, II, and III are correct; (B) if I and III are correct; (C) if II & IV are correct; (D) if none is correct.
- (I) Energy of the reactants is approximately 65 J.
 (II) Energy change for the reaction is approximately -37 J .
 (III) Activation energy for the reaction is approximately 73 J.
 (IV) Energy of the activated complex is approximately 73 J.
15. Consider the diagram in the question depicting the change in concentration of products and reactants with time for the following first order decomposition reaction $A_3 \longrightarrow 3A$

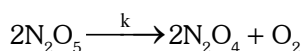


The time corresponding to point of intersection of the two concentration curves pertains to

- (A) $t_{1/2}$ (B) $t_{1/3}$ (C) $t_{1/4}$ (D) $t_{1/5}$
16. The first order decomposition of dinitrogen pentaoxide, N_2O_5 , was studied in chloroform solution at 45°C and the results are expressed in the following diagram.

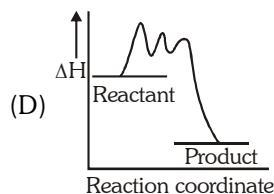
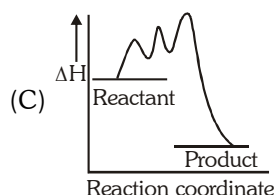
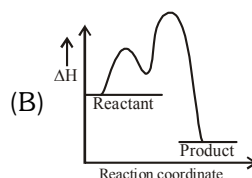
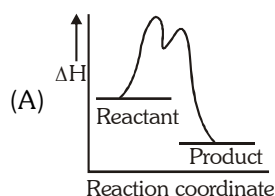


From the diagram, The expected value for the specific rate constant k for the reaction is

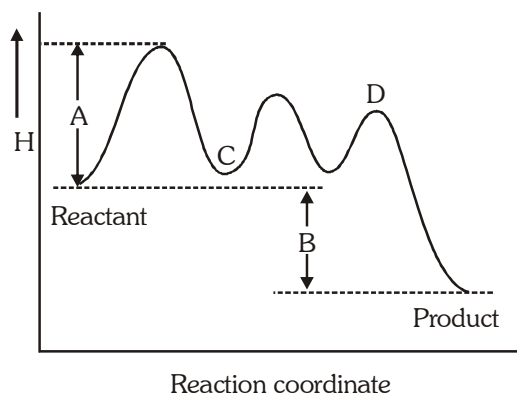


- (A) $3.1 \times 10^{-4} \text{ s}^{-1}$ (B) $9.7 \times 10^{-5} \text{ s}^{-1}$ (C) $8.7 \times 10^{-3} \text{ s}^{-1}$ (D) $5.4 \times 10^{-6} \text{ s}^{-1}$

17. Which one of the following reaction energy diagrams best represents a reaction in the forward direction; that is spontaneous/exothermic and complete in two consecutive steps in which the second step is little more difficult than the first step?



18. Given the following potential energy diagram :

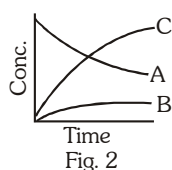
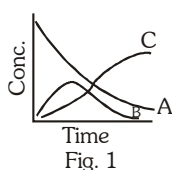


Which of the following is correctly labeled and be specific ?

- (A) $A = E_a$; $B = \Delta H$; C is 1st transition state and D is 3rd intermediate.
 (B) $A = \Delta H$; $B = E_a$; C is 1st intermediate and D is 3rd transition state.
 (C) $A = E_a$; $B = \Delta H$; C is 1st intermediate and D is 1st transition state.
 (D) $A = E_a$; $B = \Delta H$; C is 1st intermediate and D is 3rd transition state.

ONE OR MORE THAN ONE CORRECT OPTIONS

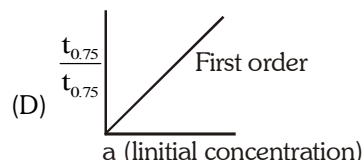
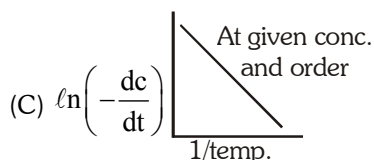
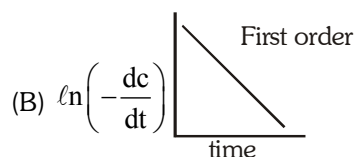
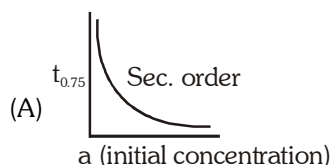
19. For the consecutive reaction $A \xrightarrow{k_1 (\text{time}^{-1})} B \xrightarrow{k_2 (\text{time}^{-1})} C$ following curves were obtained depending upon the relative values of k_1 & k_2



Now much of the following is the correct match

- (A) figure 1 - ($k_1 < k_2$) (B) figure 2 - ($k_1 < k_2$)
 (C) figure 2 - ($k_1 >> k_2$) (D) figure 1 - ($k_1 >> k_2$)

20. Which is **correct** graph :



21. For a certain reaction $A \longrightarrow$ products, the $t_{1/2}$ as a function of $[A]_0$ is given as below :

$[A]_0$ (M)	0.1	0.025
$t_{1/2}$ (min.)	100	50

Which of the following is true :

(A) The order is $1/2$

(B) $t_{1/2}$ would be $100\sqrt{10}$ min for $[A]_0 = 1\text{ M}$

(C) The order is 1

(D) $t_{1/2}$ would be 100 min for $[A]_0 = 1\text{ M}$

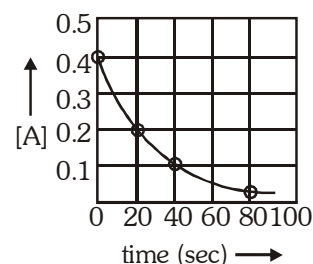
22. A certain reaction $A \longrightarrow B$ follows the given concentration (Molarity) - time graph. Which of the following statement (s) is/are true ?

(A) The reaction is second order with respect to **A**

(B) The rate of this reaction at 20 second will be $7 \times 10^{-3} \text{ M s}^{-1}$

(C) The rate for this reaction at 80 second will be $1.75 \times 10^{-3} \text{ M s}^{-1}$

(D) The $[B]$ will be 0.35 M at $t = 60$ second



23. For the reaction $2A + B \longrightarrow C$ with the rate law $\frac{d[C]}{dt} = k[A]^1[B]^{-1}$ and started with A and B in stoichiometric proportion which is/are true ?

(A) unit of k is $\text{Mole litre}^{-1} \text{ s}^{-1}$

(B) $[A]$, $[B]$ and $[C]$ all will be linear functions of time

(C) $[C] = 2kt$

(D) $[C] = kt$

24. The polarimeter readings in an experiment to measure the rate of inversion of cane sugar (1st order reaction) were as follows

Time (min)	0	30	∞
angle (degree)	30	20	-15

Identify the **true** statement (s) $\log 2 = 0.3$, $\log 3 = 0.48$, $\log 7 = 0.84$, $\log_e 10 = 2.3$

(A) The half life of the reaction is 75 min

(B) The solution is optically inactive at 120 min.

(C) The equimolar mixture of the products is dextrorotatory

(D) The angle would be 7.5° at half time

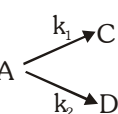
25. The ozone layer in the earth's upper atmosphere is important in shielding the earth from very harmful ultraviolet radiation. The ozone, O_3 , decomposes according to the equation $2O_3(g) \rightarrow 3O_2(g)$. The mechanism of the reaction is thought to proceed through an initial fast, reversible step and then a slow second step.

Step 1 : Equilibrium step $O_3(g) \rightleftharpoons O_2(g) + O(g)$

Step 2 : Slow $O_3(g) + O(g) \rightarrow 2O_2(g)$

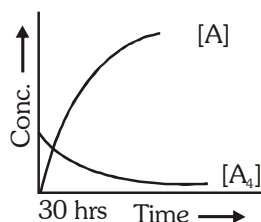
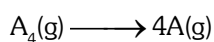
The which of the following is correct ?

- (A) Step 2 is rate determining step
- (B) the rate expression for step 2 is $-\frac{d}{dt}[O_3] = k[O_3][O]$
- (C) For Step 1, molecularity is 1
- (D) For Step 2, molecularity if 2

26. Consider the following case of competing 1st order reactions A
- 

After the start of the reaction at $t = 0$ with only A, the $[C]$ is equal to the $[D]$ at all times. The time in which all three concentrations will be equal is given by

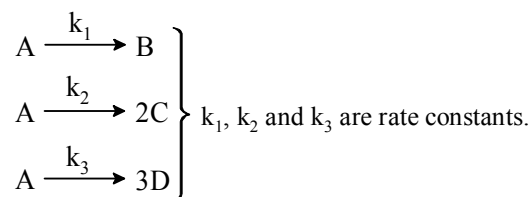
- (A) $t = \frac{1}{2k_1} \ln 3$ (B) $t = \frac{1}{2k_2} \ln 3$ (C) $t = \frac{1}{3k_1} \ln 2$ (D) $t = \frac{1}{3k_2} \ln 2$
27. Consider the following first order decomposition and the accompanying concentration-time plot :



Which of the following statements regarding the decomposition reaction is/are true ?

- (A) At 30 hrs, only 20% reaction is complete.
- (B) Half-life of the reaction is 90 hrs (approximately).
- (C) Rate of decomposition decreasing linearly with time.
- (D) Changing initial concentration will change the time at which the two curves are intersecting.
28. Consider the following first order general reaction :
- $$A \xrightarrow{k} P$$
- If 'a' represents initial concentration of reactant and 'x' represent instantaneous concentration of product P, then which of the following statement(s) is/are correct ?
- (A) If concentration of both 'A' and 'P' are plotted on the same concentration time curve, two curves will intersect at half-life.
- (B) If $(dx/dt)_{t=0}$ is plotted against 'a', a straight line results with slope k.
- (C) At any time 't', fraction of A remaining = e^{-kt} .
- (D) At any time 't', amount of P, i.e. $x = a(1 - e^{-kt})$

29. A chemical reaction of unknown order : $A \longrightarrow P$, is started with only 'A' at an initial concentration of 2×10^{-3} M and after 30 min, concentration of A was found to be 10^{-3} M, while after 90 min from beginning concentration of A was found to be 5×10^{-4} M. Which of the following is/are true statement(s)
- (A) The order of reaction is two
 (B) After 3.0 hr from beginning, concentration of A will decrease to 2.5×10^{-4} M.
 (C) A plot of [A] versus 't' will be a straight line [A] is concentration of A remaining at any time 't'
 (D) The half-life of reaction is $(k[A]_0)^{-1}$.
30. Consider the following parallel reactions, all are first order :



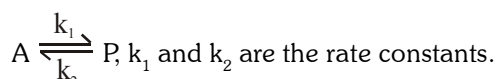
Which of the following statements are incorrect ?

- (A) B, C and D are produced in 1 : 2 : 3 molar ratio.
 (B) Concentration of neither B, C and D ever exceeds A.
 (C) If $k_1 > k_2 > k_3$, at any time 't', the order of concentrations are : $[B] > [C] > [D]$
 (D) At any time 't' the ratio of concentrations $[B] : [C] : [D]$ is $k_1 : k_2 : k_3$.

31. Consider a successive reaction (all first order) $A \xrightarrow{k_1} B \xrightarrow{k_2} C \xrightarrow{k_3} D$

The correct statement(s) is/are

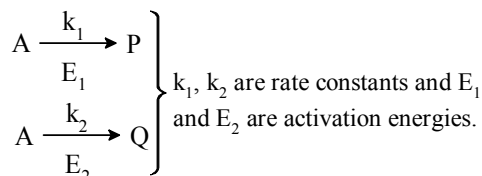
- (A) Concentration of A decreases exponentially with time.
 (B) Concentration of both B and C first increases, reaches a maxima then decreases
 (C) If $k_1 < k_2$ and $k_2 < k_3$, $[B]_{\max}$ will be greater than $[C]_{\max}$
 (D) None of these
32. Consider the following first order process : $A \longrightarrow P$
- At 27°C , addition of a catalyst lowers the half-life to one-tenth of the original half-life. Which of the following is (are) true deduction ?
- (A) Activation energy of catalyzed path is 5744 (rounded off) joule less than the activation energy of uncatalyzed path
 (B) If temperature is increased to 127°C and then a catalyst is added, half-life will again decrease to one-tenth of uncatalyzed half-life.
 (C) Addition of a catalyst will have greater effect at lower temperatures.
 (D) A catalyst increases the rate of a reaction by a certain factor independent of experimental temperature
33. For a reversible reaction (first order in both directions)



If equilibrium constant (K) is greater than unity, then

- (A) Rate of forward reaction is always greater than rate of backward reaction.
 (B) Amount of product of equilibrium is greater than the amount of reactant.
 (C) Activation energy for the forward direction (E_{af}) is greater than activation energy for backward reaction (E_{ab}).
 (D) Addition of a catalyst will increase both k_1 and k_2 by same factor.

34. Consider the following parallel, first order processes



The correct deduction(s) is (are)

(A) At any time, the ratio of concentrations of P and Q is $k_1 : k_2$ if reaction starts with pure reactant.

(B) If a catalyst is added, it will decrease both E_1 and E_2 but by same amount.

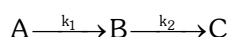
(C) If 'E' represents the overall activation energy for decay of A, it will be $\left(\frac{E_1 k_1 + E_2 k_2}{k_1 + k_2} \right)$

(D) If $k_1 > k_2$, E_1 must be smaller than E_2 if both path have same pre-exponential factor.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

The reactions in which initial reactant converts into final products through lot of intermediate products is known as sequential reactions or consecutive reactions. For example : if A converts into B and B into C and so many such reactions are known as sequential or consecutive reactions. For the following first order sequential reaction A



Initially only A was present, not B and C. B can only exist in this sequential reaction, when its rate of formation must be higher than that of its rate of decomposition, which is governed by its rate constant of formation and deformation i.e., k_1 and k_2 respectively. Higher the value of k_1 , higher the rate of formation of B and higher the value of k_2 , higher the rate of decomposition of B. The stage of the reaction, where the rate of formation of B becomes equal to its rate of decomposition is known as steady state of B.

Now from the above sequential or consecutive reaction, it is very clear that

$$-\frac{d[A]}{dt} = k_1[A]_t \quad \text{or} \quad [A]_t = [A]_{0e^{-k_1 t}} \dots\dots\dots (i)$$

$$+\frac{d[B]}{dt} = k_1[A]_t - k_2[B]_t \quad \text{or} \quad \frac{d[B]}{dt} + k_2[B]_t = k_1[A]_t = k_1[A]_{0e^{-k_1 t}} \dots\dots\dots (ii)$$

Solving first order differential equation (ii), with proper integrating factor, we get

$$[B]_t = \frac{k_1[A]_0}{k_2 - k_1} [e^{-k_1 t} - e^{-k_2 t}]$$

In sequential reaction, the time required for maximum concentration of B is known as $t_{\max}$, which is obtained by the slope of B versus time.

Please keep in mind that if B be initially present, then the remaining concentration of B is taken along with produced concentration of B for A.

Answer the following questions on the basis of the above write up :

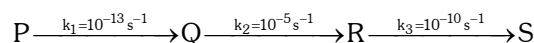
35. If the initial concentration of A with which the reaction started is 2 M with $k_1 = 10^{-6} \text{ s}^{-1}$ and $k_2 = 10^{-1} \text{ s}^{-1}$ for the above sequential reaction, when what is the concentration of C formed if the reaction is carried out for 4.606×10^6 seconds ?

(A) 0.45 (B) 0.55 (C) 0.9 (D) 1.98

36. For the steady state of the above consecutive reaction at any time t, if the half life of B is 10 minutes, then what will be the half-life of A ? Given that concentration of A at time t is four times of B.

(A) 10 minutes (B) 20 minutes (C) 25 minutes (D) 40 minutes

37. If the above sequential reaction be replaced by the following sequential



then which of the following will be rate determining step ?

- (A) $P \longrightarrow Q$ (B) $Q \longrightarrow R$ (C) $R \longrightarrow S$ (D) $P \longrightarrow S$

38. For the sequential reaction in question number (37), what will be the rate of formation of S ?

- (A) $k_2 [Q]_t$ (B) $k_3 [P]_t$ (C) $k_3 [R]_t$ (D) $k_1 [P]_t$

39. In the following sequential reaction, $A \xrightarrow{k_1=10^{-1} \text{ s}^{-1}} B \xrightarrow{k_2=10^{-2} \text{ s}^{-1}} C$

then what will be the value of t_{max} maximum concentration of B ?

- (A) 100 seconds (B) 60 seconds (C) 45 seconds (D) 25.5 seconds

40. If the above sequential reaction be carried out with $k_1 = 10^{-1} \text{ s}^{-1}$ and $k_2 = 10^{-2} \text{ s}^{-1}$ with the initial concentration of A is 10 M, then what will be the value of maximum concentration of B at time t ?

- (A) $10 \times \frac{1}{(10)^{1/9}} \text{ M}$ (B) $10 \times \frac{1}{(10)^{1/8}} \text{ M}$ (C) $10 \times \frac{1}{(10)^{1/5}} \text{ M}$ (D) $\sqrt{10} \text{ M}$

41. For the sequential chain reaction $X \xrightarrow{k_1=10^{-1} \text{ s}^{-1}} Y \xrightarrow{k_2=10^{-2} \text{ s}^{-1}} Z$, what will be the value of the time required for maximum rate of change of concentration of Y ?

- (A) 100 seconds (B) 30 seconds (C) 40 seconds (D) 51 seconds

42. For the following first order sequential reaction (where $k_1 > k_2$), $A \xrightarrow{k_1} B \xrightarrow{k_2} C$

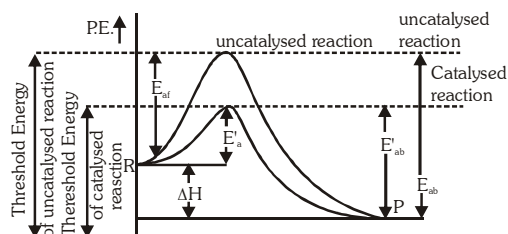
if initially A with conc. $[A]_0$ and B with conc. $[B]_0$ where present, then what will be the concentration of B at time t ?

- (A) $\frac{k_1[A]_0}{k_2 - k_1} [e^{-k_1 t} - e^{-k_2 t}] + [B]_0 e^{-k_1 t}$ (B) $\frac{k_2[A]_0}{k_2 - k_1} [e^{-k_1 t} - e^{-k_2 t}] + [B]_0 e^{-k_1 t}$
 (C) $\frac{k_1[A]_0}{k_1 - k_2} [e^{-k_1 t} - e^{-k_2 t}] + [B]_0 e^{-k_1 t}$ (D) $\frac{k_1[A]_0}{k_2 - k_1} [e^{-k_1 t} - e^{-k_2 t}] + [B]_0 e^{-k_2 t}$

Comprehension # 2

The energy profile diagrams are very important tool through which, we predict lot of things, like the amount of activation energy of the catalysed and uncatalysed reaction as well as for the backward reaction and forward reaction. Threshold energy of the catalysed and uncatalysed reaction.

Following is the graph which represents the energy profile diagrams for the catalysed and the uncatalysed reaction. The graph of the lower energy barrier is the energy profile diagram for the catalysed reaction and of the catalysed reaction and of the higher energy barrier is the energy profile diagram of uncatalysed reaction.



From the graph, it is very clear that E_{af} and E_{ab} are the activation energy of the uncatalysed forward and backward reactions respectively. E'_{af} and E'_{ab} are the activation energy of the catalysed forward and backward reaction respectively.

The reaction which occurs by the absorption of heat is known as endothermic reaction and by release of heat, or energy is known exothermic reaction. ΔH is known as change in enthalpy of the reaction.

For exothermic reaction, ΔH is negative and for endothermic reaction, ΔH is positive from Arrhenius equation,

we know that $k = Ae^{\frac{-E_a}{RT}}$

where k is rate constant, which signifies effective number of collisions.

A is known as Arrhenius constant, or maximum rate constant or, frequency factor which signifies maximum number of collision with proper orientation of the reacting species per mol per second. While,

Equilibrium constant $= k_{\text{equm.}} = Ae^{\frac{-\Delta H}{RT}}$

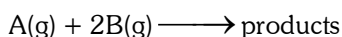
The catalysed reaction occurs at 500 K and the uncatalysed reaction occurs at 800 K in such a way so that the rate of catalysed reaction and uncatalysed reaction becomes equal as well as the catalysed lower down the energy barrier by 60 J.

Answer the following questions on the basis of above write up :

43. What will be the value of activation energy to the uncatalysed reaction ?
 (A) 120 J (B) 100 J (C) 130 J (D) 160 J
44. From given above information, what will be the threshold energy for the uncatalysed reaction, if the normal energy of the reactant is 40 J ?
 (A) 90 J (B) 120 J (C) 170 J (D) 200 J
45. From the given above information, what will be threshold energy for the catalysed reaction, if the normal energy of the reactant is 40 J ?
 (A) 170 J (B) 200 J (C) 140 J (D) 180 J
46. From the above information given in comprehension, if the normal energy of the reactant is 50 J and that of product is 20 J, then what will be the activation energy of the uncatalysed backward reaction ?
 (A) 170 J (B) 180 J (C) 160 J (D) 190 J
47. From the above information given in comprehension as well as from question no. 46 What will be the value of activation energy of the catalysed backward reaction ?
 (A) 150 J (B) 130 J (C) 160 J (D) 190 J
48. On the basis of question no. 46, what will be the value of threshold energy of the uncatalysed reaction
 (A) 180 J (B) 210 J (C) 170 J (D) 190 J
49. On the basis of questions no. 46, what will be the value of threshold energy of the catalysed reaction ?
 (A) 150 J (B) 190 J (C) 170 J (D) 200 J
50. Which of the following statements are correct about the catalysed and uncatalysed reaction ?
 (A) For catalysed and uncatalysed reaction, ΔH of the reaction becomes same.
 (B) At constant temperature, equilibrium constant for catalysed reaction and for the uncatalysed remains same.
 (C) Even at constant temperatures the equilibrium constant of the catalysed and uncatalysed reactions are not same.
 (D) Both (A) and (B).
51. Which of the following statements are correct for the catalysed and uncatalysed reactions ?
 (A) At constant temperature, the ΔG° of the catalysed and uncatalysed reactions are same
 (B) Even at constant temperature, the ΔG° of the catalysed and the uncatalysed reactions are different.
 (C) At constant temperature, for the catalysed and uncatalysed reactions ΔS (entropy change) are same.
 (D) Both (A) and (C).
52. What percentage fraction of the molecule will cross over the energy barrier at 2000 K temperature for 36.848 kJ activation energy. (Given $R = 8 \text{ J mol}^{-1} \text{ K}^{-1}$)
 (A) 10% (B) 20% (C) 90% (D) 80%

Comprehension # 3

For a particular reaction at constant temperature,



Following rate information were recorded :

Exp. No.	Initial [A]	Initial [B]	Initial rate ($M \text{ min}^{-1}$)
1	1.00	1.00	1.00
2	2.00	4.00	8.00
3	3.00	9.00	27.00
4	4.00	2.00	?

Answer the following questions based on the above observation.

53. Which of the following is the correct rate law for the given reaction ?
 (A) $R = k[A][B]^2$ (B) $R = k[A]^2[B]$ (C) $R = k[A]^2[B]^{1/2}$ (D) $R = k[A]^{1/2}[B]^2$
54. What is the value of rate constant ?
 (A) $1.00 M^{-3/2} \text{ min}^{-1}$ (B) $4.00 M^{-3/2} \text{ min}^{-1}$ (C) $9.00 M^{-3/2} \text{ min}^{-1}$ (D) $5.6 M^{-3/2} \text{ min}^{-1}$
55. What is the rate of reaction in the 4th experiment ?
 (A) $15.3 M \text{ min}^{-1}$ (B) $22.4 M \text{ min}^{-1}$ (C) $29.4 M \text{ min}^{-1}$ (D) $36 M \text{ min}^{-1}$

MATCH THE COLUMN

56. **Column - I**

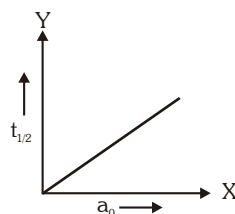
(Order of reaction)

Column - II

(Its corresponding graph with its initial conc, a_0)

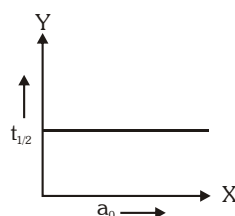
(A) First

(p)



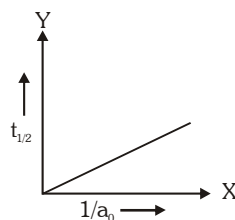
(B) Second

(q)



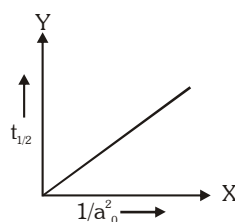
(C) Third

(r)



(D) Zero

(s)



(A) (A- q), (B - r); (C-s); (D-p)

(B) (A- r), (B - q); (C-s); (D-p)

(C) (A- q), (B - r); (C-p); (D-s)

(D) (A- s), (B - r); (C-q); (D-p)

57. Column - I**Column - II**

- | | |
|--|---|
| (A) Half life for 1 st order reaction | (p) $t_{1/2} = \frac{\ln 2}{k}$ |
| (B) Half life for 2 nd order reaction | (q) $t_{1/2} = \frac{1}{k \times a}$ |
| (C) Half life for zero order reaction | (r) $t_{1/2} = \frac{a}{2k}$ |
| (D) Half life for 3 rd order reaction | (s) $t_{1/2} = \frac{3}{2k \times a^2}$ |
| (A) (A- p), (B - q); (C-r); (D-s) | (B) (A- q), (B - p); (C-r); (D-s) |
| (C) (A- r), (B - q); (C-p); (D-s) | (D) (A- s), (B - q); (C-r); (D-p) |

58. Match the following fraction of time for the 1st order reaction**Column - I****Column - II**

- | | |
|-----------------------------------|-----------------------------------|
| (A) $t_{63/64}$ | (p) $6t_{1/2}$ |
| (B) $t_{15/16}$ | (q) $2t_{3/4}$ |
| (C) $t_{31/32}$ | (r) $1.67 \times t_{7/8}$ |
| (D) $t_{255/256}$ | (s) $2.t_{15/16}$ |
| (A) (A- p), (B - q); (C-r); (D-s) | (B) (A- q), (B - p); (C-r); (D-s) |
| (C) (A- r), (B - q); (C-p); (D-s) | (D) (A- s), (B - q); (C-r); (D-p) |

59. Match the following :**Column - I****Column-II**

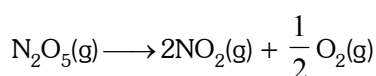
- | | |
|--------------------------------------|---|
| (A) $k_f > k_b$ | (p) Collision of reactants has greater probability of forming product than collision of product forming reactants |
| (B) $\Delta H > 0$ | (q) E_a (forward) < E_a (backward) |
| (C) $A_f > A_b$ | (r) E_a (forward) > E_a (backward) |
| (D) $\Delta H < 0$ | (s) Backward reaction is thermodynamically more difficult than the forward reaction. |
| (A) (A- p), (B - q); (C-r); (D-s) | (B) (A- q), (B - p); (C-r); (D-s) |
| (C) (A- p), (B - r); (C-p); (D-q, s) | (D) (A- s), (B - q); (C-r); (D-p) |

LEVEL-1**SUBJECTIVE TYPE QUESTIONS**

- At 27°C it was observed during a reaction of hydrogenation that the pressure of hydrogen gas decreases from 2 atmosphere to 1.1 atmosphere in 75 minutes. Calculate the rate of reaction (in M sec⁻¹) and rate of reaction in terms of pressure.
- For the elementary reaction $2A + B_2 \longrightarrow 2AB$. Calculate how much the rate of reaction will change if the volume of the vessel is reduced to one third of its original volume?
- For the reaction $3BrO^- \rightarrow BrO_3^- + 2Br^-$ in an alkaline aqueous solution, the value of the second order (in BrO⁻) rate constant at 80°C in the rate law for $-\frac{\Delta[BrO^-]}{\Delta t}$ was found to be 0.056 L mol⁻¹ s⁻¹. What is the rate of constant when the rate law is written for ,

(a) $\frac{\Delta[BrO_3^-]}{\Delta t}$, (b) $\frac{\Delta[Br^-]}{\Delta t}$?

- Dinitropentaoxide decomposes as follows :



Given that

$$-d[N_2O_5]/dt = k_1[N_2O_5]$$

$$d[NO_2]/dt = k_2[N_2O_5]$$

$$d[O_2]/dt = k_3[N_2O_5]$$

What is the relation between k_1 , k_2 and k_3 ?

- Suppose that the Sun consists entirely of hydrogen atom and releases the energy by the nuclear reaction, $4 {}^1_1H \longrightarrow {}^4_2He$ with 26 MeV/atom of energy released. If the total output power of the Sun is assumed to 26 remain constant at 3.9×10^{26} W, find the time it will take to burn all the hydrogen. Take the mass of the Sun as 1.7×10^{30} kg.
- In the following reaction, rate constant is $1.2 \times 10^{-2} \text{ M s}^{-1}$ $A \longrightarrow B$. What is concentration of B after 10 and 20 min., if we start with 10 M of A.
- From the following data for the zero order reaction $A \longrightarrow \text{products}$. Calculate the value of k.

Time (min.)	[A]
0	0.10 M
1	0.09 M
2	0.08 M

- A certain substance A is mixed with an equimolar quantity of substance B. At the end of an hour A is 75% reacted. Calculate the time when A is 10% unreacted. (Given: order of reaction is zero)
- A first order reaction is 75% completed in 72 min.. How long time will it take for
(i) 50% completion (ii) 87.5% completion
- A drug is known to be ineffective after it has decomposed 30%. The original concentration of a sample was 500 units/ml. When analyzed 20 months later, the concentration was found to be 420 units/ml. Assuming that decomposition is of I order, what will be the expiry time of the drug?
- A viral preparation was inactivated in a chemical bath. The inactivation process was found to be first order in virus concentration. At the beginning of the experiment 2.0 % of the virus was found to be inactivated per minute . Evaluate k for inactivation process.
- At 800° C the rate of reaction $2NO + H_2 \longrightarrow N_2 + H_2O$
Changes with the concentration of NO and H₂ are

S.No.	[NO] in M	[H ₂] in M	$-\frac{1}{2} \frac{d[NO]}{dt}$ in M sec ⁻¹
(i)	1.5×10^{-4}	4×10^{-3}	4.4×10^{-4}
(ii)	1.5×10^{-4}	2×10^{-3}	2.2×10^{-4}
(iii)	3.0×10^{-4}	2×10^{-3}	8.8×10^{-4}

- (a) What is the order of this reaction? (b) What is the rate equation for the reaction?
 (c) What is the rate when.
 $[H_2] = 1.5 \times 10^{-3} \text{ M}$ and $[NO] = 1.1 \times 10^{-3} \text{ M}$?

- 13.** The catalytic decomposition of N_2O by gold at 900°C and at an initial pressure of 200 mm is 50% complete in 53 minutes and 73% complete in 100 minutes.
- (i) What is the order of the reaction?
- (ii) Calculate the velocity constant.
- (iii) How much of N_2O will decompose in 100 min. at the same temperature but at initial pressure of 600 mm?

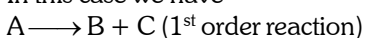
- 14.** The pressure of a gas decomposing at the surface of a solid catalyst has been measured at different times and the results are given below

t (sec)	0	100	200	300
P_g (Pascal)	4×10^3	3.5×10^3	3×10^3	2.5×10^3

Determine the order of reaction, its rate constant.

- 15.** The half life period of decomposition of a compound is 50 minutes. If the initial concentration is halved, the half life period is reduced to 25 minutes. What is the order of reaction?

- 16.** In this case we have



Time	t	∞
Total pressure of A + B + C	P_2	P_3

Find k.

- 17.** $\text{A} \longrightarrow \text{B} + \text{C}$ (1st order reaction)

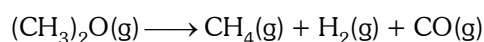
Time	t	∞
Total pressure of (B + C)	P_2	P_3

Find k.

- 18.** The reaction $\text{AsH}_3(\text{g}) \longrightarrow \text{As}(\text{s}) + \frac{3}{2} \text{H}_2(\text{g})$ was followed at constant volume at 310°C by measuring the gas pressure at intervals Show from the following figures that reaction is of first order.

Time (in hrs)	0	5	7.5	10
Total pressure (in mm)	758	827	856	882

- 19.** The thermal decomposition of dimethyl ether as measured by finding the increase in pressure of the reaction



at 500°C is as follows:

Time (sec.)	390	1195	∞
Pressure increase (mm Hg)	96	250	624

the initial pressure of ether was 312 mm Hg. Write the rate equation for this reaction and determine the rate constant of reaction.

- 20.** From the following data show that decomposition of H_2O_2 in aqueous solution is first order.

Time (in minutes)	0	10	20
Volume (in c.c. of KMnO_4)	22.8	13.3	8.25

- 21.** The following data were obtained in experiment on inversion of cane sugar.

Time (minutes)	0	60	120	180	360	∞
Angle of rotation (degree)	13.1	11.6	10.2	9	5.87	-3.8

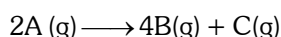
Show that the reaction is of first order. After what time would you expect a zero reading in polarimeter?

- 22.** The decomposition of N_2O_5 according to the equation $2 \text{N}_2\text{O}_5(\text{g}) \longrightarrow 4 \text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ is a first order reaction. After 30 min. from start of decomposition in a closed vessel the total pressure developed is found to be 284.5 mm Hg. On complete decomposition, the total pressure is 584.5 mm Hg. Calculate the rate constant of the reaction.

23. A definite volume of H_2O_2 under going spontaneous decomposition required 22.8 c.c. of standard permanganate solution for titration. After 10 and 20 minutes respectively the volumes of permanganate required were 13.8 and 8.25 c.c.

(a) Find order of reaction. How may the result be explained?
 (b) Calculate the time required for the decomposition to be half completed.
 (c) Calculate the fraction of H_2O_2 decomposed after 25 minutes.

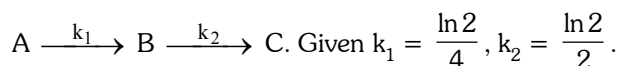
24. The reaction given below, rate constant for disappearance of A is $7.48 \times 10^{-3} \text{ sec}^{-1}$. Calculate the time required for the total pressure in a system containing A at an initial pressure of 0.1 atm to rise to 0.145 atm and also find the total pressure after 100 sec.



25. For a parallel first order reaction $x \begin{cases} \xrightarrow{k_1} y \\ \xrightarrow{k_2} z \end{cases}$, calculate value of ratio, $\frac{[x]_t}{[y]_t + [z]_t}$ at any given instant t.

26. For a parallel first order reaction $A \begin{cases} \xrightarrow{k_1} B \\ \xrightarrow{k_2} C \end{cases}$, $k_1 = x \text{ hr}^{-1}$; $k_1 : k_2 = 1 : 10$. Calculate $\frac{[C]}{[A]}$ after one hour from the start of the reaction. Assuming only A was present in the beginning.

27. How much time would be required for the B to reach maximum concentration for the given sequential first order reaction



28. In gaseous reactions important for understanding the upper atmosphere, H_2O and O react bimolecularly to form two OH radicals. ΔH for this reaction is 72 kJ at 500 K and $E_a = 77 \text{ kJ mol}^{-1}$, then calculate E_a for the bimolecular recombination of 2OH radicals to form H_2O & O at 500 K

29. The energy of activation of a first order reaction is $104.5 \text{ kJ mol}^{-1}$ and pre-exponential factor (A) is $5 \times 10^{13} \text{ sec}^{-1}$. At what temperature, will the reaction have a half life of 1 minute?

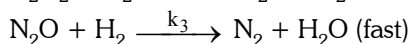
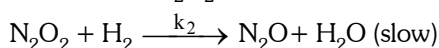
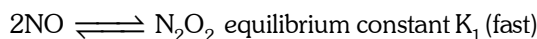
30. The specific rate constant for a reaction increases by a factor of 4, if the temperature is changed from 27°C to 47°C . Find the activation energy for the reaction.

31. A catalyst lowers the activation energy for a certain reaction from 75 kJ to 25 kJ mol^{-1} . What will be the effect on the rate of reaction at 25°C , other things being equal.

32. Given that the temperature coefficient for the saponification of ethyl acetate by NaOH is 1.75. Calculate activation energy for the saponification of ethyl acetate.

33. At 380°C , the half-life period for the first order decomposition of H_2O_2 is 360 min. The energy of activation of the reaction is 200 kJ mol^{-1} . Calculate the time required for 75% decomposition at 450°C .

34. For the reaction $2\text{H}_2 + 2\text{NO} \longrightarrow \text{N}_2 + 2\text{H}_2\text{O}$, the following mechanism has been suggested :

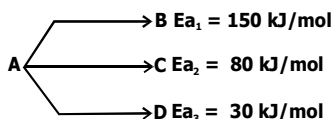


Establish the rate law for given reaction.

35. The acid catalysed hydrolysis of an organic compound A at 30°C has a time for half change of 100min. when carried out in a buffer solution at pH = 5, and 10 min. , when carried out at pH = 4. Both times of half-change are independent of the initial concentration of A. If the rate constant k is given by

$$\frac{-d[A]}{dt} = k [A]^a [H^+]^b, \text{ what are the values of } (a + b) ?$$

36. For a parallel reaction :



Mole percentage of B, C and D after one hour was found to be 10, 25 and 15 respectively. Determine the overall activation energy (in kJ mol^{-1}) for this reaction.

37. For a zero order reaction $2A \xrightarrow{k} B$
 variation of conc. as a function of time is given as

Calculate time to reduce concentration of 'A' from 4 M to 2 M in seconds.

38. For two chemical reactions : (I) $A \longrightarrow B$ (II) $C \longrightarrow D$

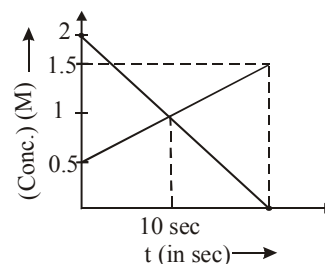
$$\text{at } 30^\circ\text{C} \quad \frac{k_I}{k_{II}} = \frac{1}{2} \quad \text{and} \quad \frac{\alpha(I)}{\alpha(II)} = \frac{2}{1}$$

(where k_I = rate constant for Ist reaction at 30°C , k_{II} = rate constant for IInd reaction at 30°C)

Calculate $\frac{k'_I}{k'_{II}}$ at 50°C .

(where k'_I = rate constant for Ist reaction at 50°C , k'_{II} = rate constant for IInd reaction at 50°C).

(Where $\alpha(I)$ = temperature coefficient for Ist reaction, $\alpha(II)$ = temperature coefficient for IInd reaction).



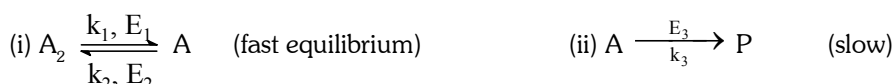
39. The reaction $A(aq) \longrightarrow B(aq) + C(aq)$ is monitored by measuring optical rotation of reaction mixture at different time interval. The species A, B and C are optically active with specific rotation 20° , 30° and -40° per unit concentration respectively. Starting with pure A if the value of optical rotation was found to be $+2.5^\circ$ after **0.693 min** and optical rotation was -5° after **infinite time**. The rate constant for first order conversion of A into B and C in units of min^{-1} . (Use : $\log 2 = 0.3010$)

40. For a **zero order** reaction $A \xrightarrow{k} B$, time to reduce concentration of 'A' from 2 M to 1 M was found to be 10 sec. Then calculate time (in sec.) to reduce concentration of 'A' from 8 M to 0.5 M?

41. For **Ist order** reaction calculate the ratio of $\frac{t_{99.99\%}}{t_{90\%}} = ?$

42. For a chemical reaction $A \xrightarrow{k} B$, reaction starts with pure 'A'. After 10 sec concentration of 'B' was found to be 3 times that of concentration of 'A' at that time (**in sec**). Calculate time to complete 87.5% reaction. (Assume reaction follows first order kinetics).

43. For a reaction $A_2 \rightarrow P$ reaction mechanism is given as

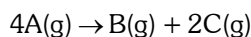


If $E_3 = 100 \text{ kJ/mole}$, $E_1 = 60 \text{ kJ/mole}$, $E_2 = 55 \text{ kJ/mole}$

Apparent activation energy for the overall reaction (in kJ/mole) will be ?

44. In a first order reaction time to reduce concentration of reactant from 800 mole dm^{-3} to 50 mole dm^{-3} was found to be 40 seconds. At the same temperature for same reaction calculate time (in seconds) to reduce concentration of reactant from 20 M to 5 M.

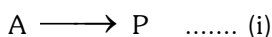
45. Decomposition of A follows first order kinetics by the following equation



If initially, total pressure was 800 mm of Hg and after 10 minutes it is found to be 650 mm of Hg. What is half-life of A (in minutes) ? (Assume only A is present initially)

46. For a 1st order reaction $A(g) \xrightarrow{k} B(g)$ at 27°C . Reaction starts with pure A. After 0.0693 second and 0.1386 second from the start of the reaction, concentration of B is found to be 2 M and 3 M respectively. Calculate rate constant k in sec^{-1} . (Use $\ln 2 = 0.693$)

47. Consider the following two first order reactions :



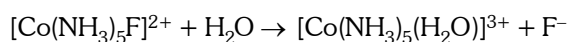
Reaction (i) is 75% complete in 4 hrs while reaction (ii) takes 16 hrs for same 75% completion of reaction under identical conditions. By how many hours, half-life of (ii) is greater than the half-life of (i) ?

Level -2

SUBJECTIVE TYPE QUESTIONS

1. A solution of A is mixed with an equal volume of a solution of B containing the same number of moles, and the reaction $A+B=C$ occurs. At the end of 1h, A is 75 % reacted. How much of A will be left unreacted at the end of 2 h if the reaction is (a) first order in A and zero order in B; (b) first order in both A and B ; and (c) zero order in both A and B ?

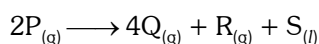
2. The complex $[\text{Co}(\text{NH}_3)_5\text{F}]^{2+}$ reacts with water according to the equation.



The rate of the reaction = rate const. $\times [\text{complex}]^a [\text{H}^+]^b$. The reaction is acid catalysed i.e. $[\text{H}^+]$ does not change during the reaction. Thus rate = $k'[\text{Complex}]^a$ where $k' = k[\text{H}^+]^b$, calculate 'a' and 'b' given the following data at 25°C.

[Complex]M	$[\text{H}^+]$ M	$T_{1/2}$ hr	$T_{3/4}$ hr
0.1	0.01	1	2
0.2	0.02	0.5	1

3. The decomposition of a compound P, at temperature T according to the equation



is the first order reaction. After 30 minutes from the start of decomposition in a closed vessel, the total pressure developed is found to be 317 mm Hg and after a long period of time the total pressure observed to be 617 mm Hg. Calculate the total pressure of the vessel after 75 minute, if volume of liquid S is supposed to be negligible. Also calculate the time fraction $t_{7/8}$.

Given : Vapour pressure of S (l) at temperature T = 32.5 mm Hg.

4. A certain reactant B^{n+} is getting converted to $\text{B}^{(n+4)+}$ in solution. The rate constant of this reaction is measured by titrating a volume of the solution with a reducing reagent which only reacts with B^{n+} and $\text{B}^{(n+4)+}$. In this process, it converts B^{n+} to $\text{B}^{(n-2)+}$ and $\text{B}^{(n+4)+}$ to $\text{B}^{(n-1)+}$. At $t=0$, the volume of the reagent consumed is 25 ml and at $t = 10$ min, the volume used up is 32 ml. Calculate the rate constant of the conversion of B^{n+} to $\text{B}^{(n+4)+}$ assuming it to be a first order reaction.

5. For the reaction,



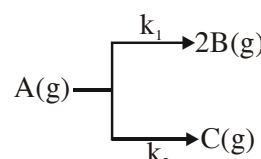
where $k = 5 \times 10^{-5} (\text{mol/L})^{-2} \text{min}^{-1}$

Determine the **time (in minutes)** in which concentration of 'A' becomes half of its initial concentration. If initial concentration of A and B are 0.2 M and 2×10^3 M respectively.

6. The reaction $\text{cis-Cr}(\text{en})_2(\text{OH})_2^+ \xrightleftharpoons[k_2]{k_1} \text{trans-Cr}(\text{en})_2(\text{OH})_2^+$ is first order in both directions. At 25°C the equilibrium constant is 0.16 and the rate constant k_1 is $3.3 \times 10^{-4} \text{s}^{-1}$. In an experiment starting with the pure cis form, how long would it take for half the equilibrium amount of the trans isomer to be formed ?

7. For the following first order gaseous reaction

The initial pressure in a container of capacity V litres is 1 atm. Pressure at time $t = 10$ sec is 1.4 atm and after infinite time it becomes 1.5 atmosphere. Find the rate constant k_1 and k_2 for the appropriate reactions.

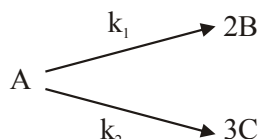


8. At room temperature (20°C) orange juice gets spoilt in about 64 hours. In a refrigerator at 3°C juice can be stored three times as long before it gets spoilt. Estimate (a) the activation energy of the reaction that causes the spoiling of juice. (b) How long should it take for juice to get spoilt at 40°C?

9. For the mechanism $A + B \xrightleftharpoons[k_2]{k_1} C$; $C \xrightarrow{k_3} D$

- Derive the rate law using the steady-state approximation to eliminate the concentration of C.
- Assuming that $k_3 \ll k_2$, express the pre-exponential factor A and E_a for the apparent second-order rate constant in terms of A_1, A_2 and A_3 and E_{a1}, E_{a2} and E_{a3} for the three steps.

10. For a parallel 1st order reaction



reaction starts with pure 'A', at 300 K, mole percentage of B and C is found to be 40% and 60% respectively any time. Calculate mole percentage of B in total product at 320 K.

(Given that temperature coefficient for $A \rightarrow 2B$ and $A \rightarrow 3C$ are 3 and 2 respectively)

SINGLE CHOICE QUESTIONS

- 1.(B) 2.(D) 3.(A) 4.(D) 5.(C) 6.(D) 7.(C) 8.(D) 9.(B) 10.(D)
 11.(A) 12.(A) 13.(C) 14.(D) 15.(C) 16.(A) 17.(B) 18.(D)

ONE OR MORE THAN ONE CORRECT QUESTIONS

- 19.(A,B) 20.(ABC) 21.(A,B) 22. (B,D) 23. (A,B,C)
 24. (A,B,D) 25. (ALL) 26. (A,B) 27. (A,B) 28. (A,B,C)
 29. (A,D) 30. (A,B,C,D) 31. (A,B,C) 32. (A,C) 33. (B,D)
 34. (A,C,D)

COMPREHENSION QUESTIONS

35. (D) 36. (D) 37. (A) 38. (C) 39. (D) 40.(A) 41. (D) 42. (D) 43. (D) 44. (D)
 45. (C) 46. (D) 47. (B) 48. (B) 49. (A) 50. (D) 51. (D) 52. (A) 53. (C) 54. (A)
 55. (B)

MATCH THE COLUMN

- 56.(A) 57.(A) 58.(A) 59.(C)

ANSWERKEY - Level - 1

1. $8.12 \times 10^{-6} \text{ Ms}^{-1}$, $0.012 \text{ atm min}^{-1}$
 2. rate increase by 27 times
 3. (a) $0.019 \text{ mol L}^{-1} \text{ s}^{-1}$, (b) $0.037 \text{ mol L}^{-1} \text{ s}^{-1}$
 4. $2k_1 = k_2 = 4k_3$
 5. $8/3 \times 10^{18} \text{ sec}$
 6. (i) 7.2 M, (ii) Think
 7. $K = 0.01 \text{ M min}^{-1}$
 8. 1.2 hr
 9. (i) 36 min., (ii) 108 min.
 10. expiry time = 41 months
 11. $3.3 \times 10^{-4} \text{ s}^{-1}$
 12. (a) Third order, (b) $r = k[\text{NO}]^2[\text{H}_2]$, (c) $8.85 \times 10^{-3} \text{ M sec}^{-1}$.
 13. (i) first order (ii) $k = 1.308 \times 10^{-2} \text{ min}^{-1}$ (iii) 73%
 14. (i) Zero order, (ii) $K = 5 \text{ Pa/s}$
 15. Zero order
 16. $k = \frac{1}{t} \ln \frac{P_3}{2(P_3 - P_2)}$
 17. $k = \frac{1}{t} \ln \frac{P_3}{(P_3 - P_2)}$
 18. First order
 19. (i) $r = K[(\text{CH}_3)_2\text{O}]$, $0.000428 \text{ sec}^{-1}$
 20. First order
 21. 966 min
 22. $k_1 = 2.605 \times 10^{-3} \text{ min}^{-1}$

23. (a) first order, (b) 13.75 minutes, (c) 0.716 24. 0.180 atm, 47.69 sec
25. $\frac{1}{e^{(K_1+K_2)t}-1}$
26. $\frac{[C]}{[A]} = \frac{10}{11} (e^{11x} - 1)$ 27. $t = 4 \text{ min}$
28. 5 kJ mol^{-1} 29. 349.1 K
30. $55.33 \text{ kJ mole}^{-1}$ 31. rate of reaction increases 5.81×10^8 times
32. $10.757 \text{ k cal mol}^{-1}$ 33. $t = 20.4 \text{ minutes}$
34. $r = K [\text{NO}]^2 [\text{H}_2]$, where $K = k_2 \times K_1$
35. (0002) 36. (0079) 37. (0020) 38. (0002)
39. (0001) 40. (0075) 41. (0004)
42. (0015) 43. (0105) 44. (0020) 45. (0005)
46. (0010) 47. (0006)

Level - 2

1. (a) 6.25 ; (b) 14.3 ; (c) 0%
2. $a = b = 1$
3. $P_t = 379.55 \text{ mm Hg}$, $t^{7/8} = 399.96 \text{ min}$
4. 0.0207 min^{-1}
5. 0050.00
6. 4.83 mins
7. 0.0805
8. (a) 43.46 kJmol^{-1} , (b) 20.47 hour
9. (a) $\frac{d(D)}{dt} = \frac{k_1 k_3 (A)(B)}{k_2 + k_3}$; (b) $E_a = E_{a1} + E_{a3} - E_{a2}$. $A = \frac{A_1 A_3}{A_2}$
10. (0060)

RADIOACTIVITY**SINGLE CHOICE CORRECT**

- ${}_{83}\text{Bi}^{214}$ decays to A by α -emission ; A then decays to B by beta emission, which decays to C by another beta emission. Element C decays to D by still another beta emission, and D decays by α -emission to a stable isotope E. What is an element E ?
 (A) ${}_{81}\text{Tl}^{207}$ (B) ${}_{80}\text{Hg}^{206}$ (C) ${}_{79}\text{Au}^{206}$ (D) ${}_{82}\text{Pb}^{206}$
- The activity of a radioactive nuclide (X^{100}) is 6.023 curie at a certain time ' t '. If its disintegration constant is $3.7 \times 10^4 \text{ s}^{-1}$ the mass of X after t sec is :
 (A) $6.022 \times 10^6 \text{ g}$ (B) 10^{-13} g (C) 10^{-15} g (D) 10^{-17} g
- Activity of a radioactive substance is A_1 at time t_1 and A_2 at time t_2 ($t_2 > t_1$), then the ratio of $\frac{A_2}{A_1}$ is :
 (A) $e^{\lambda(t_2+t_1)}$ (B) $e^{\lambda(t_1-t_2)}$ (C) $e^{-\lambda(t_1+t_2)}$ (D) $\frac{t_2}{t_1}$
- The half-life of ${}_{6}\text{C}^{14}$ is 5730 year. What fraction of it's original C^{14} would left after 22920 year of storage?
 (A) 0.50 (B) 0.25 (C) 0.125 (D) 0.0625
- A radioactive sample had an initial activity of 56 dpm. After 69.3 minutes, it was found to have an activity of 28 dpm. Find the number of atoms in a sample having an activity of 100 dpm.
 (A) 693 (B) 100 (C) 1000 (D) 10,000
- A radioactive sample has initial activity of 28 dpm 30 minutes later its activity 14 dpm. How many atoms of nuclide were present initially ?
 (A) 2800 (B) 1217 (C) 528 (D) 2802
- The half-life of Co^{60} is 5.27 year ($\lambda = 2.5 \times 10^{-7} \text{ min}^{-1}$). The activity of 2.0 gm of the sample is nearly :
 (A) $5 \times 10^5 \text{ dpm}$ (B) $2.5 \times 10^{10} \text{ dpm}$ (C) $5 \times 10^{15} \text{ dpm}$ (D) 10^{10} dpm
- Half-life ($t_{1/2}$) for a radioactive decay is 6930 sec. The time required to fall the rate of decay to $\left(\frac{1}{100}\right)^{\text{th}}$ of it's initial value is
 (A) 69.3 sec (B) 20,000 sec (C) 23030 sec (D) none of these
- A sample of radioactive substance is found 90% of it's initial amount after one day. What % of the original sample can be found after 3 days?
 (A) 81 (B) 72.9 (C) 25 (D) 65.61
- If time t is required for a radioactive substance to one third of it's initial amount, what fraction would be left after $0.5 t$?
 (A) $\frac{1}{2}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{3}$ (D) $\sqrt{\frac{2}{3}}$
- The activity of the hair of Egyptian mummy is 1.75 dpm. $t_{1/2}$ of ${}_{6}\text{C}^{14}$ is 5770 year and disintegration rate of fresh sample of C^{14} is 14 dpm. Find out age of mummy.
 (A) 23080 year (B) 138480 year (C) 11998.3 year (D) 17313.6 year

12. The amount of ${}^6\text{C}^{14}$ isotope in a piece of wood is found to one fourth ($1/4$) of that present in a fresh piece of wood. Calculate the age of the piece of wood ($t_{1/2}$ of ${}^6\text{C}^{14} = 5770$ years)
 (A) 7999 year (B) 11540 year (C) 16320 year (D) 23080 year
13. A radioactive element undergoing decay is left 20% of its initial weight after certain period of time t , how many such periods should elapse from the start for 50% of the element to be left over?
 (A) 3 (B) 4 (C) 5 (D) None of these
14. Indium-112 is radioactive and has a very short half-life ($t_{1/2} = 14$ min). Its decay constant and average life are respectively :
 (A) 0.0495 min^{-1} , 9.7 min (B) 0.495 min^{-1} , 20.2 min
 (C) 9.7 min^{-1} , 20.2 min (D) 0.0495 , 20.2 min
15. The half-life of a radioactive element is 100 minutes. The time interval between the stages to 50% and 87.5% decay will be :
 (A) 100 min (B) 50 min (C) 200 min (D) 25 min
16. The half-life of Tc^{99} is 6.0 hr. The total residual activity in a patient 30 hr after receiving an injection containing Tc^{99} must be more than $0.01 \mu\text{Ci}$. What is the minimum activity (in μCi) that the sample injected an have?
 (A) 0.16 (B) 0.32 (C) 0.64 (D) 0.08
17. A pure radio-chemical preparation was observed to disintegrate at the rate of 2140 counts/minutes at 12:35 PM. at 3:35 PM. of the same day, the disintegration rate of the sample was only 535 count/minutes. What is the half-life of the material?
 (A) 50 min (B) 100 min (C) 200 min (D) None of these
18. A certain radioactive isotope ${}_Z\text{X}^A$ ($t_{1/2} = 100$ days) decays to ${}_{Z-4}\text{Y}^{A-8}$. If 1 mole of ${}_Z\text{X}^A$ is kept in sealed container, how much He gas will accumulate at STP in 200 days?
 (A) 11.2 litre (B) 33.6 litre (C) 22.4 litre (D) 44.8 litre
19. A radioactive substance decays 25% in 10 minute. If at start there are 4×10^{20} atoms present, after what time will the number of atoms be reduced to 10^{20} atoms? (given $\ln 3 = 1.098$)
 (A) 10.98 min (B) 21.97 min (C) 48.19 min (D) None of these
20. The time of decay for a nuclear reaction is given by $t = 4t_{1/2}$. The relation between the mean life (T) and time of decay (t) is given by :
 (A) $2T \ln 2$ (B) $4T \ln 2$ (C) $2T^4 \ln 2$ (D) $1/T^2 \ln 2$
21. Two radio isotopes A and B of atomic weight X and Y are mixed in equal amount by weight. After 20 days, their weight ratio is found to be 4 : 1. Isotope A has a half-life of 1 day. The half-life of isotope B is :
 (A) $1.11 \frac{Y}{X}$ day (B) $0.11 \frac{X}{Y}$ day (C) 0.6237 day (D) None of these
22. Two radioactive nuclides A and B have half-lives 50 min and 10 min respectively. A fresh sample contains the nuclide of B to be eight time that of A. How much time should elapse so that the number of nuclides of A becomes double of B?
 (A) 30 (B) 40 (C) 50 (D) 100
23. A radioactive nuclide is produced at a constant rate of α per second. Its decay constant is λ . If N_0 be the no. of nuclei at time $t = 0$, then max. no. of nuclei possible are :
 (A) N_0 (B) α/λ (C) $N_0 + \frac{\alpha}{\lambda}$ (D) $\frac{\lambda}{\sigma} + N_0s$
24. An analysis of the rock shows that the relative number of Sr^{87} and Rb^{87} ($t_{1/2} = 4.7 \times 10^{10}$ year) atoms is 0.05. What is the age of the rock? Assume all Sr^{87} to be formed from Rb^{87} only.
 (A) 7.62×10^9 year (B) 1.43×10^9 year (C) 3.28×10^9 year (D) 4.32×10^8 year

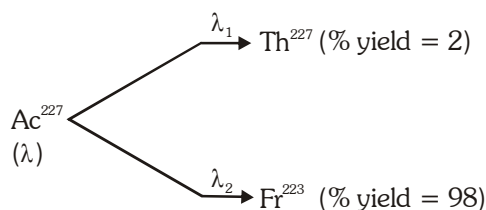
25. There are two radio nuclei A and B. A is a α -emitter and B is β -emitter, their disintegration constant are in the ratio of 1 : 2. What should be the number of atoms of A and B at time $t = 0$, so that probability of getting of α and β -particles are same at time $t = 0$:

(A) 2 : 1 (B) 4 : 1 (C) 1 : 2 (D) 1 : 4

26. A radioactive substance (parent) decays to it's daughter element, the age of radioactive substance (t) is related to the daughter (D)/parent (p) ratio by the equation :

(A) $t = \frac{1}{\lambda} \ln \left(1 + \frac{p}{d} \right)$ (B) $t = \frac{1}{\lambda} \ln \left(1 + \frac{d}{p} \right)$ (C) $t = \frac{1}{\lambda} \ln \left(\frac{d}{p} \right)$ (D) $t = \frac{1}{\lambda} \ln \left(\frac{p}{d} \right)$

27. Ac^{227} has a half-life of 22 years. The decays follows two parallel paths



What are the decay constants (λ) in year^{-1} for Th and Fr respectively ?

(A) 0.03087, 0.00063 (B) 0.00063, 0.03087 (C) 0.02, 0.98 (D) None of these

28. ${}_{84}\text{Po}^{218}$ ($t_{1/2} = 183$ sec) decay to ${}_{82}\text{Pb}^{214}$ ($t_{1/2} = 161$ sec) by α -emission, while Pb^{214} is a β -emitter. In an experiment starting with 1 mole of pure Po^{218} , how much time would be required for the number of nuclei of ${}_{82}\text{Pb}^{214}$ to reach maximum?

(A) 147.5 (B) 247.5 (C) 182 (D) 304

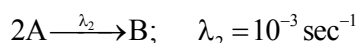
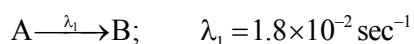
29. Upon irradiating californium with neutrons, a scientist discovered a new nuclide having mass number of 250 and a half-life of 30 min. 90 min. after the irradiation, the observed radioactivity due to nuclide was 100 dis/min. How many atoms of the nuclide were prepared initially ?

(A) 2.4×10^4 (B) 3.46×10^4 (C) 1900 (D) 800

30. A radioactive isotope X with half-life of 6.93×10^9 years decay to Y which is stable. A sample of rock from the moon was found to contain both the elements X and Y in the mole ratio 1 : 7. What is the age of the rock?

(A) 2.079×10^{10} years (B) 1.94×10^{10} years (C) 1.33×10^9 years (D) 10^{10} years

31. The average (mean) life of a radio nuclide which decays by parallel path is



(A) 52.63 sec (B) 500 sec (C) 50 sec (D) none of these

32. The radioactive decay ${}_{83}\text{Bi}^{211} \longrightarrow {}_{81}\text{Tl}^{207}$, takes place in 100 L closed vessel at 27°C . Starting with 2 moles of ${}_{83}\text{Bi}^{211}$ ($t_{1/2} = 130$ sec), the pressure development in the vessel after 520 sec will be :

(A) 1.875 atm (B) 0.2155 atm (C) 0.4618 atm (D) 4.618 atm

33. A fresh radioactive mixture containing short lived species A and B. Both collectively emitting α -particles initially at the rate of 8000 α -particles per minute. 20 minutes later, they emits at the rate of 3500 α -particles per minute. If the half-lives of the species A and B are 10 minutes and 500 hours respectively, then the ratio of activities of A to B in the initial mixture was :

(A) 4 : 6 (B) 6 : 4 (C) 3 : 4 (D) 3 : 1

34. In order to determine the volume of blood in an animal, a 1.0 mL sample of solution of 10^3 dpm of ${}^3\text{H}$ is injected into the animal blood stream. After sufficient time for circulatory equilibrium to be established, 2 mL of blood is found to have activity of 10 dpm. The volume of blood in animal is :

(A) 199 mL (B) 198 mL (C) 200 mL (D) 20 mL

- 35.** The ratio of activities of two radio nuclides X and Y in a mixture at time $t = 0$ was found to be 4:1. After two hours, the ratio of activities become 1 : 1. If the $t_{1/2}$ of radio nuclide X is 20 min then $t_{1/2}$ [in minutes] of radio nuclide Y is :
- (A) 10 (B) 20 (C) 30 (D) 40
- 36.** Find the age of an ancient Egyptian wooden article (in years) from the given information.
- (i) Activity of 1 g of carbon obtained from ancient wooden article = 7 counts/min/g
 (ii) Activity of 1 g carbon obtained from fresh wooden sample = 15.4 counts per min/g
 (iii) Percentage increase in level of C^{14} due to nuclear explosions in past 100 years is 10%
 (iv) $t_{1/2}$ of ${}^6C^{14} = 5770$ years
- (A) 5.770×10^3 (B) 16.87×10^3 (C) 2488 (D) none of these
- 37.** ${}^{27}_{13}\text{Al}$ is a stable isotope. ${}^{29}_{13}\text{Al}$ is expected to disintegrate by :
- (A) α emission (B) ${}^0_{-1}\beta$ emission (C) Positron emission (D) Proton emission
- 38.** Loss of a β -particle is equivalent to :
- (A) Increase of one proton only (B) Decrease of one neutron only
 (C) Both (A) and (B) (D) None of these
- 39.** Two radioactive material A_1 and A_2 have decay constants of $10\lambda_0$ and λ_0 . If initially they have same number of nuclei, then after time $\frac{1}{9\lambda_0}$ the ratio of number of their undecayed nuclei will be :
- (A) $\frac{1}{e}$ (B) $\frac{1}{e^2}$ (C) $\frac{1}{e^3}$ (D) $\frac{\sqrt{e}}{1}$
- 40.** The half-life of a radioactive isotope is three hours. If the initial mass of the isotope were 256 g, the mass of it remaining undecayed after 18 hours would be :
- (A) 16.0 g (B) 4.0 g (C) 8.0 g (D) 12.0 g
- 41.** Consider the following nuclear reactions :
- $${}^{238}_{92}\text{M} \rightarrow {}^x_{\text{Y}}\text{N} + 2 {}^4_2\text{He}; \quad {}^x_{\text{Y}}\text{N} \rightarrow {}^A_{\text{B}}\text{L} + 2\beta^+$$
- The number of neutrons in the element L is :
- (A) 142 (B) 144 (C) 140 (D) 146
- 42.** The half-life of a radioisotope is four hours. If the initial mass of the isotope was 200 g, the undecayed mass remaining after 24 hours is :
- (A) 1.042 g (B) 2.084 g (C) 3.125 g (D) 4.167 g
- 43.** A radioactive element gets spilled over the floor of a room. Its half-life period is 30 days. If the initial activity is ten times the permissible value, after how many days will it be safe to enter the room ?
- (A) 1000 days (B) 300 days (C) 10 days (D) 100 days
- 44.** Which of the following nuclear reactions will generate an isotope ?
- (A) neutron particle emission (B) positron emission
 (C) α -particle emission (D) β -particle emission
- 48.** The analysis of a mineral of uranium reveals that ratio of mole of ${}^{206}\text{Pb}$ and ${}^{238}\text{U}$ in sample is 0.2. If effective decay constant of process ${}^{238}\text{U} \longrightarrow {}^{206}\text{Pb}$ is 1 then age of rock is :
- (A) $\frac{1}{\lambda} \ln \frac{5}{4}$ (B) $\frac{1}{\lambda} \ln \left(\frac{5}{1} \right)$ (C) $\frac{1}{\lambda} \ln \frac{4}{1}$ (D) $\frac{1}{\lambda} \ln \left(\frac{6}{5} \right)$
- 49.** Which of the following processes represents a gamma-decay ?
- (A) ${}^A\text{X}_Z + \gamma \longrightarrow {}^A\text{X}_{Z-1} + a + b$ (B) ${}^A\text{X}_Z + {}^1_0\text{n} \longrightarrow {}^{A-3}\text{X}_{Z-2} + c$
 (C) ${}^A\text{X}_Z \longrightarrow {}^A\text{X}_Z + f$ (D) ${}^A\text{X}_Z + e_{-1} \longrightarrow {}^A\text{X}_{Z-1} + g$

50. Select the incorrect statement(s) :

- (A) The emission of gamma radiation involves transition between energy levels within the nucleus.
 (B) ${}^4_2\text{He}$ is formed due to emission of beta particle from tritium ${}^3_1\text{H}$.
 (C) When positron (${}^0_{+1}e$) is emitted, $\frac{n}{p}$ ratio increases.
 (D) None of these

51. A sample contains 0.1 gram-atom of radioactive isotope ${}^A_Z\text{X}$ ($t_{1/2} = 5$ days). How many number of atoms will decay during eleventh day ? [$N_A = \text{Avogadro's number}$]

- (A) $0.1 \left(-e^{-\frac{0.693 \times 11}{5}} + e^{-\frac{0.693 \times 10}{5}} \right)$
 (B) $0.1 \left(-e^{-\frac{0.693 \times 11}{5}} + e^{-\frac{0.693 \times 10}{5}} \right)$
 (C) $0.1 \left(-e^{-\frac{0.693 \times 11}{5}} + e^{-\frac{0.693 \times 10}{5}} \right) N_A$
 (D) $0.1 \left(-e^{-\frac{0.693 \times 11}{5}} + e^{-\frac{0.693 \times 10}{5}} \right) N_A$

52. A sample of ${}^{14}\text{CO}_2$ was to be mixed with ordinary CO_2 for a biological tracer experiment. In order that 10 cm³ of diluted gas should have 10⁴ dis/min, what activity (in μCi) of radioactive carbon is needed to prepare 60 L of diluted gas at STP. [$1 \text{ Ci} = 3.7 \times 10^{10} \text{ dps}$]

- (A) 270 μCi (B) 27 μCi (C) 2.7 μCi (D) 2700 μCi

53. The radioactivity of a sample is R_1 at a time T_1 and R_2 at a time T_2 . If the half life of the specimen is T , the number of atoms that have disintegrated in the time $(T_2 - T_1)$ is equal to :

- (A) $(R_1 T_1 - R_2 T_2)$ (B) $(R_1 - R_2)$ (C) $(R_1 - R_2)/T$ (D) $\frac{(R_1 - R_2) \times T}{0.693}$

54. C^{14} is a beta active nucleus. A sample of C^{14}H_4 gas kept in a closed vessel shows increase in pressure with time. This is due to :

- (A) the formation of N^{14}H_3 and H_2 (B) the formation of B^{11}H_3 and H_2
 (C) the formation of $\text{C}^{14}_2\text{H}_4$ and H_2 (D) the formation of C^{12}H_3 , N^{14}H_2 and H_2

COMPREHENSION QUESTIONS

Comprehension # 1

Nuclei of a radioactive element 'A' are being produced at a constant rate, α . The element has a decay constant, λ . At time, $t = 0$, there are N_0 nuclei of the element.

55. The number of nuclei of A at time 't' is :

- (A) $\frac{\alpha}{\lambda} (1 - e^{-\lambda t})$ (B) $N_0 \cdot e^{-\lambda t}$
 (C) $\frac{1}{\lambda} [\alpha - (\alpha - \lambda N_0) e^{-\lambda t}]$ (D) $\frac{N_0 \cdot \alpha}{\lambda} \left[1 - \left(1 - \frac{\lambda}{\alpha} \right) e^{-\lambda t} \right]$

56. If $\alpha = 2N_0\lambda$, the number of nuclei of A after $t = \frac{\ln 2}{\lambda}$ becomes

- (A) zero (B) $2N_0$ (C) $1.5 N_0$ (D) $0.5 N_0$

57. If $\alpha = 4N_0\lambda$, the number of nuclei of A when A has reached steady state :

- (A) zero (B) $4N_0$ (C) $2N_0$ (D) N_0

LEVEL-1 (SUBJECTIVE TYPE QUESTIONS)

- Classify each of the following nuclides as "beta (${}^0_{-1}\beta$) emitter", or "positron (${}^0_{+1}\beta$) emitter":
 ${}^{49}_{20}\text{Ca}$ ${}^{195}_{80}\text{Hg}$ ${}^8_5\text{B}$ ${}^{150}_{67}\text{Ho}$ ${}^{30}_{13}\text{Al}$ ${}^{94}_{36}\text{Kr}$. Note: ${}^{84}_{36}\text{Kr}$ ${}^{200}_{80}\text{Hg}$ and ${}^{165}_{67}\text{Ho}$ are stable
- Of the three isobars ${}^{114}_{48}\text{Cd}$ ${}^{114}_{49}\text{In}$ and ${}^{114}_{50}\text{Sn}$, which is likely to be radioactive? Explain your choice.
- Complete the following nuclear equations:
 (A) ${}^{14}_7\text{N} + {}^4_2\text{He} \rightarrow {}^{17}_8\text{O} + \dots$ (B) ${}^9_4\text{Be} + {}^4_2\text{He} \rightarrow {}^{12}_6\text{C} + \dots$
 (C) ${}^9_4\text{Be} (p, \alpha) \dots$ (D) ${}^{30}_{15}\text{P} \rightarrow {}^{30}_{14}\text{Si} + \dots$
 (E) ${}^3_1\text{H} \rightarrow {}^3_2\text{He} + \dots$ (F) ${}^{43}_{20}\text{Ca} (\alpha, \dots) {}^{46}_{21}\text{Sc}$
- The activity of the radioactive sample drops to 1/64 of its original value in 2 hr find the decay constant (λ).
- In the nuclear reaction $X^{200} \longrightarrow A^{110} + B^{90}$. If the binding energy per nucleon for X, A and B is 7.4 MeV, 8.2 MeV and 8.2 MeV respectively, Find energy released.
- Consider the following reaction; ${}^2_1\text{H} + {}^2_1\text{H} = {}^4_2\text{He} + Q$.
 Mass of the deuterium atom = 2.0141 u ; Mass of the helium atom = 4.0024 u. Find 'Q'
- When ${}^a_c\text{X}$ changes to ${}^b_d\text{Y}$. Find number of α and β particle.
- Find the α -activity of 1 gm ${}^{226}\text{Ra}$. ($t_{1/2} = 1620$ year).
- The half life of the nuclide Rn^{220} is 54.5 sec. Find mass (in kg) of radon having activity equivalent to 1 millicurie.
- A radioactive substance decays 20% in 10 min if at start there are 5×10^{20} atoms present, find time after which the number of atoms will be reduced to 10^{18} .
- The binding energies per nucleon for ${}^2_1\text{H}$ and ${}^4_2\text{He}$ are 1.1 MeV and 7.0 MeV respectively. Find the energy released when two deuterons fuse to form a helium nucleus.
- The half life period of ${}^{125}_{53}\text{I}$ is 60 days. What % of radioactivity would be present after 240 days.
- At any given time a piece of radioactive material ($t_{1/2} = 30$ days) contains 10^{12} atoms. Calculate the activity of the sample in dps.
- At a given instant there are 25 % undecayed radioactive nuclei in a sample. After 10 sec the number of undecayed nuclei remains 12.5 % . Calculate :
 (i) mean – life of the nuclei and
 (ii) The time in which the number of undecayed nuclear will further reduce to 6.25 % of the reduced number.
- Calculate the energy released in joules and MeV in the following nuclear reaction:
 ${}^2_1\text{H} + {}^2_1\text{H} \longrightarrow {}^3_2\text{He} + {}^1_0\text{n}$
 Assume that the masses of ${}^2_1\text{H}$, ${}^3_2\text{He}$ and neutron (n) respectively are 2.020, 3.0160 and 1.0087 in amu.

LEVEL-2 (SUBJECTIVE TYPE QUESTIONS)

- (A) Calculate number of α and β -particles emitted when ${}_{92}\text{U}^{238}$ changes into radioactive ${}_{82}\text{Pb}^{206}$.

(B) Th^{234} disintegrates and emits 6β and 7α -particles to form a stable element. Find the atomic number and mass number of the stable product. Also identify the element.
- One of the hazards of nuclear explosion is the generation of Sr^{90} and its subsequent incorporation in bones. This nuclide has a half life of 28.1 year. Suppose one microgram was absorbed by a new-born child, how much Sr^{90} will remain in his bones after 20 years?
- ${}_{84}\text{Po}^{210}$ decays with α -particle to ${}_{82}\text{Pb}^{206}$ with a half life of 138.4 day. If 1.0 g of ${}_{84}\text{Po}^{210}$ is placed in a sealed tube, how much helium will accumulate in 69.2 day? Express the answer in cm^3 at 1 atm & 273 K. Also report the volume of He formed if 1 g of $\text{Po}^{210} \text{O}_2$ is used.
- A sample of U^{238} (half life = 4.5×10^9 yr) ore is found to contain 23.8 g of U^{238} and 20.6 g of Pb^{206} . Calculate the age of the ore.
- Ac^{227} has a half life of 22 year w.r.t radioactive decay. The decay follows two parallel paths, one leading to Th^{227} and the other leading to Fr^{223} . the percentage yields of these two daughters nuclides are 2% and 98% respectively. What is the rate constant in yr^{-1} , for each of the separate paths?
- Calculate the age of a vegetarian beverage whose tritium content is only 15% of the level in living plants.
(Given $t_{1/2}$ for ${}^3_1\text{H}$ = 12.3 years)
- An isotopes of Potassium ${}^{40}_{19}\text{K}$ has a half life of 1.4×10^9 year and decays to Argon ${}^{40}_{18}\text{Ar}$ which is stable.
 - Write down the nuclear reaction representing this decay.
 - A sample of rock taken from the moon contains both potassium and argon in the ratio 1/3. Find age of rock.

ANSWER KEY

SINGLE CHOICE CORRECT

1. (D) 2. (C) 3. (B) 4. (D) 5. (D) 6. (B) 7. (C) 8. (D) 9. (B) 10. (B)
11. (D) 12. (B) 13. (D) 14. (D) 15. (C) 16. (B) 17. (D) 18. (B) 19. (C) 20. (B)
21. (D) 22. (C) 23. (B) 24. (C) 25. (A) 26. (B) 27. (B) 28. (B) 29. (B) 30. (A)
31. (C) 32. (C) 33. (D) 34. (A) 35. (C) 36. (A) 37. (B) 38. (C) 39. (A) 40. (B)
41. (B) 42. (C) 43. (D) 44. (A) 45. (A) 46. (C) 47. (A) 48. (D) 49. (C) 50. (B)
51. (C) 52. (B) 53. (D) 54. (A)

COMPREHENSION BASED QUESTIONS

55. (C) 56. (C) 57. (B)

LEVEL-1 (SUBJECTIVE TYPE QUESTIONS)

- beta emitter : ^{49}Ca , ^{30}Al , ^{94}Kr , positron emitter : ^{195}Hg , ^8B , ^{150}Ho
- $^{114}_{49}\text{In}$, odd number of nucleons
- (a) ^1_1H , (b) ^1_0n , (c) ^6_3Li , (d) $^0_{+1}\text{e}$ (e) $^0_{-1}\text{e}$, (f) p (proton)
- $1 = 2.079 \text{ hr}^{-1}$
- 160 MeV.
- 24
- $a = \frac{a-b}{4}$; $b = d + \frac{(a-b)}{2} - c$
- $2.16 \times 10^{12} \text{ dpm.}$
- 1.06×10^{-15}
- 4.65 hr
- 23.6 MeV
- 6.25 %
- $2.67 \times 10^5 \text{ dps}$
- (i) $t_{\text{means}} = 14.43 \text{ s}$ (ii) 40 seconds
- DE = 14.25 MeV

LEVEL-2 (SUBJECTIVE TYPE QUESTIONS)

- (a) No. of a-particles = 8, No. of b-particles = 6; (b) $^{206}_{82}\text{Pb}$
- $6.13 \times 10^{-7} \text{ g}$
- 31.25 cm^3 , 27.104 cm^3
- $4.5 \times 10^9 \text{ year}$
- $6.30 \times 10^{-4} \text{ yr}^{-1}$, $3.087 \times 10^{-2} \text{ yr}^{-1}$
- 33.67 years
- (i) $^{40}_{19}\text{K} \longrightarrow ^{40}_{18}\text{Ar} + ^0_{+1}\text{b} + \nu$
(ii) $2.8 \times 10^9 \text{ years}$

QUIZ # 1**LIQUID SOLUTION****PHYSICAL CHEMISTRY****SINGLE OPTION CORRECT**

- There are some of the characteristics of the supersaturated solution
 I : Equilibrium exists between solutions and solid solute
 II : If a crystal of solute is added to supersaturated solution, crystallisation occurs rapidly
 III : Supersaturated solutions contain more solute than they should at a particular temperature
 Correct characteristics of supersaturated solutions are :
 (A) I, II, III (B) II, III (C) I, III (D) I, II
- When KCl dissolves in water (assume endothermic dissolution) :
 (A) $\Delta H = +ve, \Delta S = +ve, \Delta G = +ve$ (B) $\Delta H = +ve, \Delta S = -ve, \Delta G = -ve$
 (C) $\Delta H = +ve, \Delta S = +ve, \Delta G = -ve$ (D) $\Delta H = -ve, \Delta S = -ve, \Delta G = +ve$
- Which statement best explains the meaning of the phrases "like dissolves like" ?
 (A) A Solute will easily dissolve a solute of similar mass
 (B) A solvent and solute with similar intermolecular forces will readily form a solution
 (C) The only true solutions are formed when water dissolves a non-polar solute
 (D) The only true solutions are formed when water dissolves a polar solute
- The dissolving process is exothermic when :
 (A) The energy released in solvation exceeds the energy used in breaking up solute-solute and solvent-solvent interactions
 (B) The energy used in solvation exceeds the energy released in breaking up solute-solute and solvent-solvent interactions
 (C) The energy released in solvation is about the same as the energy used in breaking up solute-solute and solvent-solvent interactions
 (D) The energy used in solvation is about the same as the energy used in breaking up solute-solute and solvent-solvent interactions
- Consider the following four liquids :

1. Water : highly polar ; H-bonding	2. Hexanol : slightly polar ; some H-bonding
3. Chloroform : slightly polar ; no H-bonding	4. Octane : non-polar ; no H-bonding

 Which pair of liquids is immiscible ?
 (A) Water and octane (B) Water and hexanol
 (C) Hexanol and chloroform (D) Chloroform and octane
- An ionic compound that attracts atmospheric water so strongly that a hydrate is formed is said to be :
 (A) Dilute (B) Hygroscopic (C) Immiscible (D) Miscible
- Colligative properties have many practical uses, some of them may be :
 I : Melting of snow by salt
 II : Desalination of sea water
 III : Determination of molar mass
 IV : Determination of melting point and boiling point of solvent
 Actual practical uses are :
 (A) I, II (B) III, IV (C) I, II, III (D) II, III, IV
- A complex containing K^+ , $Pt(IV)$ and Cl^- is 100% ionised giving $i = 3$. Thus, complex is :
 (A) $K_2[PtCl_4]$ (B) $K_2[PtCl_6]$ (C) $K_3[PtCl_5]$ (D) $K[PtCl_3]$

9. In which case, van't Hoff factor i remains unchanged ? [Assume common complexes of these ions]
 (A) PtCl_4 reacts with KCl
 (B) aq. ZnCl_2 reacts with aq. NH_3
 (C) aq. FeCl_3 reacts with aq. $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (D) KMnO_4 reduced to MnO_2 in alkaline medium (MnO_2 a black ppt)
10. If $\text{p}K_a = -\log K_a = 4$, and $K_a = Cx^2$ then van't Hoff factor for weak monobasic acid when $C = 0.01 \text{ M}$ is :
 (A) 1.01 (B) 1.02 (C) 1.10 (D) 1.20
11. Consider following terms (m = molality) :

I : mK_b ;

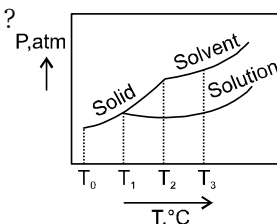
II : $mK_b i$

III : $\frac{\Delta T_b}{i}$

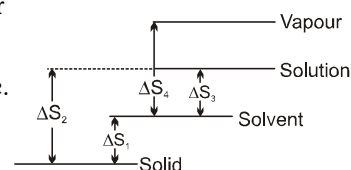
IV : K_b

Terms which can be expressed in degree (temperature) are

- (A) III, IV (B) I, II (C) I, II, III (D) I, III
12. What is the normal freezing point of the solution represented by the phase diagram ?
 (A) T_1 (B) T_2
 (C) T_3 (D) T_0



13. Select correct statement :
 (A) Solution has more molecular randomness than a pure solvent has, the entropy change between solution and solid is larger than the entropy change between pure solvent and solid
 (B) Heats of fusion of solution and solvent are similar since similar forces of intermolecular forces are involved
 (C) Sugar containing solution freezes at a lower temperature than pure water
 (D) All are correct statements
14. Some entropy change are represented in figure. Select correct entropy change.



- (A) $\Delta S_1, \Delta S_2, \Delta S_3$ (B) $\Delta S_1, \Delta S_2, \Delta S_4$ (C) $\Delta S_1, \Delta S_2, \Delta S_3, \Delta S_4$ (D) ΔS_2 and ΔS_4
15. If 'A' contains 2% NaCl and is separated by a semipermeable membrane from 'B' which contains 10% NaCl, which event will occur ?
 (A) NaCl will flow from 'A' to 'B' (B) NaCl will flow from 'B' to 'A'
 (C) Water will flow from 'A' to 'B' (D) Water will flow from 'B' to 'A'

16. Match the terms with definitions

Term	Definition
I : Hemolysis	A : the shrinking of a red blood cell when placed in a solution of greater osmolarity than then cell itself
II : Crenation	B : refers to two solutions of same osmolarity
III : Hygroscopic	C : refers to a solution having a greater osmolarity than other solution.
IV : Hypertonic	D : an ionic compound which attracts water molecules from the atmosphere
V : Isotonic	E : the swelling and bursting of a red blood cell when placed in an solution of lower osmolarity than the cell itself.

Matching codes are :

	I	II	III	IV	V
(A)	A	B	C	D	E
(B)	E	D	C	B	A
(C)	E	D	B	C	A
(D)	E	A	D	C	B

17. Total vapour pressure of mixture of 1 mol of volatile component A ($p_A^\circ = 100$ mmHg) and 3 mol of volatile component B ($p_B^\circ = 60$ mmHg) is 75 mm. For such case :
- (A) there is positive deviation from Raoult's law
 (B) boiling point has been lowered
 (C) force of attraction between A and B is smaller than that between A or between B and B
 (D) all the above statements are correct
18. Relative decrease in vapour pressure of an aqueous NaCl is 0.167. Number of moles of NaCl present in 180g of H_2O is :
- (A) 2 mol (B) 1 mol (C) 3 mol (D) 4 mol
19. Select correct statement :
- (A) Osmosis, like all colligative properties, results from an increase in entropy as pure solvent passes through the membrane and mixes with the solution
 (B) Desalination of sea-water is done by reverse osmosis
 (C) Both are correct statements
 (D) None is correct statement
20. Van't Hoff factors of aqueous solutions of X, Y, Z are 1.8, 0.8 and 2.5. Hence, their (assume equal concentrations in all three cases)
- (A) b.p. : $X < Y < Z$ (B) f. p. $Z < X < Y$
 (C) osmotic pressure : $X = Y = Z$ (D) v. p. : $Y < X < Z$
21. The fundamental cause of ΔT (depression) is :
- (A) higher entropy of the solution relative to that of pure solvent
 (B) lower entropy of the solution relative to that of pure solvent
 (C) higher enthalpy of the solution relative to that of pure solvent
 (D) lower enthalpy of the solution relative to that of pure solvent
22. When an ideal binary solution is in equilibrium with its vapour, molar ratio of the two components in the solution and in the vapour phase is :
- (A) Same
 (B) Different
 (C) May or may not be same depending upon volatile nature of the two components
 (D) None of the above
23. The melting points of most of the solid substances increases with an increase of pressure acting on them. However, ice melts at a temperature lower than its usual melting point, when the pressure increases. This is because :
- (A) Ice is less denser than water (B) Pressure generates heat
 (C) The bonds break under pressure (D) Ice is not a true solid
24. Which characterises the weak intermolecular forces of attraction in a liquid ?
- (A) High boiling point (B) High vapour pressure
 (C) High critical temperature (D) High heat of vaporization

Comprehension (Q.No. 25 to Q. No. 26)

Addition of non-volatile solute to a solvent always increases the colligative properties such as osmotic pressure, ΔP , ΔT_b and ΔT_f . All these colligative properties are directly proportional to molality if solutions are dilute. The increases in colligative properties on addition of non-volatile solute is due to increase in number of solute particles.

25. For different aqueous solutions of 0.1 N urea, 0.1 N Na_2SO_4 and 0.1 N Na_3PO_4 solution at 27°C , the correct statements are:

1. The order of osmotic pressure is, $\text{NaCl} > \text{Na}_2\text{SO}_4 > \text{Na}_3\text{PO}_4 > \text{urea}$

2. $\pi = \frac{\Delta T_b}{K_b} \times \text{ST}$ for urea solution

3. Addition of salt on ice increases its melting point

4. Addition of salt on ice brings in melting of ice earlier

(A) 2, 3, 4

(B) 1, 2, 4

(C) 1, 2, 3

(D) 3, 4

26. 1 g mixture of glucose and urea present in 250 mL aqueous solution shows the osmotic pressure of 0.74 atm at 27°C . Assuming solution to be dilute, which are correct?

1. Percentage of urea in mixture is 17.6

2. Relative lowering in vapour pressure of this solution is 5.41×10^{-4}

3. The solution will boil at 100.015°C , if K_b of water is $0.5 \text{ K molality}^{-1}$

4. If glucose is replaced by same amount of sucrose, the solution will show higher osmotic pressure at 27°C

5. If glucose is replaced by same amount of NaCl, the solution will show lower osmotic pressure at 27°C

(A) 1, 2, 3

(B) 1, 2, 3, 5

(C) 2, 4, 5

(D) 1, 4, 5

Multiple Correct Choice Type

27. Freezing point lowering expression is $\Delta T_f = K_f m$ (molality)

Which of the following assumptions are considered for the validity of above equation?

(A) The solution is dilute

(B) The ΔH_f (latent heat of fusion of solvent) is independent of temperature between the actual and normal freezing point

(C) The solid-phase consists of pure solvent when solution is allowed to cool

(D) None of these

28. Select correct statements :

(A) Gases which have high value of van der Waals constant 'a' are easily liquefied

(B) Easily liquefied gases are water soluble

(C) Ions forming gases in a solvent are soluble in that solvent

(D) None of these

29. Select incorrect statement :

(A) Higher the value of k_H (Henry's law constant) at a given pressure, the lower is the solubility of the gas in the liquid

(B) Solubility of a gas in a liquid decreases with increases in temperature and pressure

(C) to minimise the painful effects accompanying the decomposition of deep sea divers, O_2 diluted with less soluble He gas is used as breathing gas

(D) The solubility of a gas in a liquid is governed by Henry's law

30. Which combination of (I) vapour pressure, (II) intermolecular forces and (III) ΔH_{vap} (latent heat of vaporisation) is matched correctly

	I	II	III
(A)	high	weak	small
(B)	high	strong	large
(C)	low	weak	large
(D)	low	strong	small

31. Consider following cases :

I : 2M CH_3COOH solution in benzene at 27°C where there is dimer formation to the extent of 100%

II : 0.5 M KCl aq. solution at 27°C , which ionises 100% which is/are true statement(s) ?

(A) Both are isotonic

(B) I is hypertonic

(C) II is hypertonic

(D) None is correct

32. Which one of the statements given below concerning properties of solutions, describes a colligative effect ?
- (A) Boiling point of pure water decreases by the addition of ethanol
 (B) Vapour pressure of pure water decreases by the addition of nitric acid
 (C) Vapour pressure of pure benzene decreases by the addition of naphthalene
 (D) Boiling point of pure benzene increases by the addition of toluene
33. A graph plotted between $\frac{P}{d}$ vs. d (where P is osmotic pressure of solution of a solute of mol. wt. m and d is its density at temperature T). Pick out the correct statements about the plots :
- (A) $\left[\frac{P}{d}\right]_{d \rightarrow 0} = \frac{ST}{m}$ (B) The intercept of the plot $\frac{ST}{m}$
 (C) The intercept of the plot $= \left[\frac{P}{d}\right]_{d \rightarrow 0}$ (D) $\left[\frac{P}{d}\right]_{d \rightarrow 0}$ is independent of temperature
34. Which of the following statements are correct for van't Hoff factor 'i' for a solution of weak electrolyte A_xB_y ?
- (A) $i = 1 - \alpha + x\alpha + y\alpha$
 (B) $i > 1$ at normal dilution
 (C) i increases more rapidly with dilution and attains a limiting value of $(x + y)$ at infinite dilution
 (D) The increase in 'i' with dilution is due to increases in molality of solution with dilution
35. For a dilute solution having molality m of a given solute in a solvent of mol.wt. M , b.pt. T_b and heat of vaporisation per mole ΔH ; $\left[\frac{\partial T_b}{\partial m}\right]_{m \rightarrow 0}$ is equal to :
- (A) Molal elevation constant (B) $\frac{RT_b^2 M}{\Delta_{\text{vap}} H}$; where M in kg $\Delta_{\text{vap}} H$ and R in J mol^{-1}
 (C) $\frac{RT_b M}{\Delta_{\text{vap}} S}$; where M in kg; $\Delta_{\text{vap}} S$ and R in J mol^{-1} (D) $\frac{RT_b^2 M}{1000 \Delta_{\text{vap}} H}$; where M in g; R and $\Delta_{\text{vap}} H$ expressed in same unit of heat.
36. Which facts are true when we use van't Hoff equation $PV = CST$ for osmotic pressure P of dilute solutions ?
- (A) The equation is identical to that of ideal gas equation
 (B) The solute particles in solution are analogous to the gas molecules and the solvent is analogous to the empty space between the gas molecules
 (C) Solute molecules are dispersed in the solvent the way the gas molecules are dispersed in empty space
 (D) The equation is not identical to that of ideal gas equation
37. Which statements are correct about antifreeze mixtures to use to melt ice or snow on roads ?
- (A) Antifreeze mixture of CaCl_2 + water (f.pt. -50°C) is preferred over KCl + water (f.pt. -10°C)
 (B) The low f.pt. of CaCl_2 + water is due to the formation of $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$ at low temperature
 (C) The use of antifreeze for salt solutions causes major problems of corrosion of steel car bodies and reinforcement bars in concrete road structures.
 (D) More is the amount of salt spreaded on road, easier is melting of ice.

Subjective-Type

38. It is found that elevation in boiling point of a given aq. NaCl solution is equal to depression in freezing point of 0.25 molal aq. Na_2CO_3 solution. If $K_f = 1.86^\circ\text{mol}^{-1}\text{kg}$ and $K_b = 0.52^\circ\text{mol}^{-1}\text{kg}$, calculate molality of NaCl solution, assume complete ionisation of NaCl and Na_2CO_3 in aqueous solution.

39. The freezing point of an aqueous solution of KCN containing $0.1892 \text{ mol kg}^{-1}$ was -0.704°C . On adding 0.095 mole $\text{Hg}(\text{CN})_2$, the freezing point of the solution was -0.530°C . If complex formation takes place according to the equation, $\text{Hg}(\text{CN})_2 + m \text{ KCN} \rightleftharpoons \text{K}_m [\text{Hg}(\text{CN})_{m+2}]$, what is the formula of the complex? $K_f (\text{H}_2\text{O})$ is $1.86 \text{ kg mol}^{-1} \text{ K}$.
40. For equimolal aq. solution of K_2SO_4 and NaHSO_4 value of $\lim_{m \rightarrow 0} \left(\frac{\Delta T_b(\text{K}_2\text{SO}_4)}{\Delta T_b(\text{NaHSO}_4)} \right)$ is :

ANSWER-KEY (QUIZ # 01)

1. (B) 2. (C) 3. (B) 4. (A) 5. (A) 6. (B) 7. (C) 8. (B) 9. (B)
10. (C) 11. (C) 12. (A) 13. (D) 14. (C) 15. (C) 16. (D) 17. (D) 18. (B)
19. (C) 20. (B) 21. (A) 22. (C) 23. (A) 24. (B) 25. (B) 26. (A) 27. (A,B,C)
28. (A,B,C) 29. (B) 30. (A) 31. (A) 32. (B,C) 33. (A,B,C) 34. (A,B,C)
35. (All) 36. (A,B,C) 37. (All) 38. (1,3,4) 39. $(\text{K}_2 [\text{Hg}(\text{CN})_4])$ 40. (1)

QUIZ # 2**SINGLE OPTION CORRECT**

1. Silver chloride has a solubility of 0.007 g/litre in water at 27°C . What is the osmotic pressure (in atm) of a saturated solution of AgCl ? **Given : Formula mass of AgCl $143.5 \approx \frac{1000}{7}$ & $R = \frac{1}{12} \text{ litre atm mol}^{-1} \text{ K}^{-1}$**
- (A) 2.45×10^{-3} (B) 1.225×10^{-3} (C) 2.205×10^{-4} (D) None
2. On cooling a sugar of 1 M to -2°C .
- (A) Sugar and water both freezes. (B) Only water freezes and ice separates out
- (C) Ice containing sugar and water separates out (D) Sugar is separated out from solution.
3. When 10.84 gm of HgCl_2 is dissolved in 1 kg water the freezing point of the solution is -0.0744°C while when 3.24 gm of $\text{Hg}(\text{NO}_3)_2$ is dissolved in 1 kg of water the freezing point of the solution is found to be -0.0558°C . Then
- (Use : $\text{Hg} = 200$, $\text{U} = 35.5$, $\text{N} = 14$, $\text{O} = 16$)
- (A) HgCl_2 is a non-electrolyte (B) $\text{Hg}(\text{NO}_3)_2$ is a non-electrolyte
- (C) HgCl_2 and $\text{Hg}(\text{NO}_3)_2$ both are non-electrolysts (D) HgCl_2 , $\text{Hg}(\text{NO}_3)_2$ both are strong electrolytes
4. pH of a 0.1 M weak monobasic acid is found to be 2, Hence it's osmotic pressure at the given temperature TK is (where S is solution constant)
- (A) 1.1 ST (B) 0.1 ST (C) 0.11 ST (D) 0.01 ST
5. An electrolyte XY_2 has same freezing point as the solution of glucose of twice the molality of XY_2 . The degree of dissociation XY_2 is
- (A) 0.25 (B) 0.5 (C) 0.33 (D) 0.67
6. **STATEMENT-1** : Binary ideal solution boils at temperature which is lower than normal boiling point of more volatile component and higher than the normal boiling point of less volatile component.
- STATEMENT-2** : For any composition of ideal mixture, its boiling temperature is in between the normal boiling points of pure component.
- (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
- (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
- (C) Statement-1 is True, Statement-2 is False
- (D) Statement-1 is False, Statement-2 is True

7. The normal boiling point of water is 100°C . What will be its standard boiling point.
 (A) 100°C (B) Much greater than 100°C
 (C) Less than 100°C (D) Slightly greater than 100°C
8. Pure benzene freezes at 5.3°C , A solution of 0.223g of phenylacetic acid ($\text{C}_6\text{H}_5\text{CH}_2\text{COOH}$) in 4.4g of benzene ($K_f = 5.12 \text{ Kkg mol}^{-1}$) freezes at 4.47°C . From this observation, one can conclude that
 (A) Phenyl acetic acid exists as such in benzene
 (B) Phenyl acetic acid undergoes partial ionization in benzene
 (C) Phenyl acetic acid undergoes complete ionization in benzene
 (D) Phenyl acetic acid associated in benzene
9. On diluting saturated solution of AgCl in water select the correct observations :
 (A) Solubility of AgCl increases. (B) Freezing point of solution decreases.
 (C) Boiling point of solution decreases. (D) None of these
10. A urea solution in 250 g of water freezes at -0.744°C . This solution was cooled to some temperature where some ice is formed. Solution was decanted-off and heated to 100°C where the vapour pressure was found to be 752.4 mm of Hg. K_f of water is 1.86 K kg/mol .
 (I) What mass of ice will be formed ?
 (A) 69 gm (B) 71.8 gm (C) 80 gm (D) 80.2 gm
 (II) What is the temperature to which solution is cooled
 (A) -1°C (B) -2°C (C) -1.04°C (D) -1.4°C
11. An aqueous solution containing 15 g mixture of urea and glucose boils at 100.27°C . Addition of a further 9.0 g glucose to the above solution causes it to boil at 100.36°C . Determine mass percentage of urea in the original mixture.
 (A) 40% (B) 60% (C) 70% (D) 80%
12. A one litre solution is prepared by dissolving some solid lead-nitrate in water. The solution was found to boil at 100.15°C . To the resulting solution 0.2 mole NaCl was added. The resulting solution was found to freeze at -0.7998°C . Determine solubility product of PbCl_2 . Given $K_b = 0.5$ and $K_f = 1.86$. Assume molality to be equal to molarity in all case.
 (A) 4×10^{-4} (B) 4×10^{-5} (C) 4×10^{-6} (D) 10^{-6}
13. A mixture of MgCl_2 and urea of combined mass 21.5 g is dissolved in enough water to make up a 250 mL solution. The osmotic pressure of the solution is 48 atm at 27°C . Calculate the mass percentage of MgCl_2 in the mixture.
 (Use : $R = 0.08 \text{ litre-atm mol}^{-1} \text{ K}^{-1}$)
 (A) 42% (B) 46% (C) 42.8% (D) 44.18%
14. A glucose solution in 100 g of water boils at 100.26°C . If this solution is heated to 101°C , determine the mass of water left at equilibrium.
 (A) 36 gm (B) 20 gm (C) 26 gm (D) 16 gm
15. A 100 g 12% by weight urea solution is placed together with a 200 g glucose solution in a big jar and left for a very long time. The vapour pressure in the jar at equilibrium was found to be 23.52 mm of Hg. If the vapour pressure of pure water at experimental temperature is 24 mm of Hg, determine mass of glucose in its solution.
 (A) 18 gm (B) 18.9 gm (C) 20 gm (D) 19.5 gm
16. The vapour pressure of a solution of methanol and ethanol at 20°C was found to be 78 mm of Hg. Adding 12 g of urea to 112 g of this solution lowers the vapour pressure to 72.8 mm of Hg. Determine the mass of methanol in the original solution.
 (A) 73.6 gm (B) 38.4 gm (C) 42 gm (D) 83.7 gm

17. Vapour pressure of an equimolar mixture of benzene and toluene was found to be 80 torr. If the vapour above the liquid phase is condensed in a beaker, vapour pressure of this condensate was found to be 100 torr. vapour pressure of pure benzene and pure toluene in the given condition respectively.

(A) 110 torr, 50 torr (B) 120 torr, 40 torr (C) 100 torr, 60 torr (D) None of these

18. The vapour pressure of pure perfluorocyclohexane (C_6F_{12}) are given by the following equations :

$$\text{Solid } C_6F_{12} : \log_{10} p(\text{mm Hg}) = 8.68 - \frac{1890}{T}, \quad \text{Liquid } C_6F_{12} : \log_{10} p(\text{mm Hg}) = 7.60 - \frac{1530}{T}$$

The molar enthalpy of fusion of solid C_6F_{12} is approximately

(A) 1658 cal (B) 360 cal (C) 3420 cal (D) 720 cal

Paragraph for Question nos 19 to 21.

1 mole of liquid A and 9 moles of liquid B are mixed to form a solution. If $P_B^0 = 400$ mm of Hg and

$P_A^0 = 20$ mm of Hg at a temperature 'T' and normal boiling point of liquid B is 300K then answer the questions that follow.

Given data : K_b for B = $2.7 \text{ K Kg.mol}^{-1}$, Molar mass of B = 100

19. It is observed that pressure of vapour above the solution at 'T' Kelvin is 350 mm of Hg. The true statement is :

(A) The liquids form ideal solution
 (B) The solute-solvent interactions are weaker than solvent-solvent or solute-solute interactions.
 (C) The volume of final solution will be lower than sum of individual volumes.
 (D) The enthalpy change due to mixing will be zero

20. If 'A' is assumed to be perfectly non volatile then what will be normal boiling point of the solution.

(A) 300 (B) 300.3 (C) 303 (D) 300.003

21. If 'A' is perfectly non volatile and it dimerises to an extent of 60% then what will be the vapour pressure of the solution.

(A) 360 mm of Hg (B) $\frac{3600}{9.7}$ mm of Hg (C) $\frac{4000}{9.7}$ mm of Hg (D) 36 mm of Hg

Paragraph for question nos. 22 to 24.

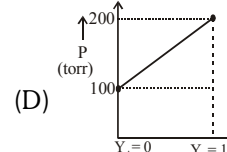
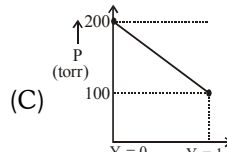
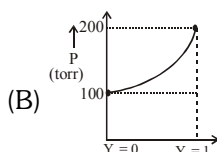
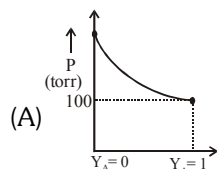
When a liquid is completely miscible with another liquid, a homogeneous solution consisting of a single phase is formed. If such a solution is placed in a closed evacuated vessel, the total pressure exerted by the vapour, after the system attained equilibrium will be equal to the sum of partial pressures of the constituents. A solution is said to be ideal if its constituents follow Raoult's law under all conditions of concentrations. i.e., the partial pressures of each and every constituent is given by

$$p_o = x_i p_i^0$$

where p_i is the partial pressures of the pure constituent.

An ideal solution contains two volatile liquid A ($P^0 = 100$ torr) and B ($P^0 = 200$ torr) and mixture contain 1 mole of A and 4 mole of B.

22. graph is correct representation of total pressure with mole fraction of A in vapour phase (y_A).



23. If 1 mole non volatile and non-electrolyte solid solute is added in original mixture then what will be new vapour pressure of solution.
 (A) 180 torr (B) 150 torr (C) 210 torr (D) None
24. In above question if liquid 'A' is completely polymerise into insoluble solid leaving behind liquid 'B' and 1 mole non volatile solid then :
 (A) vapour pressure of mixture get increase (B) vapour pressure of mixture get decrease.
 (C) boiling point of mixture get increase (D) vapour pressure and boiling point will remain same

Integer-Type

25. For $[\text{CrCl}_3 \cdot x\text{NH}_3]$ elevation in boiling pointing of one molal aqueous solution is double of one molal aqueous solution of glucose, find out x, if complex is 100% ionized.
 (Note :- Co-ordination no. of Cr is 6)
26. If NaHSO_4 is dissolved in water and form a solution having concentration tending to zero and abnormal molar mass of NaHSO_4 is expressed as '10 X' then value of X will be :

ANSWER-KEY (QUIZ # 2)

1. (A) 2. (B) 3. (A) 4. (C) 5. (B) 6. (D) 7. (D) 8. (D) 9. (D)
 10. (I) (B) (II) (C) 11. (A) 12. (C) 13. (D) 14. (C) 15. (B) 16. (B) 17. (B)
 18. (A) 19. (C) 20. (C) 21. (B) 22. (A) 23. (B) 24. (A) 25. (0004)
 26. (0004)

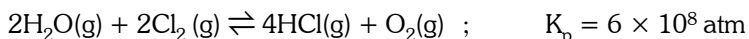
QUIZ # 3

Marking Scheme : Single Correct & Comprehension (+4, -2), Integer Type (+4, -1).

SINGLE OPTION CORRECT

1. Pressure over ideal binary liquid mixture containing 10 moles each of liquid A and B is gradually decreased isothermally. If $P_A^0 = 200$ mm Hg and $P_B^0 = 100$ mm Hg, find the pressure at which half of the liquid is converted into vapour.
 (A) 150 mm Hg (B) 166.5 mm Hg (C) 133 mm Hg (D) 141.4 mm Hg
2. For an ideal binary liquid solution with $P_A^0 > P_B^0$, which relation between X_A (mole fraction of A in liquid phase) and Y_A (mole fraction of A in vapour phase) is correct?
 (A) $Y_A < Y_B$ (B) $X_A > X_B$ (C) $\frac{Y_A}{Y_B} > \frac{X_A}{X_B}$ (D) $\frac{Y_A}{Y_B} < \frac{X_A}{X_B}$
3. The vapour pressure of an aqueous solution is found to be 750 torr at certain temperature 'T'. If 'T' is the temperature at which pure water boils under atmospheric pressure and same solution show elevation in boiling point $\Delta T_b = 1.04$ K, find the atmospheric pressure ($K_b = 0.52$ K kg mol⁻¹)
 (A) 777 (B) 779 (C) 782 (D) 746
4. The freezing point depression of a 0.10 M solution of formic acid is -0.2046°C . What is the equilibrium constant for the reaction at 298 K?
 $\text{HCOO}^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{HCOOH}(\text{aq}) + \text{OH}^-(\text{aq})$
 (Given : $K_f(\text{H}_2\text{O}) = 1.86$ K kg mol⁻¹, Molarity = molality)
 (A) 1.1×10^{-3} (B) 9×10^{-12} (C) 9×10^{-13} (D) 1.1×10^{-11}

5. At certain temperature (T) for the gas phase reaction



If Cl_2 , HCl & O_2 are mixed in such a manner that the partial pressure of each is 1 atm and the mixture is brought into contact with excess of liquid water. What would be approximate partial pressure of Cl_2 when equilibrium is attained at temperature (T)?

[Given : Vapour pressure of water is 380 mm Hg at temperature (T)]

- (A) 8.1×10^{-7} atm (B) 10^{-4} atm (C) 3.6×10^{-3} atm (D) 9×10^{-4} atm
6. The molar volume of X(l) ($d = 0.9$ g/ml) increases by a factor of 3000 as it vaporises at 27°C and that of Y(l) ($d = 0.88$ g/ml) increases by a factor of 8000 at 27°C . A miscible liquid solution of X and Y at 27°C has a vapour pressure of 50 torr. Find the mole fraction of Y in solution is :

[Given : $R = 0.082$ atm/L / mol/K ; Molar mass of X = 75 ; Molar mass of Y = 88]

- (A) 0.48 (B) 0.52 (C) 0.62 (D) 0.247

COMPREHENSION

Paragraph for question nos. 7 to 9.

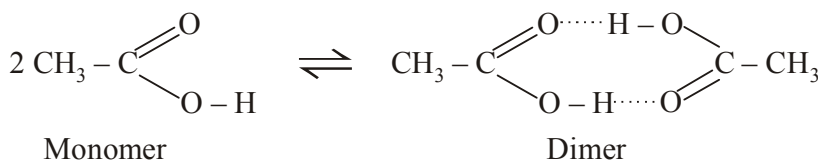
10 mole of liquid 'A' and 20 mole of liquid 'B' is mixed in a cylindrical vessel containing a piston arrangement. Initially a pressure of 2 atm is maintained on the solution. Now, the piston is raised slowly and isothermally. Assume ideal behaviour of solution and A and B are completely miscible.

$$P_A^\circ = 0.6 \text{ atm and } P_B^\circ = 0.9 \text{ atm}$$

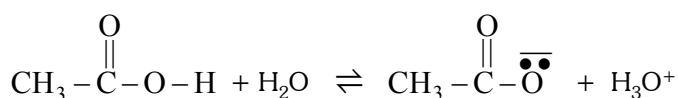
7. The pressure below which the evaporation of liquid solution will start, is
 (A) 0.6 atm (B) 0.8 atm (C) 0.77 atm (D) 0.9 atm
8. The pressure at which $\frac{1}{3}$ th of the total amount (by mol) of liquid solution taken initially, will be present in the vapour form, is
 [Given : $\sqrt{57} = 7.55$]
 (A) 0.783 (B) 0.800 (C) 0.600 (D) 0.652
9. The minimum pressure for the existence of liquid solution is
 (A) 0.6 atm (B) 0.8 atm (C) 0.77 atm (D) 0.9 atm
10. Which of the following is correct for a non-ideal solution of liquids A and B, showing negative deviation?
 (A) $\Delta H_{\text{mix}} = -ve$ (B) $\Delta V_{\text{mix}} = -ve$ (C) $\Delta S_{\text{mix}} = +ve$ (D) $\Delta G_{\text{mix}} = -ve$

Paragraph for question nos. 11 to 13

Acetic acid tends to form dimer due to the formation of intermolecular hydrogen bonding.



The equilibrium constant for this reaction is $1.5 \times 10^2 \text{ M}^{-1}$ in benzene solution and $3.6 \times 10^{-2} \text{ M}^{-1}$ in water. In benzene, monomer does not dissociate but in water, monomer dissociates simultaneously with acid dissociation constant $2.0 \times 10^{-5} \text{ M}$.



Dimer does not dissociate in benzene as well as in water.

[Given : $\log 2 = 0.3$]

11. The molar ratio of dimer to monomer for 0.1 M acetic acid in benzene is equal to
 (A) 150 : 1 (B) 1 : 150 (C) 5 : 2 (D) 2 : 5
12. The molar ratio of dimer to monomer for 0.1 M acetic acid in water (neglecting the dissociation of acetic acid in water) is equal to
 (A) 250 : 1 (B) 1 : 250 (C) 9 : 2500 (D) 2500 : 9
13. The pH of 0.1 M acetic acid solution in water, considering the simultaneous dimerisation of acid is
 (A) 1 (B) 2.85 (C) 5.7 (D) 3.42

MULTIPLE QUESTIONS

14. Acetone and carbon disulphide form binary liquid solution showing positive deviation from Raoult law. The normal boiling point (T_b) of pure acetone is less than that of pure CS_2 . Pick out the **incorrect** statements among the following.
- (A) Boiling temperature of mixture is always less than boiling temperature of acetone.
 (B) Boiling temperature of Azeotropic mixture is always less than boiling temperature of pure CS_2 .
 (C) When a small amount of CS_2 (less volatile component) is added to excess of acetone boiling point of resulting mixture increases.
 (D) A mixture of CS_2 and CH_3COCH_3 can be completely separated by simple fractional distillation.

SUBJECTIVE QUESTIONS

15. A very dilute saturated solution of a sparingly soluble salt A_3B_4 has a vapour pressure of 20 mm of Hg at temperature T, while pure water exerts a pressure of 20.0126 mm Hg at the same temperature. Calculate the solubility product constant of A_3B_4 at the same temperature.
16. The Henry law constant for $O_2(g)$ and $N_2(g)$ in water at $0^\circ C$ are 2.5×10^4 bar and 5.0×10^4 bar respectively. Calculate ΔT_f of solution made by dissolving air at 1 bar pressure.
 Given : mole % of O_2 in air = 20 & mole % of N_2 in air = 80
 K_f of $H_2O = 1.86 \text{ K kg mol}^{-1}$

ANSWER-KEY (QUIZ # 3)

1. (D) 2. (C) 3. (A) 4. (B) 5. (D) 6. (A) 7. (B) 8. (A,B) 9. (C)
 10. (A,B,C,D) 11. (C) 12. (C) 13. (B) 14. (A,C,D)
 15. (5.4×10^{-13}) 16. [Ans. 2.48×10^{-3}]

LIQUID SOLUTION

SINGLE CHOICE QUESTIONS

- Three solutions are prepared by adding 'w' gm of 'A' into 1 kg of water, 'w' gm of 'B' into another 1 kg of water and 'w' gm of 'C' in another 1 kg of water (A, B, C are non electrolytic). Dry air is passed from these solutions in sequence (A $\longrightarrow$ B $\longrightarrow$ C). The loss in weight of solution A was found to be 2 gm while solution B gained 0.5 gm and solution C lost 1 gm. Then the relation between molar masses of A, B and C is :

(A) $M_A : M_B : M_C = 4 : 3 : 5$ (B) $M_A : M_B : M_C = \frac{1}{4} : \frac{1}{3} : \frac{1}{5}$

(C) $M_C > M_A > M_B$ (D) $M_B > M_A > M_C$
- How many m moles of sucrose should be dissolved in 500 gms of water so as to get a solution which has a difference of 103.57°C between boiling point and freezing point. ($K_f = 1.86 \text{ K Kg mol}^{-1}$, $K_b = 0.52 \text{ K Kg mol}^{-1}$)

(A) 500 mmoles (B) 900 mmoles (C) 750 mmoles (D) 650 mmoles
- Select the correct statement:

(A) The semipermeable membrane is the cause of osmotic pressure.

(B) Osmosis results from decrease in entropy.

(C) Osmotic pressure depends on temperature and concentration but is independent of the nature of the membrane.

(D) The passage of solvent molecules occur only in one direction through a semipermeable membrane.
- The incorrect statement is :

(A) Vapour pressure of a liquid always increases by increasing temperature.

(B) Vapour pressure only depends on temperature and not on the nature of substance.

(C) Vapour pressure does not depend on the quantity of the liquid taken and the surface area of the liquid.

(D) Vapour pressure is not a colligative property & is independent of the concentration of the liquid.
- The molar heat of vapourization of toluene is ΔH_v . If its vapour pressure at 315 K is 60 torr & that at 355 K is 300 torr then $\Delta H_{\text{vap}} = ?$ ($\log 2 = 0.3$)

(A) 37.5 kJ/mole (B) 3.75 kJ/mole (C) 37.5 J/mol (D) 3.75 J/mole
- Which of the following has been arranged in order of decreasing freezing point?

(A) $0.05 \text{ M KNO}_3 > 0.04 \text{ M CaCl}_2 > 0.140 \text{ M sugar} > 0.075 \text{ M CuSO}_4$

(B) $0.04 \text{ M BaCl}_2 > 0.140 \text{ M sucrose} > 0.075 \text{ M CuSO}_4 > 0.05 \text{ M KNO}_3$

(C) $0.075 \text{ M CuSO}_4 > 0.140 \text{ M sucrose} > 0.04 \text{ M BaCl}_2 > 0.05 \text{ M KNO}_3$

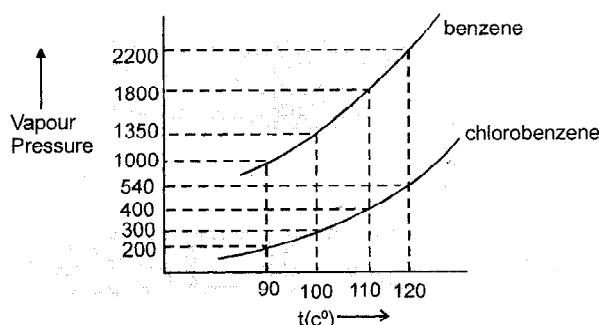
(D) $0.075 \text{ M CuSO}_4 > 0.05 \text{ M NaNO}_3 > 0.140 \text{ M sucrose} > 0.04 \text{ M BaCl}_2$
- A sample of air is saturated with benzene (vapor pressure = 100 mm Hg at 298 K) at 298K, 750mm Hg pressure. If it is isothermally compressed to one third of its initial volume, the final pressure of the system is

(A) 2250 torr (B) 2150 torr (C) 2050 torr (D) 1950 torr
- Available are 1L of 0.1 M NaCl and 2L of 0.2 M CaCl_2 solutions. Using **only these two solutions** what maximum volume of a solution can be prepared having $[\text{Cl}^-] = 0.34 \text{ M}$ exactly. Both electrolytes are strong

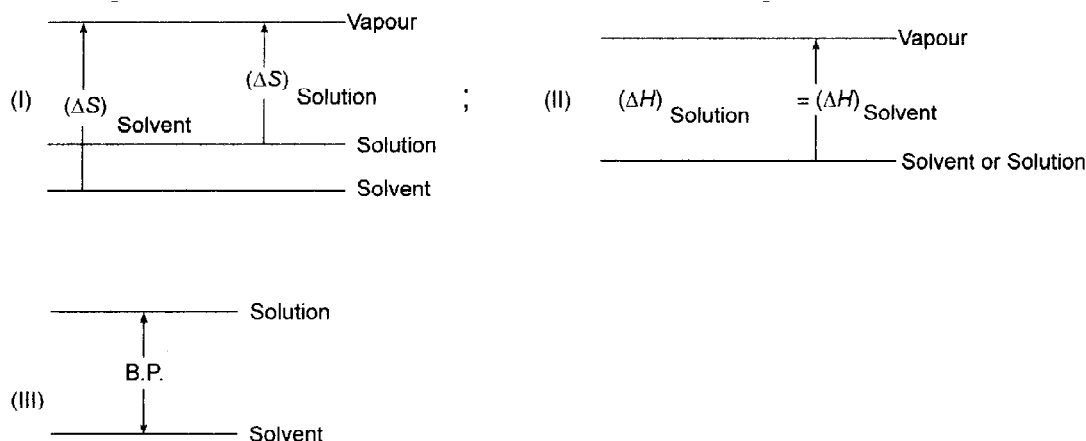
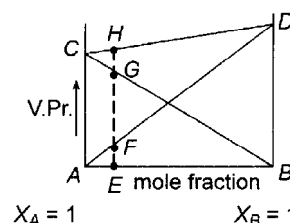
(A) 2.5 L (B) 2.4 L (C) 2.3 L (D) None of these
- A solute 'S' undergoes a reversible trimerization when dissolved in a certain solvent. The boiling point elevation of its 0.1 molal solution was found to be identical to the boiling point elevation in case of a 0.08 molal solution of a solute which neither undergoes association nor dissociation. To what percent had the solute 'S' undergone trimerization ?

(A) 30% (B) 40% (C) 50% (D) 60%

10. Assuming the formation of an ideal solution, determine the boiling point of a mixture containing 1560 g benzene (molar mass = 78) and 1125 g chlorobenzene (molar mass = 112.5) using the following against an external pressure of 1000 Torr.

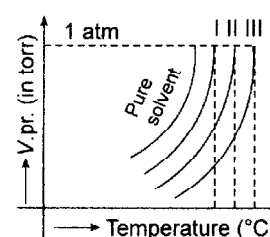


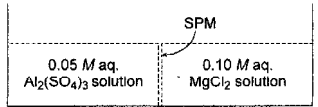
- (A) 90°C (B) 100°C (C) 110° (D) 120°C
11. Based on the given diagram, which of the following statements regarding the homogenous solutions of two volatile liquids are correct ?
- (1) Plots AD and BC show that Raoult's law is obeyed for the solution in which B is a solvent and A is the solute and as well as for that in which A is solvent and B is solute.
- (2) Plot CD shows that Dalton's law of partial pressures is obeyed by the binary solution of components A and B.
- (3) $EF + EG = EH$; and AC and BD correspond to the vapour pressures of the pure solvents A and B respectively. Select the correct answer using the options given below:
- (A) Only 1 (B) 2 and 3 (C) 1 and 3 (D) All
12. Which represents correct difference when non-volatile solute is present in an ideal solution ?



- (A) I, II, III (B) I, III (C) II, III (D) I, II
13. The vapour pressure curves of the same solute in the same solvent are shown. The curves are parallel to each other and do not intersect. The concentrations of solutions are in order of :

- (A) $I < II < III$
 (B) $I = II = III$
 (C) $I > II > III$
 (D) $I > III > II$



14. If two liquids A ($P_A^\circ = 100$ torr) and B ($P_B^\circ = 200$ torr) are completely immiscible with each other, each one will behave independently of the other, are present in a closed vessel. The total vapour pressure of the system will be:
- (A) less than 100 torr (B) less than 200 torr
(C) between 100 to 200 torr (D) 300 torr
15. When a liquid that is immiscible with water was steam distilled at 95.2°C at a total pressure of 748 torr, the distillate contained 1.25 g of the liquid per gram of water. The vapour pressure of water is 648 torr at 95.2°C , what is the molar mass of liquid?
- (A) 7.975 g/mol (B) 166 g/mol (C) 145.8 g/mol (D) None of these
16. One molal solution of a carboxylic acid in benzene shows the elevation of boiling point of 1.518 K . The degree of association for dimerization of the acid in benzene is (K_b for benzene = $2.53\text{ K kg mol}^{-1}$):
- (A) 60% (B) 70% (C) 75% (D) 80%
17. Consider 0.1 M solutions of two solutes X and Y. The solute X behaves as a univalent electrolyte while the solute Y dimerises in solution. Which of the following statements are correct regarding these solutions?
- (1) The boiling point of the solution of X will be higher than that of Y
(2) The osmotic pressure of the solution of Y will be lower than that of X
(3) The freezing point of the solution of X will be lower than that of Y
(4) The relative lowering of vapour pressure of both the solutions will be the same. Select the correct answer from the option given below:
- (A) 1, 2 and 3 (B) 2, 3 and 4 (C) 1, 2 and 4 (D) 1, 3 and 4
18. If M_{normal} is the normal molecular mass and α is the degree of ionization of $K_3[\text{Fe}(\text{CN})_6]$, then the abnormal molecular mass of the complex in the solution will be:
- (A) $M_{\text{normal}}(1 + 2\alpha)^{-1}$ (B) $M_{\text{normal}}(1 + 3\alpha)^{-1}$ (C) $M_{\text{normal}}(1 + \alpha)^{-1}$ (D) equal to M_{normal}
19. A 5% (w/V) solution of cane sugar (molecular weight = 342) is isotonic with 1% (w/V) solution of a substance X. The molecular weight of X is:
- (A) 34.2 (B) 171.2 (C) 68.4 (D) 136.8
20. Two beaker A and B present in a closed vessel. Beaker A contains 152.4 g aqueous solution of urea, containing 12 g of urea. Beaker B contains 196.2 g glucose solution, containing 18 g of glucose. Both solutions allowed to attain the equilibrium. Determine wt. % of glucose in its solution at equilibrium:
- (A) 6.71 (B) 14.49 (C) 16.94 (D) 20
21. An industrial waste water is found to contain 8.2% Na_3PO_4 and 12% MgSO_4 by weight in solution. If % ionisation of Na_3PO_4 and MgSO_4 are 50 and 60 respectively then its normal boiling point is [$K_b(\text{H}_2\text{O}) = 0.50\text{ K kg mol}^{-1}$]:
- (A) 102.3°C (B) 103.35°C (C) 101.785°C (D) None of these
22. Ratio of $\frac{\Delta T_b}{K_b}$ of 10 g AB_2 and 14 g A_2B per 100 g of solvent in their respective, solution (AB_2 and A_2B both are non-electrolytes) is 1 mol/kg in both cases. Hence, atomic wt. of A and B are respectively:
- (A) 100, 40 (B) 60, 20 (C) 20, 60 (D) None of these
23. 100 g of $\text{C}_6\text{H}_{12}\text{O}_6$ (aq.) solution has vapour pressure is equal to 40 torr at certain temperature. Vapour pressure of $\text{H}_2\text{O}(l)$ is 41.8 torr at same temperature. If this solution is cooled to -9.3°C , what mass of ice will be separated out? ($K_f = 1.86\text{ K kg mol}^{-1}$)
- (A) 95.5 g (B) 4.5 g (C) 34.6 g (D) 47.8 g
24. Study the following figure and choose the correct options. Assuming complete dissociation of electrolyte:
- (A) There will be no net moment of any substance across the membrane
(B) MgCl_2 will flow towards the $\text{Al}_2(\text{SO}_4)_3$ solution
(C) $\text{Al}_2(\text{SO}_4)_3$ will flow towards the MgCl_2 solutions
(D) The π (osmotic pressure) of 0.1 M MgCl_2 is higher than the π of 0.05 M $\text{Al}_2(\text{SO}_4)_3$
- 

ONE OR MORE THAN ONE CORRECT OPTIONS

25. In which of the following pairs of solutions will the values of the vant Hoff factor be the same?
- (A) 0.05 M $K_4[Fe(CN)_6]$ and 0.10 M $FeSO_4$
 (B) 0.10 M $K_4[Fe(CN)_6]$ and 0.05 M $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$
 (C) 0.20 M NaCl and 0.10 M $BaCl_2$
 (D) 0.05 M $FeSO_4 \cdot (NH_4)_2SO_4 \cdot 6H_2O$ and 0.02 M $KCl \cdot MgCl_2 \cdot 6H_2O$
26. Vapour pressure of solution containing 6g of a non-volatile solute in 180 g water is 20.0 Torr. If 1 mole water is further added vapour pressure increases by 0.02 Torr. Which of the following is true?
- (A) The molecular weight of solute is $54 g mol^{-1}$
 (B) The vapour pressure of pure water is 20.22 Torr
 (C) Addition of more water in the solution will further raise the vapour pressure of solution.
 (D) The vapour pressure of pure water is 22.22 Torr
27. For chloroform and acetone or for a solution of chloroform and acetone if p_s (observed (actual)) is compared with p_s (Theoretical (Raoult)) then which of the following is /are true?
- (A) $p_s(\text{actual}) < p_s(\text{Raoult})$ (B) $\lim_{x_{\text{chloroform}} \rightarrow 0} (p_{\text{acetone}}^0 - p_s^{\text{actual}}) = 0$
 (C) $\lim_{x_{\text{acetone}} \rightarrow 0} (p_{\text{chloroform}}^0 - p_s(\text{actual})) = 0$ (D) None of these
28. For exact determination of molecular mass through colligative properties measurement:
- (A) solute must be volatile
 (B) solution must be very dilute
 (C) solute must not be dissociated or associated
 (D) none of these
29. For the depression of freezing point experiment. The correct statement(s) is/are:
- (A) Vapour pressure of pure solvent is more than that of solution
 (B) Vapour pressure of pure solvent is less than that of solution
 (C) Only solute molecules solidify at the freezing point
 (D) Only solvent molecules solidify at the freezing point
30. Which of the following statements is (are) incorrect?
- (A) 0.1 M KCl solution will have the same osmotic pressure as 0.1 M glucose solution
 (B) 0.1 M KCl solution will have the same boiling point as 0.1 M urea solution
 (C) 0.1 M glucose and 0.1 m urea are isotonic
 (D) 0.1 M $MgCl_2$ solution will have less relative lowering of vapour pressure than 0.1 m NaCl

COMPREHENSION BASED QUESTIONS

Comprehension # 1

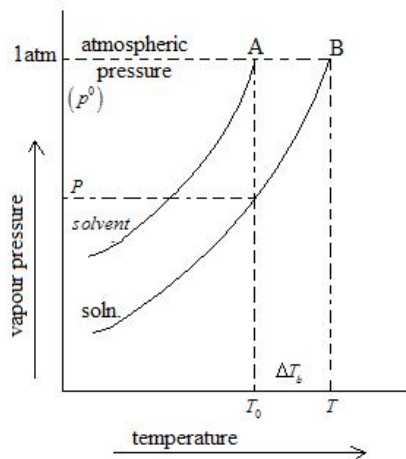
At any fixed temperature, the vapour phase is always richer in more volatile component as compared to the solution phase. In other words, mole fraction of more volatile component is always greater in vapour phase than in solution phase. We can also say that vapour phase is relatively richer in the component whose addition to liquid mixture results in an increase in total vapour pressure.

31. If 2 moles of A and 3 moles of B are mixed to form an ideal solution, vapour pressures of A and B are 120 and 180 mm of Hg respectively. The total vapour pressure of solution will be :
- (A) 48 mm Hg (B) 108 mm Hg (C) 156 mm Hg (D) 15.6 mm Hg

32. From the statement in question number 33, the composition of A and B in the vapour phase when the first traces of vapours are formed is :
- (A) A = 0.407 ; B = 0.592 (B) A = 0.8 ; B = 0.1
 (C) A = 0.109 ; B = 0.791 (D) A = 0.307 ; B = 0.693
33. From the statement in question number 33, at what pressure will the last traces of liquid disappear?
- (A) 100 mm Hg (B) 130 mm Hg (C) 140 mm Hg (D) 150 mm Hg

Comprehension # 2

Figure explains elevation in boiling point when a non-volatile solute is added to a solvent



34. Elevation in boiling point of an aqueous urea solution is 0.52° (K_b of water is $0.52 \text{ K Kg mol}^{-1}$). Hence mole fraction of urea in this solution is
- (A) 0.982 (B) 0.0567 (C) 0.943 (D) 0.018
35. A complex of iron and cyanide ions is 100% ionized at one molal solution. If the elevation in boiling point is 2.08°C (K_b for water $0.52 \text{ K Kg mol}^{-1}$) then the complex is
- (A) $K_3[Fe(CN)_6]$ (B) $Fe(CN)_2$ (C) $K_4[Fe(CN)_6]$ (D) $[Fe(CN)_4]^-$
36. Ratio of $\frac{\Delta T_b}{K_b}$ of 6% AB_2 and 9% A_2B (both are non electrolytes) is one mole/Kg in both the cases. Hence atomic masses of A and B are respectively.
- (A) 60, 90 (B) 40, 40 (C) 40, 10 (D) 10, 40

Comprehension # 3

When a volatile substance A ($P_A^0 = 400 \text{ mm}$) is added to another volatile substance B ($P_B^0 = 500 \text{ mm}$), the vapour pressure of the resulting solution is 470 mm of Hg. To this solution a third volatile solvent C ($P_C^0 = 600 \text{ mm}$) is added and vapour pressure of final solution is 496 mm of Hg.

37. Mole fraction of A in the final solution
- (A) 0.20 (B) 0.22 (C) 0.24 (D) 0.26
38. Mole fraction of A in the vapour phase
- (A) 0.2 (B) 0.22 (C) 0.24 (D) 0.26
39. Mole fraction of C in the vapour phase
- (A) 0.2 (B) 0.22 (C) 0.24 (D) 0.26

LEVEL - 1**SUBJECTIVE TYPE QUESTIONS**

- At 90°C, the vapour pressure of toluene is 400 torr and that of σ -xylene is 150 torr. What is the composition of the liquid mixture that boils at 90°C, when the pressure is 0.50 atm? What is the composition of vapour produced?
- The vapour pressure of water is 17.54 mm Hg at 293 K. Calculate vapour pressure of 0.5 molal solution of a solute in it.
- Benzene and toluene form two ideal solution A and B at 313 K. Solution A (total pressure P_A) contains equal mole of toluene and benzene. Solution B contains equal masses of both (total pressure P_B). The vapour pressure of benzene and toluene are 160 and 60 mm Hg respectively at 313 K. Calculate the value of P_A/P_B .
- The cryoscopic constant for acetic acid is 3.6 K kg/mol. A solution of 1 g of a hydrocarbon in 100 g of acetic acid freezes at 16.14°C instead of the usual 16.60°C. The hydrocarbon contains 92.3% carbon. What is the molecular formula?
- The osmotic pressure of a solution of a synthetic polyisobutylene in benzene was determined at 25°C. A sample containing 0.20 g of solute/100 cm³ of solution developed a rise of 2.4 mm at osmotic equilibrium. The density of the solution was 0.88 g/cm³. What is the molecular weight of the polyisobutylene?
- 10 gm of solute A and 20 gm of solute B are both dissolved in 500 ml water. The solution has the same osmotic pressure as 6.67 gm of A and 30 gm of B dissolved in the same amount of water at the same temperature. What is the ratio of molar masses of A and B?
- A storage battery contains a solution of H₂SO₄ 38% by weight. What will be the Van't Hoff factor if the $\Delta T_{f(\text{experiment})}$ is 29.08. [Given $K_f = 1.86 \text{ mol}^{-1} \text{ Kg}$]
- A 1.2% solution (w/v) of NaCl is isotonic with 7.2% solution (w/v) of glucose. Calculate degree of ionization and Van't Hoff factor of NaCl.
- x g of urea was dissolved in 500 g of water and cooled upto -0.5°C whereby 128 g of ice separates out from the solution. If cryoscopic constant for water be 1.86°C/m. Calculate the value of x.
- The vapour pressure of CS₂ at 50°C is 854 torr and a solution of 2.0 g sulphur in 100 g of CS₂ has vapour pressure 848.9 torr. If the formula of sulphur molecule is S_n, then calculate the value of n. (at. mass of S = 32).
- The molal freezing point depression constant for benzene is 4.9 K kgmol⁻¹. Selenium exists as a polymer of the type S_n. When 3.26 g selenium is dissolved in 226 g of benzene, the observed freezing point is 0.112°C lower than that of pure benzene. If atomic mass of Se is 78.8, find the value of n.
- An aqueous solution of glucose (m.w. = 180) containing 12 g dissolved in 100 g of water was found to boil at 100.34°C while the boiling point of pure water was 100°C. Calculate If molal elevation constant is expressed as $x \times 10^{-2}$ then value of x will be :
- Two elements A and B form compounds having molecular formulae AB₂ and AB₄. When dissolved in 20.0 g of benzene, 1.0 g of AB₂ lowers the freezing point by 2.3 °C, whereas 1.0 g of AB₄ lowers the freezing point by 1.3°C. The molal depression constant for benzene in 1000 g is 5.1. Calculate the atomic masses of A and B.
- 2 g of benzoic acid dissolved in 25 g of C₆H₆ shows a depression in freezing point equal to 1.62 K. Molal depression constant of C₆H₆ is 4.9 K mol⁻¹ kg. What is the percentage association of acid if it forms double molecule in solution ?
- A decimolar solution of potassium ferrocyanide is 50% dissociated at 300 K. Calculate the osmotic pressure of the solution. (S = 8.314 J K⁻¹ mol⁻¹)
- The degree of dissociation of Ca(NO₃)₂ in a dilute aqueous solution containing 7 g of salt per 100 g of water at 100°C is 70%. If the vapour pressure of water is 760 mm, calculate the vapour pressure of solution.
- Given that ΔT_f is the depression in freezing point of the solvent in a solution of a

non-volatile solute of molality 'm', the quantity $\lim_{m \rightarrow 0} \left(\frac{\Delta T_f}{m} \right)$ is equal to

18. The freezing point of a solution containing 0.2 g of acetic acid in 20.0g of benzene is lowered by 0.45°C . Calculate the degree of association of acetic acid in benzene. Assuming acetic acid dimerizes in benzene.
 K_f for benzene = $5.12 \text{ K mol}^{-1} \text{ kg}$.
19. 0.85 % aqueous solution of NaNO_3 is apparently 90 % dissociated at 27°C . Calculate its osmotic pressure. ($R = 0.082 \text{ atm K}^{-1} \text{ mol}^{-1}$)
20. The molar volume of liquid benzene (density = 0.877 g ml^{-1}) increases by a factor of 2750 as it vaporizes at 20°C and that of liquid toluene (density = 0.867 g ml^{-1}) increases by a factor of 7720 at 20°C . A solution of benzene & toluene at 20°C has a vapour pressure of 46.0 torr. Find the mole fraction of benzene in the vapour above the solution.
21. 1.22 g $\text{C}_6\text{H}_5\text{COOH}$ is added into solvent and data of ΔT_b and K_b are given as –
 (1) In 100 g of CH_3COCH_3
 $\Delta T_b = 0.17$ and $K_b = 1.7 \text{ kg Kelvin/mol}$
 (2) In 100 g benzene
 $\Delta T_b = 0.13$ and $K_b = 2.6 \text{ kg Kelvin/mol}$.
 (i) Find out the molecular weight of $\text{C}_6\text{H}_5\text{COOH}$ in both the cases
 (ii) What do you deduce out of it in terms of structure of benzoic acid ?

LEVEL -2**SUBJECTIVE TYPE QUESTIONS**

1. At 100°C , benzene & toluene have vapour pressure of 1375 & 558 Torr respectively. Assuming these two form an ideal binary solution, calculate the composition of the solution that boils at 1 atm & 100°C . What is the composition of vapour issuing at these conditions?
2. At 25°C , 1 mol of A having a vapor pressure of 100 torr and 1 mol of B having a vapor pressure of 300 torr were mixed. The vapor at equilibrium is removed, condensed and the condensate is heated back to 25°C . The vapors now formed are again removed, recondensed and analyzed. What is the mole fraction of A in this condensate?
3. Vapour pressure of C_6H_6 and C_7H_8 mixture at 50°C is given by $P (\text{mm Hg}) = 180 X_B + 90$, where X_B is the mole fraction of C_6H_6 . A solution is prepared by mixing 936g benzene and 736g toluene and if the vapours over this solution are removed and condensed into liquid and again brought to the temperature of 50°C , what would be mole fraction of C_6H_6 in the vapour state?
4. The vapour pressure of a certain liquid is given by the equation:
 $\log_{10} P = 3.54595 - \frac{313.7}{T} + 1.40655 \log_{10} T$ where P is the vapour pressure in mm and T = Kelvin Temperature.
 Determine the molar latent heat of vaporisation as a function of temperature. Calculate the its value at 80 K.
5. The vapour pressure of two pure liquids, A and B that form an ideal solution are 300 and 800 torr respectively, at temperature T. A mixture of the vapour of A and B for which the mole fraction of A is 0.25 is slowly compressed at temperature T. Calculate
 (A) the composition of the first drop of the condensate,
 (B) the total pressure when this drop is formed,
 (C) the composition of the solution whose normal boiling point is T,
 (D) the pressure when only the last bubble of vapour remains, and
 (E) the composition of the last bubble.
6. When the mixture of two immiscible liquids (water and nitrobenzene) boils at 372 K and the vapour pressure at this temperature are 97.7 kPa (H_2O) and 3.6 kPa ($\text{C}_6\text{H}_5\text{NO}_2$). Calculate the weight % of nitrobenzene in the vapour.

7. 30 ml of CH_3OH ($d = 0.7980 \text{ gm cm}^{-3}$) and 70 ml of H_2O ($d = 0.9984 \text{ gm cm}^{-3}$) are mixed at 25°C to form a solution of density $0.9575 \text{ gm cm}^{-3}$. Calculate the freezing point of the solution. $K_f(\text{H}_2\text{O})$ is $1.86 \text{ kg mol}^{-1} \text{ K}$. Also calculate its molarity.
8. An aqueous solution containing 288 gm of a non-volatile compound having the stoichiometric composition $\text{C}_x\text{H}_{2x}\text{O}_x$ in 90 gm water boils at 101.24°C at 1.00 atmospheric pressure. What is the molecular formula?
 $K_b(\text{H}_2\text{O}) = 0.512 \text{ K mol}^{-1} \text{ kg}$ $T_b(\text{H}_2\text{O}) = 100^\circ\text{C}$
9. The addition of 3 gm of substance to 100 gm CCl_4 ($M = 154 \text{ gm mol}^{-1}$) raises the boiling point of CCl_4 by 0.60°C of $K_b(\text{CCl}_4)$ is $5.03 \text{ kg mol}^{-1} \text{ K}$. Calculate
 (A) the freezing point depression (B) the relative lowering of vapour pressure
 (C) the osmotic pressure at 298 K (D) the molar mass of the substance
 Given $K_f(\text{CCl}_4) = 31.8 \text{ kg mol}^{-1} \text{ K}$ and ρ (density) of solution = 1.64 gm/cm^3
10. A 10% w/w solution of cane sugar has undergone partial inversion according to the reaction:
 Sucrose + Water $\longrightarrow$ Glucose + Fructose. If the boiling point of solution is 100.27°C .
 (a) What is the average mass of the dissolved materials?
 (b) What fraction of the sugar has inverted?
 $K_b(\text{H}_2\text{O}) = 0.512 \text{ K mol}^{-1} \text{ kg}$
11. Sea water is found to contain 5.85% NaCl and 9.50% MgCl_2 by weight of solution. Calculate its normal boiling point assuming 80% ionisation for NaCl and 50% ionisation of MgCl_2 [$K_b(\text{H}_2\text{O}) = 0.51 \text{ kg mol}^{-1} \text{ K}$].
12. The latent heat of fusion of ice is 80 calories per gram at 0°C . What is the freezing point of a solution of KCl in water containing 7.45 grams of solute in 500 grams of water, assuming that the salt is dissociated to the extent of 95%?
13. Calculate the boiling point of a solution containing 0.61g of benzoic acid in 50g of carbon disulphide assuming 84% dimerization of the acid. The boiling point and K_b of CS_2 are 46.2°C and $2.3 \text{ K kg mol}^{-1}$, respectively.
14. Phenol associates in benzene to a certain extent to form a dimer. A solution containing $20 \times 10^{-3} \text{ kg}$ phenol in 1 kg of benzene has its freezing point depressed by 0.69 K. Calculate the fraction of phenol that has dimerised. K_f for benzene = $5.12 \text{ kg mol}^{-1} \text{ K}$.
15. A very dilute saturated solution of a sparingly soluble salt A_3B_4 has a vapour pressure of 20 mm of Hg at temperature T, while pure water exerts a pressure of 20.0126 mm Hg at the same temperature. Calculate the solubility product constant of A_3B_4 at the same temperature.
16. If the apparent degree of ionization of KCl ($KCl = 74.5 \text{ gm mol}^{-1}$) in water at 290 K is 0.86. Calculate the mass of KCl which must be made up to 1 dm^3 of aqueous solution to the same osmotic pressure as the 4.0% solution of glucose at that temperature.
17. An ideal solution was prepared by dissolving some amount of cane sugar (non-volatile) in 0.9 moles of water. The solution was then cooled just below its freezing temperature ($T_f = 271 \text{ K}$), where some ice get separated out. The remaining aqueous solution registered a vapour pressure of 700 torr at 373 K. Calculate the mass of ice separated out, if the molar heat of fusion of water is 6 kJ.
18. The specific conductivity of a 0.5 M a solution of monobasic acid HA at 27°C is 0.006 Scm^{-1} . It's molar conductivity at infinite dilution is $200 \text{ S cm}^2 \text{ mol}^{-1}$.
 Calculate osmotic pressure (in atm) of 0.5 M HA (aq) solution at 27°C .
 Given $R = 0.08 \frac{\text{atm L}}{\text{mol K}}$.
19. Tritium, T (an isotope of H) combines with fluorine to form weak acid TF, which ionizes to give T^+ . Tritium is radioactive and is a β -emitter. A freshly prepared aqueous solution of TF has pT (equivalent of pH) of 1.5 and freezes at -0.372°C . If 600ml of freshly prepared solution were allowed to stand for 24.8 years. Calculate
 (i) ionization constant of TF
 (ii) Number of β -particles emitted.
 (Given K_f for water = $1.86 \text{ kg mol}^{-1} \text{ K}$, $t_{1/2}$ for tritium = 12.4 years)
20. A mixture of two immiscible liquids nitrobenzene and water at 99°C has a partial vapour pressure of water 733 mm and that of nitrobenzene 27 mm. Calculate the ratio of the weights of water to nitrobenzene in the distillate.

ANSWER KEY**SINGLE CHOICE QUESTIONS**

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 1. (C) | 2. (C) | 3. (C) | 4. (B) | 5. (A) | 6. (A) |
| 7. (C) | 8. (A) | 9. (A) | 10. (B) | 11. (D) | 12. (A) |
| 13. (A) | 14. (D) | 15. (C) | 16. (D) | 17. (A) | 18. (B) |
| 19. (C) | 20. (B) | 21. (C) | 22. (B) | 23. (C) | 24. (D) |

ONE OR MORE THAN ONE CORRECT QUESTIONS

- | | | | |
|-----------|-------------|-------------|-----------|
| 25. (B,D) | 26. (A,B,C) | 27. (A,B,C) | 28. (B,C) |
| 29. (A,D) | 30. (A,B,D) | | |

COMPREHENSION BASED QUESTIONS

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 31. (C) | 32. (D) | 33. (D) | 34. (D) | 35. (A) | 36. (C) |
| 37. (C) | 38. (A) | 39. (C) | | | |

LEVEL-1

- | | |
|--|---------------------------|
| 1. 92 mol% toluene; 96.8 mol % toluene | 2. 17.38 mm |
| 3. 0.964 | 4. C_6H_6 |
| 5. 2.4×10^5 g/mol | 6. $M_A/M_B = 0.33$ |
| 7. $i = 2.5$ | 8. 0.95; 1.95 |
| 9. (0006) | 10. (0008) |
| 11. (0008) | 12. (51) |
| 13. (A = 25.57g, B = 42.645 g) | 14. ($\alpha = 99.2\%$) |
| 15. (7.483×10^5 Nm ⁻²) | 16. (746.27 mm) |
| 17. (K_f) | 18. (94.53%) |
| 19. (4.674 atm) | 20. (0.73) |
| 21. (i) $M = 122$, $M' = 244$ (ii) benzoic acid remains as it is acetone while it dimerises in benzene. | |

LEVEL-2

- | | |
|--|--|
| 1. $x_b = 0.2472$, $Y_b = 0.4473$ | |
| 2. $x_a = 0.1$ | |
| 3. 0.93 | |
| 4. ΔH_v at 80 K is 1659.1 calorie; $\Delta H_v = R [2.303 \times 313.7 + 1.40655T]$ | |
| 5. (a) 0.47, (b) 565 torr, (c) $x_A = 0.08$, $x_B = 0.92$ (d) 675 torr, (e) $x'_A = 0.11$, $x'_B = 0.89$ | |
| 6. 20.11 % | 7. -19.91°C , 7.63 M |
| 8. $C_{44}H_{88}O_{44}$ | 9. (a) 3.79°C , (b) 0.018, (c) 4.65 atm, (d) 251.5 |
| 10. (a) 210.65, (b) 62.35% | 11. $T_b = 102.3^\circ\text{C}$ |
| 12. $T_f = -0.73^\circ\text{C}$ | 13. 46.33°C |
| 14. $\alpha = 0.7333$ | 15. 5.4×10^{-13} |
| 16. 8.9 gm | 17. 12.54 g |
| 18. 12.72 | 19. (i) $K_a = 7.3 \times 10^{-3}$ (ii) 4.55×10^{22} |
| 20. (0004) | |

QUIZ # 01

SOLID STATE

PHYSICAL CHEMISTRY

Single Option Correct (+3, -1)

- Amorphous solid can be treated as
 (A) Super cooled liquid (B) Pseudo solid (C) Both (A) & (B) (D) none of these
- Which is not the property of crystalline solid
 (A) regular arrangement of lattice point in 3-D. (B) They have definite melting point.
 (C) They are anisotropic in nature (D) They are isotropic in nature.
- Quartz and silica glass are respectively :
 (A) Amorphous, Crystalline (B) Both amorphous
 (C) Both crystalline (D) Crystalline, amorphous
- Which properties are incorrectly matched :
 (A) Amorphous solid : isotropic (B) Crystalline solid : definite heat of fusion.
 (C) Amorphous solid : short range order may exist (D) Crystalline solid : isotropic
- Which crystal system have cell parameter $a = b = c$, $\alpha = \beta = \gamma$
 (i) Cubic (ii) Trigonal (iii) Orthorhombic
 (A) (i), (iii) (B) (ii), (iii) (C) (i), (ii) (D) none of these
- Select the incorrect option :
 (A) Cubic crystal system is most symmetrical
 (B) Triclinic is most general crystal system.
 (C) Orthorhombic and cubic crystal system have $\alpha = \beta = \gamma = 90^\circ$
 (D) none of these
- How many crystal system have angles $\alpha = \beta = \gamma = 90^\circ$
 (A) 4 (B) 3 (C) 5 (D) 2
- Select the incorrect pairs :
 (A) NaCl : Crystalline solid (B) P_4O_{10} : Crystalline solid
 (C) Plastics : amorphous solid (D) Quartz : amorphous solid
- End-centred unit cell is possible in :
 (i) rhombic (ii) Cubic (iii) Monoclinic
 (A) (i), (ii) (B) (i), (iii) (C) (ii) (D) Only (iii)
- Select the incorrect option :
 (A) In space lattice each lattice point is identical.
 (B) In space lattice there is regular arrangement of lattice point in 3-D.
 (C) Primitive unit cell have one lattice point per unit cell.
 (D) Cubic crystal system have only primitive unit cell.
- In given two dimensional space lattice :
 Which can act as unit cell.



- (A) 1 and 2 (B) 1 and 3 (C) 2 and 3 (D) none of these

12. In a unit cell lattice point present at centres of the one pair of the opposite faces along with at all corners, it is called
(A) body centred unit cell (B) face centred unit cell
(C) end centred unit cell (D) primitive unit cell
13. In cubic unit cell, select the incorrect one :
(i) $a = b = c$, $\alpha = \beta = \gamma = 90^\circ$
(ii) lattice points are present at edge centre.
(iii) lattice point present at body centre is shared by one unit cell
(A) (ii) (B) (i) (C) (i), (iii) (D) (i), (ii)
14. Select the correct statements :
(i) All unit cells are parallalopied
(ii) Volume of unit cell = $[\vec{a} \vec{b} \vec{c}]$ where $\vec{a}$, $\vec{b}$ and $\vec{c}$ are vector along axial distances.
(iii) C.N. equals to no. of 1st neighbours of an atom.
(A) only (i) (B) Only (i), (ii) (C) (i), (ii), (iii) (D) Only (iii)
15. If unit cell dimensions a, b, c are same in cubic and trigonal crystal system and in trigonal crystal system $\alpha = \beta = \gamma = 60^\circ$ then ratio of volume of cubic to trigonal unit cell is :
(A) $\sqrt{\frac{3}{2}}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{2}{\sqrt{3}}$ (D) $\sqrt{2}$
16. Among three unit cell of cubic system. Select the incorrect matching :
(A) Simple cubic : primitive
(B) face centred cubic : centred
(C) Body centred cubic : non-primitive
(D) face centred cubic : highest no. of elements of symmetry among all unit cells.

Paragraph for Q.17 to Q.18

Solid balls of radius 17.32 cm crystallises in bcc pattern, during one such crystallisation some oxygen gas is trapped. This trapped oxygen at 640K creates pressure of 5 atm .

Assume :

- (i) BCC arrangement is not disturbed due to trapping of gas.
(ii) Gas is uniformly distributed inside unit cell

[Take $R = 0.08 \text{ atm-litre/mole-K}$, $N_A = 6 \times 10^{23}$, Mass of a solid ball = 64 gms]

17. Calculate number of oxygen molecules present in a unit cell
(A) 24×10^{23} (B) 12×10^{23} (C) 6×10^{23} (D) 3×10^{23}
18. Calculate percentage increase in density due to trapping of gas
(A) 16.67 % (B) 33.33 % (C) 100% (D) 50%

Multiple Option Correct (+4, -2, +1 for each correct answer)

19. Cubic Crystal have :
(A) One centre of symmetry (B) 9 plane of symmetry
(C) 13 axis of symmetry (D) 9 axis of symmetry

- 20.** Select the correct options, for cubic crystal system :
- (A) 2-fold axis of symmetry : line along a body diagonal
 (B) 3-fold axis of symmetry : line along a edge diagonal
 (C) 4-fold axis of symmetry : line along a body diagonal
 (D) There is one centre of symmetry
- 21.** Select the correct options :
- (A) In a space lattice point may represent single atom, group of atoms, single molecule, group of ions.
 (B) In space lattice (crystal lattice) all lattice points are not identical.
 (C) In a solid physical properties are non-directional, then it is amorphous solid.
 (D) dry ice (solid CO_2) is a network solid.
- 22.** Calcium crystallizes in a cubic unit cell with density 3.2 g/cc. Edge-length of the unit cell is 437 picometre (pm)
- (A) The type of unit cell is FCC
 (B) The nearest neighbour distance is 309 pm
 (C) The number of nearest neighbours of a Ca atom are 12
 (D) If the metal is melted, density of the molten metal was found to be 3 g/cc. The percentage of empty space in the melted is 31%

Integer type (+3, 0)

- 23.** How many crystal system have end centred unit cells and having $\alpha = \beta = 90^\circ$?
- 24.** How many unit cells are possible in orthorhombic crystal system
- 25.** How many given solids are molecular solids ?
 NaCl, I_2 , diamond, CaCl_2 , P_4O_{10} , Rubber, Plastics.
- 26.** If x = no. of crystal system having $\alpha = \beta = \gamma$ and y = no. of crystal system having at least 2 interfacial angles equal to 90° . Then value of $(y - x)$ will be :

ANSWER KEY (QUIZ # 01)

- | | | | | | | | | |
|-------------|---------|-----------|-----------|---------|---------|---------|---------|---------|
| 1. (C) | 2. (D) | 3. (D) | 4. (D) | 5. (C) | 6. (D) | 7. (C) | 8. (D) | 9. (B) |
| 10. (D) | 11. (A) | 12. (C) | 13. (A) | 14. (C) | 15. (D) | 16. (D) | 17. (B) | 18. (D) |
| 19. (A,B,C) | 20. (D) | 21. (A,C) | 22. (ALL) | 23. (2) | 24. (4) | 25. (2) | 26. (1) | |

QUIZ # 2

Single Option Correct (+3, -1)

- 1.** A metal 'X' crystallizes in SC unit cell (edge length = a). A foreign atom 'Y' is placed at biggest possible void in unit cell. The maximum possible radius of 'Y' without creating distortion is :

- (A) $\frac{a\sqrt{3}}{4}$ (B) $\left(\frac{3-\sqrt{3}}{2}\right)a$ (C) $\left(\frac{\sqrt{3}-1}{4}\right)a$ (D) $\left(\frac{\sqrt{3}-1}{2}\right)a$

2. A pure metal 'A' can crystallizes in closest packing in 3-D (highest C.N.) then arrangement can be :
(A) CCP (B) BCC (C) HCP (D) Both (A) and (C)
3. A metallic solid crystallizes in FCC arrangement and having atomic radius $\frac{1}{\sqrt{2}} \text{ pm}$. If no. of unit cells in a cubical piece of metal having total surface area 24 cm^2 are 10^x then value of 'X' is :
(a) 15 (B) 20 (C) 30 (D) 40
4. In a solid 'AB' having the structure in which $\frac{r_+}{r_-}$ is between 0.414 and 0.732 and 'A' atoms occupy the corners of the cubic unit cell. If all the face centered atoms along one of the axis are removed, then the resultant stoichiometry of the solid is :
(A) AB_2 (B) A_2B (C) A_4B_3 (D) A_3B_4
5. A solid has a structure in which W atoms are located at the corner of a cubic lattice, O atom at the centre of edges and Na atom at centre of the cube. The formula for the compound is:
(A) NaWO_2 (B) NaWO_3 (C) Na_2WO_3 (D) NaWO_4
6. In a CCP lattice of X and Y, X atoms are present at the corners while Y atoms are at face centers. Then the formula of the compound would be if one of the X atoms from a corner is replaced by Z atoms (also monovalent) ?
(A) $\text{X}_7\text{Y}_{24}\text{Z}_2$ (B) $\text{X}_7\text{Y}_{24}\text{Z}$ (C) $\text{X}_{24}\text{Y}_7\text{Z}$ (D) XY_{24}Z

Multiple Option Correct (+4, -2, +1 for each correct answer)

7. In BCC arrangement of identical hard sphere 'A' (radius R) foreign atoms 'B' (radius = r) placed at each face centre of the unit cell (maximum possible radius) without creating distortion in host lattice then select the correct options.
(A) $\frac{r}{R} = \sqrt{2} - 1$
(B) $\frac{r}{R} = \frac{2}{\sqrt{3}} - 1$
(C) Nearest neighbours of body centred 'A' are 6B.
(D) Second neighbours of body centred 'A' are 8A.
8. Which of the following statement are true regarding tetrahedral voids in fcc lattice ?
(A) Tetrahedral voids formed by one corner spheres and three face spheres.
(B) Two tetrahedral voids can exist on one diagonal body line in fcc cubic.
(C) Tetrahedral voids are equal to number of spheres involved in body.
(D) Volume of tetrahedral voids are half of the octahedral voids.
9. The solid compound SiO_2 exist in the forms of
(A) crystalline in quartz (B) amorphous in quartz
(C) amorphous in silica glass (D) crystalline in silica glass
10. A metallic lattice crystallize in such a manner that the atoms are arranged in a hexagonal manner in one layer & the repetition is ABABAB..... of the hexagonal layers.
(A) The rank of the hexagonal unit lattice is 6
(B) the rank of the hexagonal unit lattice is 12
(C) The volume of the unit lattice is $\left[6 \times \frac{\sqrt{3}}{4} 4R^2 \right] \cdot \left[4R \frac{\sqrt{2}}{\sqrt{3}} \right]$
(D) The percentage of vacant space = 26

11. BaTiO_3 crystallizes in perovskite structure. This structure may be described as a cubic lattice, with barium ions occupying the corners, oxide ions at the face centres and titanium ions in one of the octahedral holes. Which of the following is correct about BaTiO_3 crystal lattice ?
- (A) Ti^{4+} occupies the body centred octahedral hole.
 (B) Ti^{4+} occupies the edge centred octahedral hole.
 (C) Ti^{4+} can occupy any of the octahedral holes.
 (D) $\frac{1}{4}$ th of the octahedral holes are occupied.
12. Table for some ionic radii are given

Ions	K^+	Cs^+	Mg^{+2}	Sr^{+2}	Br^-	O^{2-}	S^{2-}
Ionic radii ($\text{\AA}$)	1.33	1.69	0.66	1.12	1.96	1.40	1.86

Then predict the compound which can exist in FCC is/are

- (A) KBr (B) CsBr (C) MgO (D) SrS

Integer type (+4, -1)

13. In a planar tetra-atomic molecule AB_3 A is the centroid of an equilateral triangle formed by atoms B. If the A-B bond distance is 1, what is the square of the distance between the centres of any two B atoms.
14. AB crystallizes in a rock salt structures with A:B=1:1. In shortest distances between A and B is $Y^{1/3}$ nm. The formula mass of AB is 6.023 Y amu, where Y is arbitrary constant, find the density in kg m^{-3}
15. The minimum rotational symmetry present in a monoclinic unit cell is.
16. A body-centred cubic lattice is made-up of two types of atoms, A and B in which A occupy the body-centre, while B is at the corners. Due to some imperfections in the solid, one corner is left unoccupied per unit cell and formula of the solid is A_xB_y . Here x is ?

ANSWER-KEY (QUIZ # 02)

1. (D) 2. (D) 3. (C) 4. (D) 5. (B) 6. (B) 7. (B,C,D) 8. (A,B)
 9. (A,C) 10. (A, C, D) 11. (A, D) 12. (A, C, D) 13. (3) 14. (5)
 15. (2) 16. (8)

QUIZ # 3

SINGLE OPTION CORRECT (+3, -1)

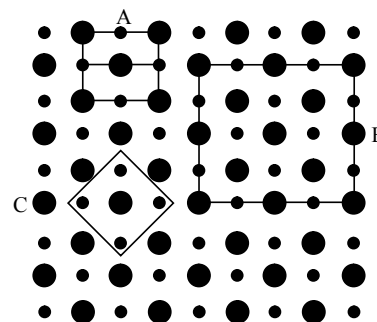
1. In diamond, carbon atom occupy FCC lattice points as well as alternate tetrahedral voids. If edge length of the unit cell is 356 pm, then radius of carbon atom is
 (A) 77.07 pm (B) 154.14 pm (C) 251.7 pm (D) 89 pm
2. Which of the following is incorrect about NaCl structure ?
 (A) Nearest neighbours of Na^+ ion is six Cl^- ion
 (B) Na^+ ion make fcc lattice
 (C) Cl^- ion pack in cubic close packed arrangement
 (D) There are eight next nearest neighbours of Na^+ ion

3. In the body-centered cubic unit cell & face centered cubic unit cell, the radius of atom in terms of edge length (a) of the unit cell is respectively:
- (A) $\frac{a}{2}, \frac{a}{2\sqrt{2}}$ (B) $\frac{a}{2\sqrt{2}}, \frac{\sqrt{3}a}{4}$ (C) $\frac{\sqrt{3}a}{4}, \frac{a}{2\sqrt{2}}$ (D) $\frac{\sqrt{3}a}{2}, \frac{a}{2\sqrt{2}}$
4. Diamond structure can be considered as ZnS (Zinc blend) structure in which each Zn^{2+} in alternate tetrahedral void and S^{2-} in cubic close pack arrangement is replaced by one carbon atom. If C-C covalent bond length in diamond is $1.5\text{\AA}$, what is the edge length of diamond unit cell ($z = 8$).
- (A) $3.46\text{\AA}$ (B) $6.92\text{\AA}$ (C) $1.73\text{\AA}$ (D) $3\text{\AA}$
5. Copper metal crystallizes in FCC lattice. Edge length of unit cell is 362 pm . The radius of largest atom that can fit into the voids of copper lattice without disturbing it.
- (A) 53 pm (B) 45 pm (C) 93 pm (D) 93 pm
6. In FCC unit cell, what fraction of edge is not covered by atoms?
- (A) 0.134 (B) 0.24 (C) 0.293 (D) None of these
7. A solid AB has the NaCl structure. If radius of the cation A^+ is 120 pm . calculate the maximum value of the radius of the anion B^-
- (A) 150 pm (B) 290 pm (C) 245 pm (D) 240 pm
8. An alloy of copper, silver and gold is found to have copper containing the ccp lattice. If silver atoms occupy the edge centres and gold is present at body centre the alloy has a formula;
- (A) $\text{Cu}_4\text{Ag}_2\text{Au}$ (B) $\text{Cu}_4\text{Ag}_4\text{Au}$ (C) $\text{Cu}_4\text{Ag}_3\text{Au}$ (D) CuAgAu
9. How many next nearest neighbours are present for Zn^{+2} ions in FCC arrangement of ZnS
- (A) 8 (B) 4 (C) 12 (D) 24
10. Choose the correct statements
- (A) All the lattice points in a Bravais lattice is equivalent.
 (B) Equivalent points in unit cells of a periodic lattice do not lie on a Bravais lattice
 (C) There are four Bravais lattices in two dimension
 (D) There are five Bravais lattices in three dimensions
11. Choose the incorrect statement:
- (A) In the lattice of fcc one of the possible unit cell is body centred tetragonal.
 (B) coordination number of fcc is 12 .
 (C) number of next nearest neighbour in fcc is 6
 (D) number of next nearest neighbour in body centred tetragonal system is 12 .
12. In a multi layered close-packed structure
- (A) there are twice as many tetrahedral holes as there are close-packed atoms
 (B) there are as many tetrahedral holes as there are closed packed atoms
 (C) there are twice as many octahedral holes as there are close-packed atoms
 (D) there are as many tetrahedral holes as there are octahedral holes
13. Given an alloy of Cu, Ag and Au in which Cu atoms constitute the CCP arrangement. If the hypothetical formula of the alloy is $\text{Cu}_4\text{Ag}_3\text{Au}$. What are the probable locations of Ag and Au atoms.
- (A) Ag-all Tetrahedral voids; Au - all Octahedral voids
 (B) Ag - $3/8^{\text{th}}$ Tetrahedral voids ; Au - $1/4^{\text{th}}$ Octahedral voids
 (C) Ag - $1/2$ Octahedral voids ; Au - $1/2$ Tetrahedral voids
 (D) Ag - all Octahedral voids ; Au - all tetrahedral voids

14. In an arrangement of type ABABA.....identical atoms of I layer A and III layer A are joined by a line passing through their centres. Suggest the correct statements -
- (A) No void is found on the line
 (B) Only Tetrahedral voids are found on the line
 (C) Only octahedral voids are found on the line
 (D) Equal number of tetrahedral and octahedral voids are found on the line

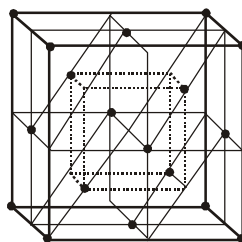
More than one correct questions (+4, -1)

15. 80.0 gm salt of weak base & strong acid XY is dissolved in water and formed 2 litre of aqueous solution. The pH of the resultant solution was found to be 5 at 298 K. If XY forms CsCl type crystal having r_{X^+} (radius of X^+) = $1.6 \text{ \AA}$ & r_{Y^-} (radius of Y^-) = $1.864 \text{ \AA}$ then select write statement(s).
 (Given : $K_b(\text{XOH}) = 4 \times 10^{-5}$; $N_A = 6 \times 10^{23}$)
- (A) Molar mass of salt is 100 g/mol
 (B) % Degree of dissociation of salt is 0.25
 (C) Edge length of AB is $4 \text{ \AA}$
 (D) Density of solid salt XY is 2 in gm/cc
16. The hcp and ccp structure of a given element. (Given radius of element is same in both structures)
- (A) have same density
 (B) have same distance between two consecutive layers (A & B)
 (C) have same co-ordination number
 (D) have same fraction of unoccupied space.
17. Which of the following two dimensional lattices are possible ?
- (A) Rhombic
 (B) Triangular
 (C) Circle
 (D) Square
18. Given here is an NaCl layer. Which of the square (s) is/are unit cells ?
- (A) A
 (B) B
 (C) C
 (D) None of these



Subjective Questions (+4, 0)

19. An element X (atomic weight = 24 gm/mole) forms a face centered cubic lattice. If the edge length of the lattice is $4 \times 10^{-8} \text{ cm}$ and the observed density is $2.40 \times 10^3 \text{ kg/m}^3$, calculate the percentage occupancy of lattice points by X element. (Given $N_A = 6 \times 10^{23}$)
20. The structure of diamond and silicon is same and can be described as atoms taking corners and faces of a cube and also taking positions of alternate tetrahedral voids. What is the value of $\frac{\text{density (diamond)}}{\text{density (silicon)}}$.



At. mass (g/mol)	Covalent Bond Length ($\text{\AA}$)
Carbon : 12	C - C = 1.50
Silicon : 28	Si - Si = 2.25

21. An element X (atomic wt. = 125) crystallises in simple cubic structure. If diameter of the largest atom which can be placed without disturbing unit cell is 366 pm. Find the density of pure element X in gm/cm^3 .

[Given : $\sqrt{3} = 1.732$, $N_A = 6 \times 10^{23}$] [Fill your answer by multiplying it with 300.]

ANSWER KEY (QUIZ # 3)

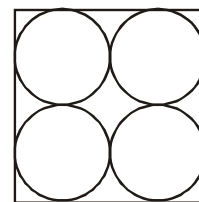
1. (A) 2. (D) 3. (C) 4. (A) 5. (A) 6. (C) 7. (B) 8. (C) 9. (C)
 10. (A) 11. (D) 12. (A) 13. (B) 14. (B) 15. (A,C) 16. (ALL) 17. (A,D)
 18. (A,B) 19. (96) 20. (1.45) 21. (500)

QUIZ # 4

Single Option Correct (+3, -1)

1. Identical 4 spheres are taken and are arranged in a layer of square packing touching each other as shown

The percentage of vacant space is

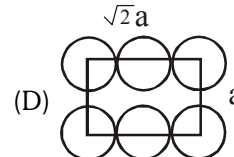
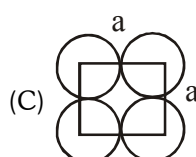
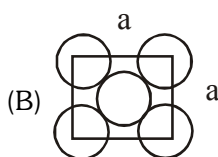
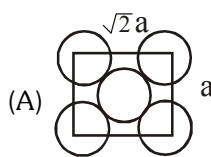


- (A) $100\left(1 - \frac{3\pi}{8}\right)$ (B) $100\left(1 - \frac{\pi}{6}\right)$ (C) $100 - \frac{3\pi}{8}$ (D) $\frac{\pi}{6}$

2. Select the incorrect statement:-

- (A) Stoichiometry of crystal remains unaffected due to Schottky defect
 (B) Frenkel defect usually shown by ionic compound having low coordination number
 (C) F-centres generation is responsible factor for imparting the colour to the crystal
 (D) Density of crystal always increases due to substitutional impurity defect

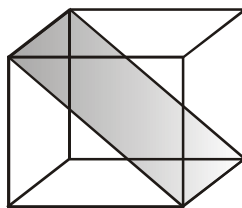
3. Which of the following represents a plane in bcc metallic crystal, where 'a' is edge length of cube -



4. Which is the property of amorphous solids.

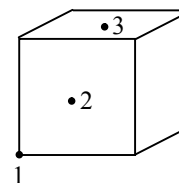
- (A) Isotropic (B) Sharp melting point
 (C) Gives clean cleavage when cut (D) None

5. Solid AB having NaCl structure, in which B atoms are at the corners. If all the atoms present in the plane shown are removed then formula of the compound left is-



- (A) AB (B) A_3B_4 (C) A_4B_3 (D) A_3B_2

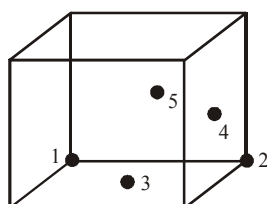
6. Which of the following defects is also known as dislocation defect ?
 (A) Frenkel defect (B) Schottky defect
 (C) Non-stoichiometric defect (D) Simple interstitial defect
7. Potassium has body-centred cubic structure with the neighbour distance 452 cm. Its relative atomic mass is 39. Its density would be-
 (A) 0.71 gm/cm³ (B) 0.81 gm/cm³ (C) 0.91 gm/cm³ (D) 1.01 gm/cm³
8. Figure shows a cube of unit cell of CCP arrangement with some atoms marked 1, 2, 3. Which of the following statements is true.



- (A) Atom 3 is twice as far from 1 as from 2
 (B) Atom 2 is equidistant from atoms 1 & 3
 (C) Atom 2 is nearer to 1 than to 3
 (D) All atoms lie on a right angled triangle
9. Consider a cube 1 of Body Centered cubic unit cell of edge length a . Now atom at the body centre can be viewed to be lying on the corner of another cube 2. Find the volume common to cube 1 and cube 2.
 (A) $\frac{a^3}{27}$ (B) $\frac{a^3}{64}$ (C) $\frac{a^3}{2\sqrt{2}}$ (D) $\frac{a^3}{8}$
10. Three lines are drawn from a single corner of an FCC unit cell to meet the other corner such that they are found to pass through exactly 1 Octahedral void only, no voids of any type and exactly 2 tetrahedral voids with exactly 1 Octahedral void only. Identify the line in the same order -
 (A) Edge, Face diagonal, Body diagonal (B) Face diagonal, Edge Body diagonal
 (C) Body diagonal, Face diagonal Edge (D) Edge, Body diagonal, Face diagonal
11. Given an alloy of Cu, Ag and Au in which Cu atoms constitute the CCP arrangement. If the hypothetical formula of the alloy is $\text{Cu}_4\text{Ag}_3\text{Au}$. What are the probable locations of Ag and Au atoms -
 (A) Ag - all Tetrahedral voids; Au - all Octahedral voids
 (B) Ag - $3/8^{\text{th}}$ Tetrahedral voids; Au - $1/4^{\text{th}}$ Octahedral voids
 (C) Ag - $1/2$ Octahedral voids; Au - $1/2$ Tetrahedral voids
 (D) Ag - all Octahedral voids; Au - all tetrahedral voids
12. Cesium chloride on heating to 760 K Changes into -
 (A) CsCl(s) (B) NaCl structure
 (C) antifluorite structure (D) ZnS structure

Multiple choice questions (+4, -1)

13. Which of the following atoms touch each other in a fcc unit cell (diagram given) -



Atom 4 is at the centre of side face

Atom 5 is at the centre of back face

Atom 3 is at the centre of bottom face

- (A) 1 & 2 (B) 1 & 3 (C) 3 & 5 (D) 3 & 4
14. An ionic crystalline solid AB having cubic unit cell, may have following arrangement(s) -
 (A) B⁻ in FCC and A⁺ occupies all tetrahedral voids.
 (B) B⁻ in CCP and A⁺ in alternate tetrahedral voids.
 (C) B⁻ at each corner and each face center and A⁺ in all octahedral voids.
 (D) B⁻ at each corner and A⁺ at each edge center.

15. The ionic radii of Cs and Cl are 0.165 nm and 0.181 nm respectively. Their atomic weights are 133 and 35.5. Then -
- (A) the lattice parameter is 0.4 nm
 (B) the lattice parameter can not be determined from this data
 (C) the density of CsCl is $4.37 \times 10^3 \text{ kg/m}^3$
 (D) the CsCl structure has a fcc structure with basis
16. In face centred cubic close packed structure octahedral sites are present at
- (A) edge centres (B) face centres (C) body centres (D) corners

Integer Questions (+4, 0)

17. Calculate the sum of number of 1st and 2nd nearest neighbours in bcc closed packing.
18. Solid AB has NaCl type structure. If the radius of A^+ and B^- are 0.5 Å and 1.5 Å respectively and formula mass of AB is 48 g/mole, what is the density of AB solid.

Take : Avogadro's number = 6×10^{23}

19. Total number of layers between a distance of $14R\sqrt{\frac{2}{3}}$ in hcp unit cell is (including first and last)

ANSWER KEY (QUIZ # 4)

1. (B) 2. (D) 3. (A) 4. (A) 5. (A) 6. (A) 7. (C) 8. (B)
9. (D) 10. (A) 11. (B) 12. (B) 13. (B,C,D) 14. (B,C) 15. (A,C)
16. (A, C) 17. (14) 18. (5) 19. (8)

QUIZ # 5**Single Option Correct (+3, -1)**

1. How many unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1.00g ?
- (A) 1.28×10^{21} unit cells (B) 1.71×10^{21} unit cells
 (C) 2.57×10^{21} unit cells (D) 5.14×10^{21} unit cells
2. The measured density of AgI is 6.94 gcm^{-3} and the theoretical density is 5.67 gm cm^{-3} . These data indicate that solid AgI has -
- (A) Schottky defect (B) Frenkel defect
 (C) Interstitial impurities (D) Both (A) and (B)
3. What is not true regarding hexagonal close packing (hcp)-
- (A) packing fraction is 0.74 (B) coordination number is 12
 (C) ABC ABC.....type packing (D) Containing both tetrahedral and octahedral voids
4. In NaCl type structure shortest distance between two nearest neighbours is 100 pm, then distance between two next nearest neighbours in the same unit cell will be
- (A) $\frac{100}{\sqrt{2}}$ pm (B) $100\sqrt{2}$ pm (C) 200 pm (D) $200\sqrt{2}$ pm
5. Percentage area of each face covered by atoms in a FCC unit cell is -
- (A) 60.4% (B) 68% (C) 74% (D) 78.5%

6. In which of the following arrangement distance between two nearest neighbours is maximum considering identical atoms in all arrangements
 (A) Simple cubic (B) bcc (C) fcc (D) equal in all
7. An atomic solid crystallizes in bcc lattice if circumference (surface) of atoms at the adjacent corners are separated by 54 pm then edge length of bcc unit cell is ($\sqrt{3} = 1.7$)
 (A) 450 pm (B) 108 pm (C) 216 pm (D) 360 pm

Comprehension (+3, -1)**Paragraph for Q. 8 to Q. 9**

In a structure of ionic compound anions are present in hcp structure while cation occupy $2/3$ part of octahedral voids.

8. Fractions of total voids occupied by cations in lattice
 (A) $1/3$ (B) $1/9$ (C) $2/9$ (D) $1/6$
9. Correct expression for packing fraction assuming anion-anion contact -

(A) $\frac{\pi}{9\sqrt{2}} \left[2 \left(\frac{r_+}{r_-} \right)^3 + 3 \right]$ (B) $\frac{\pi}{9\sqrt{2}} \left[2 \left(\frac{r_+}{r_-} \right)^3 + 1 \right]$ (C) $\frac{\pi}{3\sqrt{2}} \left[2 \left(\frac{r_+}{r_-} \right)^3 + 1 \right]$ (D) $\frac{\pi}{3\sqrt{2}} \left[2 \left(\frac{r_+}{r_-} \right)^3 + 3 \right]$

Multiple choice questions (+4, -1)

10. For a NaCl type solid ($M_w = 300$) observed density is 1.5 gm/cm^3 . What type of defect may be present in to it. (edge length of unit cell = 1 nm)
 (A) Schottky defect (B) frenkel defect
 (C) Non-stoichiometric defect (D) substitutional impurity defect
11. Select correct statements for ZnS structure -
 (A) Co-ordination number of anions is 8 (B) Co-ordination number of cation is 4
 (C) Half of voids are occupied by cations (D) $2/3$ fraction of total voids is occupied by cations
12. In a compound oxide ions arranged in CCP pattern. One-eighth of the tetrahedral holes occupied by one type of metal ion and one half of the octahedral hole occupied by another type of metal ion. If metal ions are A^{+2} & B^{+3} , then select the correct statements-
 (A) A^{+2} occupy tetrahedral holes (B) B^{+3} occupy tetrahedral holes
 (C) Formula of compound is A_2BO_4 (D) Formula of compound is AB_2O_4
13. Select correct statement -
 (A) Number of second nearest atom in primitive cubic unit cell is 12.
 (B) Minimum inter-ionic distance in Na_2O is $\sqrt{3}a/2$.
 (C) Distance between two consecutive hexagonal layer in FCC in $4r\sqrt{2/3}$.
 (D) Number of next nearest neighbour atoms for S^{2-} in ZnS type structure is 12.

Match the column (+4, 0)

- | 14. Column-I | Column-II |
|------------------|--|
| (A) Cubic | (P) Cell parameter $a = b = c$ |
| (B) Triclinic | (Q) Cell parameter $\alpha = \beta = \gamma$ |
| (C) Hexagonal | (R) Only one braivais lattic is possible |
| (D) Orthorhombic | (S) Four bravais lattices are possible |
| | (T) Type of crystal system |

- | 15. Column-I | Column-II |
|--|----------------------------|
| (A) Effective number of formula units per unit cell is 4 | (P) NaCl type |
| (B) Cations in tetrahedral voids | (Q) ZnS type |
| (C) Cations in octahedral voids | (R) CsCl type |
| (D) Cations in Cubical void | (S) Fluorite structure |
| | (T) Antifluorite structure |

Integer Questions (+4, 0)

16. Calculate the density (in gm/cm^3) of NaCl type ionic solid ($M_w = 75 \text{ gm/mol}$) if distance between two nearest cations is $250\sqrt{2} \text{ pm}$ (avogadro number = 6×10^{23})
17. Crystal AB shows zinc blende structure, where A is similar to Zn. If the radius of B^- is 650 pm then calculate the radius of A^+ ion if the edge length of unit cell is 2000 pm.
- (Given : $\sqrt{3} = 1.732$; $\sqrt{2} = 1.414$. Give your answer in pm)
- Write your answer as sum of digits. ex. {if your answer is 123 then, write your answer as (1+2+3) i.e. 6}
18. The molecular formula of a non stoichiometric oxide of vanadium is $\text{VO}_{1.8}$. The oxide contain both V^{3+} and V^{4+} ions. What is the mole % of vanadium in +3 oxidation state -
19. Calculate % volume occupied by atoms in CsCl type structure assuming anion-anion contact.

[Given $\frac{r_+}{r_-} = 0.7$; $\pi = 3$; $\sqrt{3} = 1.7$]

(Write your answer first upto nearest integer then fill in OMR sheet by adding the digits in your answer till you get single digit answer.)

20. Graphite having hcp arrangement of carbon atoms if distance between two consecutive layers is $1.6\text{\AA}$. Calculate density of graphite in gm/cm^3 ($N_A = 6 \times 10^{23}$) $\left(\frac{\sqrt{2}}{\sqrt{3}} = 0.8\right)$

(Write your answer by multiplying with $\sqrt{2}$)

21. Calculate the ratio of densities if same element undergoes fcc as well as simple cubic packing.
If your answer $x^{1/2}$ then x is.

22. Find the (A + B)

- (A) Nearest neighbour distance in NaCl type structure
(B) Nearest neighbour distance in CsCl type structure

(Assume edge length in each case is 100 unit) given $\sqrt{2} = 1.4$ and $\sqrt{3} = 1.7$

Fill your answer as sum of digits (excluding decimal places) till you get the single digit answer.

23. An element 'X' crystallizes in bcc. Find volume of unit cell in $(\text{\AA})^3$. If atomic radius is $1.73\text{\AA}$.

ANSWER KEY (QUIZ # 5)

1. (C) 2. (C) 3. (C) 4. (B) 5. (D) 6. (D) 7. (D) 8. (C) 9. (A)
10. (A,C,D) 11. (B,C) 12. (A,D) 13. (A,D)
14. (A) - P, Q, T ; (B) - R, T ; (C) - R, T ; (D) - Q, S, T 15. (A)- P,Q,S,T; (B)-Q,T; (C)-P; (D) R
16. (4) 17.(9) 18.(40) 19.(4) 20.(5) 21. (2) 22. (135) 23. (64)

SOLID STATE**SINGLE CHOICE CORRECT**

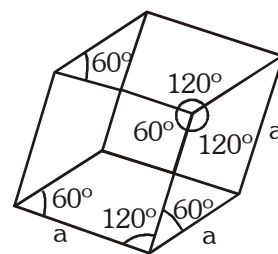
- A crystal of lead (II) sulphide has NaCl structure. In this crystal the shortest distance between a Pb^{2+} ion and S^{2-} ion is 297 pm. What is the volume of unit cell in lead sulphide ?
 (A) $209.6 \times 10^{-24} \text{ cm}^3$ (B) $207.8 \times 10^{-23} \text{ cm}^3$ (C) $22.3 \times 10^{-23} \text{ cm}^3$ (D) $209.8 \times 10^{-23} \text{ cm}^3$
- A binary solid (AB) has a rock salt structure. If the edge length is 400 pm, and radius of cation is 75 pm the radius of anion is :
 (A) 100 pm (B) 125 pm (C) 250 pm (D) 325 pm
- How many unit cells are there in 1.00 gm of cube shaped ideal crystal AB ($M = 60$) having NaCl type lattice?
 (A) 6.023×10^{23} (B) 2.5×10^{21} (C) 1.00×10^{22} (D) 6.02×10^{24}
- Select the incorrect statement:
 (A) Stoichiometry of crystal remains unaffected due to Schottky defect
 (B) Frenkel defect usually shown by ionic compounds having low coordination number
 (C) F-centre generation is responsible factor for imparting the colour to the crystal
 (D) Density of crystal always increases due to substitutional impurity defect
- In diamond, carbon atom occupy fcc lattice points as well as alternate tetrahedral voids.
 If edge length of the unit cell is 356 pm, then diameter of carbon atom is:
 (A) 77.07 pm (B) 154.14 pm (C) 251.7 pm (D) 89 pm
- A certain sample of cuprous sulphide is found to have composition $\text{Cu}_{1.8}\text{S}$, because of incorporation of Cu^{2+} ions in the lattice. What is the mole % of Cu^{2+} in total copper content in this crystal ?
 (A) 99.8% (B) 11.11% (C) 88.88% (D) None of these
- Which of the following statements for crystals having Schottky defect is not correct?
 (A) Schottky defect arises due to the absence of a cation and anion from the position which it is expected to occupy
 (B) Schottky defect are more common in ionic compounds with high co-ordination numbers
 (C) The density of the crystals having Schottky defect is larger than that of the perfect crystal
 (D) The crystal having Schottky defect is electrically neutral as a whole
- If the ratio of coordination no. of A to that of B is $x : y$, then the ratio of no. of atoms of A to no. of atoms of B in unit cell is :
 (A) $x : y$ (B) $y : x$ (C) $x^2 : y$ (D) $y : x^2$
- A rhombohedral unit cell is shown. What is its volume ? Side length = $a \text{ \AA}$.

(A) $\frac{a^3}{\sqrt{3}}$

(B) $a^3 \frac{\sqrt{3}}{2}$

(C) $\frac{a^3}{\sqrt{2}}$

(D) $a^3 \sqrt{\frac{3}{2}}$



- Diamond has face-centred cubic lattice. There are two atoms per lattice point, with the atoms at (000) and $\left(\frac{1}{4} \frac{1}{4} \frac{1}{4}\right)$ coordinates. The ratio of the carbon-carbon bond distance to the edge of the unit cell is

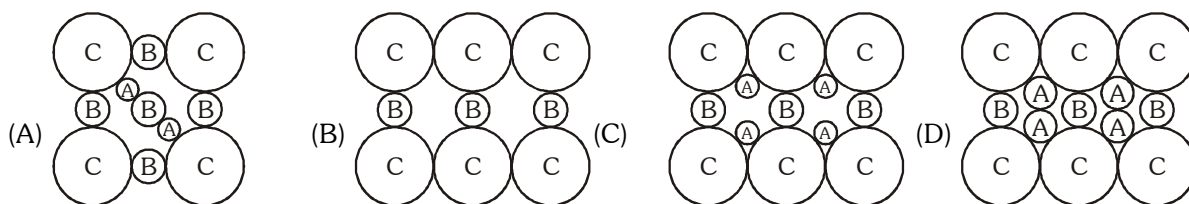
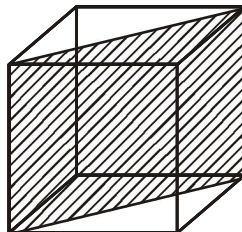
(A) $\sqrt{\frac{3}{16}}$

(B) $\sqrt{\frac{1}{4}}$

(C) $\frac{1}{4}$

(D) $\frac{1}{\sqrt{2}}$

11. In a hypothetical solid 'C' atoms are found to form cubical close packed lattice. 'A' atoms occupy all tetrahedral voids & 'B' atoms occupy all octahedral voids. 'A' and 'B' atoms are of appropriate size, so that there is no distortion in ccp lattice of 'C' atoms. Now if a plane as shown in the following figure is cut. Then the cross section of this plane will look like



12. Analysis show that nickel oxide consist of Nickle ion with 96% ions having d^8 configuration and 4% having d^7 configuration. Which amongst the following best represents the formula of the oxide.



13. Alkali metal chlorides are usually highly ionic solids and coordination number is found to be high. Which of the following information regarding them is/are false.

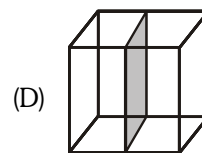
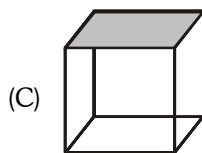
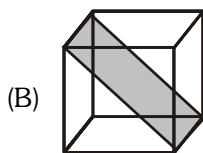
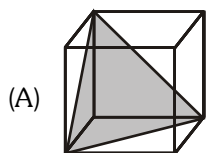
I : All chlorides have the same type of crystal structure.

II : Cl^- being large in size occupy lattice position whereas cations occupy voids.

III : High coordination number is found down the group as ratio of cationic to anionic radius increases.



14. In an f.c.c. crystal, which of the following shaded planes contains the following type of arrangement of atoms?



15. A crystal of formula AB_3 has A ions at the cube corners and B ions at the edge centres. The coordination numbers of A and B are respectively



16. In hexagonal close packing of spheres in three dimensions.

(A) In one unit cell there are 12 octahedral voids and all are completely inside the unit cell

(B) In one unit cell there are six octahedral voids and all are completely inside the unit cell

(C) In one unit cell there are six octahedral void and of which three are completely inside the unit cell and other three are from contributions of octahedral voids which are partially inside the unit cell

(D) In one unit cell there are 12 tetrahedral voids, all are completely inside the unit cell

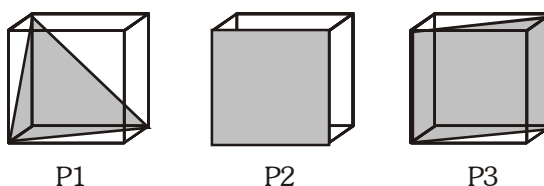
17. What is the maximum number of layers of atoms in close packed planes that will lie within two imaginary parallel planes having a distance between them of $13\sqrt{\frac{2}{3}} R$ in the copper crystal (FCC) ? Consider the atoms to be within the parallel planes if their centres are on or within the two parallel planes.
 (A) 5 (B) 6 (C) 7 (D) 8
18. In a face centred lattice of X and Y, X atoms are present at the corners while Y atoms are at face centres. Then what would be the formula of the compound if one of the X atoms from a corner is replaced by Z atoms (also monovalent) ?
 (A) $X_7Y_{24}Z_2$ (B) $X_7Y_{24}Z$ (C) $X_{24}Y_7Z$ (D) $XY_{24}Z$
19. A crystal is made of particle x, y & z, 'x' forms fcc packing, 'y' occupies all octahedral voids of 'x' and 'z' occupies all tetrahedral voids of 'x', if all the particles along one body diagonal are removed then the formula of the crystal would be :
 (A) xyz_2 (B) x_2yz_2 (C) $x_8y_4z_5$ (D) $x_5y_4z_8$
20. PCl_5 molecule has
 (A) three fold axis of symmetry (B) two fold axis of symmetry
 (C) both (D) none of these
21. A transition metal M can exist in two oxidation states +2 and +3. It forms an oxide whose experimental formula is given by M_xO where $x < 1$. Then the ratio of metal ions in +3 state to those in +2 state is given by :
 (A) $\frac{(1-x)}{(1+x)}$ (B) $1+2x$ (C) $1+x/2$ (D) $\frac{2(1-x)}{(3x-2)}$
22. The number of molecules per unit cell for a compound that crystallizes in the form of an orthorhombic end-centred lattice with a molecule at each lattice site is :
 (A) one (B) two (C) four (D) six
23. Consider the structure of rock-salt. How many Na^+ ions occupy the second-nearest neighbour location of an Na^+ ion ?
 (A) 6 (B) 12 (C) 18 (D) 24
24. In diamond, the coordination number of carbon is
 (A) four and its unit cell has eight carbon atoms (B) four and its unit cell has six carbon atoms
 (C) six and its unit cell has four carbon atoms (D) four and its unit cell has four carbon atoms
25. Zinc sulphide exists in two different forms – zinc blende and wurtzite. Both occur as 4 : 4 coordination compounds. Choose the correct option from among the following.
 (A) Zinc blende has a bcc structure and wurtzite an fcc structure.
 (B) Zinc blende has an fcc structure and wurtzite an hcp structure.
 (C) Zinc blende as well as wurtzite have a hcp structure.
 (D) Zinc blende as well as wurtzite have a ccp structure
26. Every atom or ion that forms an fcc unit cell is surrounded by
 (A) six octahedral holes and eight tetrahedral holes (B) eight octahedral holes and six tetrahedral holes
 (C) six octahedral holes and six tetrahedral holes (D) eight octahedral holes and four tetrahedral holes
27. Consider the structure of CsCl (8 : 8 coordination). How many Cs^+ ions occupy the second-nearest neighbour locations of a Cs^+ ion ?
 (A) 8 (B) 24 (C) 6 (D) 16
28. Which of the following statements is incorrect ?
 (A) A substitutional solid solution is one in which atoms of the solute metal occupy some locations that solvent metal atoms are expected to occupy.
 (B) An interstitial solid solution is one in which the solute atoms occupy the interstices (holes) between the solvent atoms.
 (C) An intermetallic compound is a compound formed between metals and metalloids.
 (D) Cu_3Zn is an intermetallic compound

- 29.** Which of the following pairs have fcc lattice structures ?
 (A) Rock salt and Wurtzite (B) Rock salt and sphalerite
 (C) CsCl and rutile (D) Polonium and fluorite
- 30.** If the three interaxial angles defining the unit cell are all equal in magnitude, the crystal cannot be
 (A) rhombohedral (B) cubic (C) hexagonal (D) tetragonal
- 31.** Which of the following statements is incorrect for the diamond structure ?
 (A) Each atom has 4 nearest neighbours and 12 next-nearest neighbours.
 (B) It is relatively empty
 (C) The proportion of the available volume which will be filled by hard spheres is only 0.34.
 (D) The proportion of the available volume which will be filled by hard spheres is only 0.46
- 32.** The molar volume of KCl and NaCl are 37.46 mL and 27.94 mL and 27.94 mL respectively. The ratio of the unit cube edges of the two crystals is :
 (A) 1.296 (B) 1.116 (C) 1.341 (D) 0.950
- 33.** The ratio of the volume of a tetragonal lattice to that of a hexagonal lattice is :
 (A) $\frac{\sqrt{3}}{2} abc$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{2}{\sqrt{3}} \frac{a^2}{b} c$ (D) $\frac{\sqrt{3}}{2} \frac{a^2}{bc}$
- 34.** An fcc lattice has a lattice parameter $a = 400$ pm. Calculate the molar volume of the lattice including all the empty space.
 (A) 10.8 mL (B) 96 mL (C) 8.6 mL (D) 9.6 mL
- 35.** Which of the following statements is incorrect for a two-dimensional hexagonal close-packed layer ?
 (A) Each sphere is surrounded by six spheres (B) Each sphere is surrounded by six voids
 (C) Each sphere has three voids (D) Each void is surrounded by three spheres
- 36.** In hcp (ABAB.....) and ccp (ABCABC.....) structures made up of spheres of equal size, the volume occupied per sphere is (r = radius of sphere)
 (A) $5.66 r^3$ (B) $1.33 r^3$ (C) $2.66 r^3$ (D) $7.40 r^3$
- 37.** Which of the following mechanisms is incorrect for a stoichiometrically pure compound AB ?
 (A) There is a displacement of A atoms to interstitial sites, leaving an equal number of A-type vacancies in the structure.
 (B) There is a displacement of B atoms to interstitial sites, accompanied by the formation of an equal number of B-type vacancies.
 (C) An equal number of A-and B-type vacancies are created by the migration of atoms to the crystal surface.
 (D) An equal number of interstitial A and B atoms, and A-and B-type vacancies, are created
- 38.** The numbers of tetrahedral and octahedral holes in a ccp array of 100 atoms are respectively
 (A) 200 and 100 (B) 100 and 200
 (C) 200 and 200 (D) 100 and 100
- 39.** Amorphous solids may be classified as
 (A) isotropic and supercooled liquids (B) anisotropic and supercooled liquids
 (C) isoenthalpic and superheated liquids (D) isotropic and superheated solids
- 40.** Three elements P, Q and R crystallize in a cubic solid lattice. The P atoms occupy the corners, Q atoms the cube centres and R atoms the edges. The formula of the compound is :
 (A) PQR (B) PQR_2 (C) PQR_3 (D) PQ_3R
- 41.** The measured density of AgI is 6.94 g cm^{-3} and the theoretical density is 5.67 g cm^{-3} . These data indicate that solid AgI has
 (A) Schottky defect (B) Frenkel defect
 (C) interstitial impurities (D) Both (A) and (B)

- 42.** Crystalline solids differ from amorphous solids in that
- all crystalline solids contain anions and cations.
 - all crystalline solids have a long range order while all amorphous solids do not.
 - all crystalline solids conduct electricity while all the amorphous solids do not.
 - all crystalline solids are hard and strong while all amorphous solids are soft and brittle.
- 43.** Choose the INCORRECT statement from the following.
- Hexagonal closest packed (HCP) and cubic closest packed (CCP) structures have coordination numbers of 12 and 6, respectively.
 - HCP and CCP structures involve ABABAB and ABCABCABC stacking, respectively.
 - HCP and CCP structures involve the same net packing density.
 - Metals can crystallize in either HCP or CCP structures, depending on the element.
- 44.** An ionic crystalline solid MX_2 , has a cubic unit cell. Which of the following arrangement of ions is consistent with the stoichiometry of the compound ?
- M^{2+} ions at the corners and at each face centres, 8 X^- ions in the tetrahedral holes.
 - M^{2+} ions at the each face centres, 8 X^- ions at the corners.
 - M^{2+} ions at the corners and X^- at each face centres 2 X^- ions completely inside the unit cell.
 - M^{2+} ions at the corners and at each face centres, 4 X^- ions completely inside the unit cell.

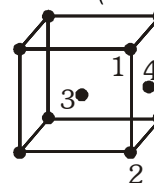
ONE OR MORE THAN ONE CORRECT

- 45.** Following three planes (P_1 , P_2 , P_3) in an FCC unit cell are shown :



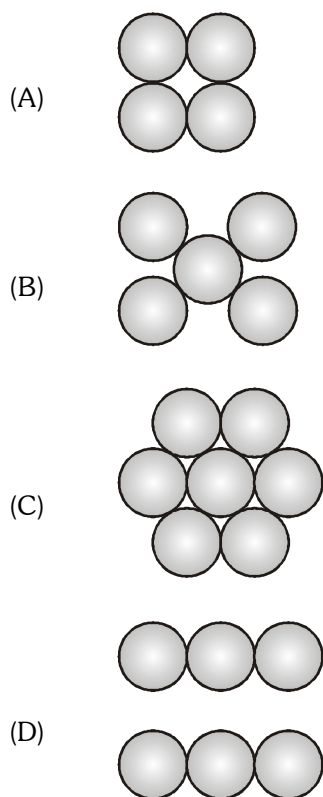
Consider the following statements and choose the correct option that follow :

- P_1 contains no voids of three dimensions.
 - P_2 contains only Octahedral voids
 - P_3 contains both Octahedral and Tetrahedral voids
 - All are true
- 46.** In the fluorite structure if the radius ratio is $\left(\sqrt{\frac{3}{2}} - 1\right)$, how many ions does each cation touch ?
- 4 anions
 - 12 cations
 - 8 anions
 - No cations
- 47.** In an f.c.c. unit cell, atoms are numbered as shown below. The atoms not touching each other are (Atom numbered 3 is face centre of front face).
- 3 & 4
 - 1 & 3
 - 1 & 2
 - 2 & 4
- 48.** The co-ordination number of fcc structure for metals is 12, since
- each atom touches 4 others in same layer, 3 in layer above and 3 in layer below
 - each atom touches 4 others in same layer, 4 in layer above and 4 in layer below
 - each atom touches 6 others in same layer, 3 in layer above and 3 in layer below
 - each atom touches 3 others in same layer, 6 in layer above and 6 in layer below

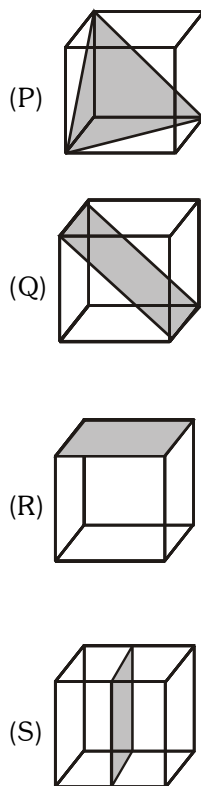


MATCH THE COLUMN

49. Column I
(Arrangement of the atoms/ions)



Column II
(Planes in fcc lattice)



50. Column I

- (A) ZnS crystal
(B) CaF_2 crystal
(C) NaCl crystal
(D) Diamond crystal

Column II

- (P) fcc
(Q) hcp
(R) Distance between closest particles is $\frac{\sqrt{3}}{4}a$
(S) Only one type of voids are occupied

51. Column I
(Arrangement in unit cell)

- (A) Cations in ccp and anions in alternate tetrahedral voids
(B) Cations in simple cubic and anions in the body centre
(C) Anions in ccp and cations in all tetrahedral voids
(D) Cations in ccp and anions in all octahedral voids

Column II
(Coordination number of cation : Anion)

- (P) 6 : 6
(Q) 4 : 4
(R) 4 : 8
(S) 8 : 8

52. Column - I
(Crystal structure)

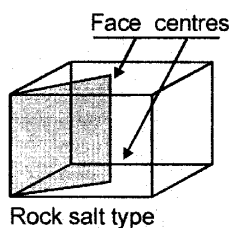
- (A) Rock salt
(B) Zinc blende
(C) Antifluorite
(D) CsCl type

Column - II
(Arrangement)

- (P) Fraction of voids occupied = $\frac{1}{3}$
(Q) Anions constitute lattice
(R) Cations in tetrahedral voids
(S) Coordination no. for cation and anion is same

LEVEL-1 (SUBJECTIVE TYPE QUESTIONS)

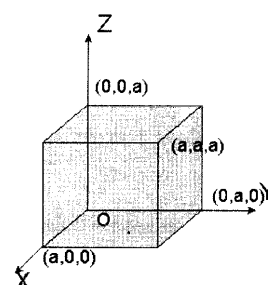
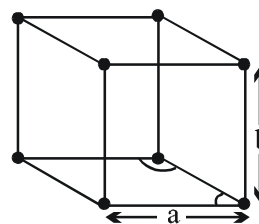
1. An element crystallizes in a structure having FCC unit cell of an edge 200 pm. Calculate the density, if 200 g of this element contains 24×10^{23} atoms.
2. What is the percent by mass of titanium in rutile, a mineral that contain Titanium and oxygen, if structure can be described as a closed packed array of oxide ions, with titanium in one half of the octahedral holes. What is the oxidation number of titanium?
3. Spinel is a important class of oxides consisting of two types of metal ions with the oxide ions arranged in CCP pattern. The normal spinel has one-eighth of the tetrahedral holes occupied by one type of metal ion and one half of the octahedral hole occupied by another type of metal ion. Such a spinel is formed by Zn^{2+} , Al^{3+} and O^{2-} , with Zn^{2+} in the tetrahedral holes. Give the formulae of spinel.
4. An element crystallizes into a structure which may be described by a cubic type of unit cell having one atom on each corner of the cube and two atoms on one of its body diagonals. If the volume of this unit cell is $24 \times 10^{-24} \text{ cm}^3$ and density of element is 7.2 g cm^{-3} , calculate the number of atoms present in 200 g of element.
5. AgCl has the same structure as that of NaCl. The edge length of unit cell of AgCl is found to be 555 pm and the density of AgCl is 5.561 g cm^{-3} . Find the percentage of sites that are unoccupied.
6. The density of KBr is 2.75 g cm^{-3} . The length of the edge of the unit cell is 654 pm. Show that KBr has face centered cubic structure.
($N = 6.023 \times 10^{23} \text{ mol}^{-1}$, At. mass : K = 39, Br = 80)
7. A crystal of lead(II) sulphide has NaCl structure. In this crystal the shortest distance between Pb^{+2} ion and S^{2-} ion is 297 pm. What is the length of the edge of the unit cell in lead sulphide? Also calculate the unit cell volume.
8. If the length of the body diagonal for CsCl which crystallises into a cubic structure with Cl^- ions at the corners and Cs^+ ions at the centre of the unit cells is $7 \text{ \AA}$ and the radius of the Cs^+ ion is $1.69 \text{ \AA}$, what is the radii of Cl^- ion?
9. Iron occurs as bcc as well as fcc unit cell. If the effective radius of an atom of iron is 124 pm. Compute the density of iron in both these structures.
10. The composition of a sample of wustite is $\text{Fe}_{0.93}\text{O}_{1.0}$. What percentage of iron is present in the form of Fe(III)?
11. In Rock Salt type structure cations (radius r) occupy Octahedral holes in the FCC of anions (radius R) such that there is no distortion in the FCC lattice. Find the perimeter of the shaded region.



12. In an ionic solid $r_{(+)} = 1.6 \text{ \AA}$ and $r_{(-)} = 1.864 \text{ \AA}$. Use the radius ratio rule to determine the edge length of the cubic unit cell in $\text{\AA}$.
13. The face diagonal length of f.c.c. cubic cell is $660\sqrt{2} \text{ pm}$. If the radius of the cation is 110 pm, What should be the radius of the anion following radius ratio rules.

LEVEL-2 (SUBJECTIVE TYPE QUESTIONS)

- A solid A^+ and B^- had NaCl type closed packed structure. If the anion has a radius of 250 pm, what should be the ideal radius of the cation? Can a cation C^+ having a radius of 180 pm be slipped into the tetrahedral site of the crystal of A^+B^- ? Give reasons for your answer.
- If the radius of Mg^{2+} ion, Cs^+ ion, O^{2-} ion, S^{2-} ion and Cl^- ion are 0.65 Å, 1.69 Å, 1.40 Å, 1.84 Å, and 1.81 Å respectively. Calculate the co-ordination numbers of the cations in the crystals of MgS, MgO and CsCl.
- KCl crystallizes in the same type of lattice as does NaCl. Given that $\frac{r_{Na^+}}{r_{Cl^-}} = 0.5$ and $\frac{r_{Na^+}}{r_{K^+}} = 0.7$ Calculate:
 - The ratio of the sides of unit cell for KCl to that for NaCl and
 - The ratio of densities of NaCl to that for KCl.
- Find the size of largest sphere that will fit in octahedral void in an ideal FCC crystal as a function of atomic radius 'r'. The insertion of this sphere into void does not distort the FCC lattice. Calculate the packing fraction of FCC lattice when all the octahedral voids are filled by this sphere.
- A cubic unit cell contains manganese ions at the corners and fluoride ions at the center of each edge.
 - What is the empirical formula of the compound?
 - What is the co-ordination number of the Mn ion?
 - Calculate the edge length of the unit cell, if the radius of Mn ion is 0.65 Å and that of F^- ion is 1.36 Å.
- Ice crystallizes in a hexagonal lattice. At the low temperature at which the structure was determined, the lattice constants where $a = 4.53$ Å, and $b = 7.60$ Å. How many molecules are contained in a given unit cell? [density (ice) = 0.92 gm/cm³]
- Potassium crystallizes in a body-centered cubic lattice with edge length, $a = 5.2$ Å.
 - What is the distance between nearest neighbours?
 - What is the distance between next-nearest neighbours?
 - How many nearest neighbours does each K atom have?
 - How many next-nearest neighbours does each K atom have?
 - What is the calculated density of crystalline potassium?
- If NaCl is doped with 10^{-3} mol % $SrCl_2$, what is the numbers of cation vacancies per mole of NaCl?
- The mineral hawleyite, one form of CdS, crystallizes in one of the cubic lattices, with edge length 5.87 Å. The density of hawleyite is 4.63 g cm⁻³.
 - In which cubic lattice does hawleyite crystallize?
 - Find the Schottky defect in g cm⁻³.
- A strong current of trivalent gaseous boron passed through a germanium crystal decreases the density of the crystal due to part replacement of germanium by boron and due to interstitial vacancies created by missing Ge atoms. In one such experiment, one gram of germanium is taken and the boron atoms are found to be 150 ppm by weight, when the density of the Ge crystal decreases by 4%. Calculate the percentage of missing vacancies due to germanium, which are filled up by boron atoms.
Atomic wt. Ge = 72.6, B = 11
- A Void Comprehension : Assume an FCC unit cell cube (edge length a) with one of cartesian coordinates. Find :
 - find the distance between two successive Tetrahedral voids
 - the distance between two successive Octahedral voids



- 12.** Four identical spheres of radius r are arranged such that centres of three of them form an equilateral triangle and the fourth one rests symmetrically above the triangle touching all of them. Find : (i) the area of the triangle thus formed.
- (ii) Perpendicular distance between the triangle and the centre of the fourth atom.
- (iii) Are all the balls identical in the arrangement.
- 13.** An ionic solid AB_2 isomorphous to the rutile structure (a tetragonal system with effective number of formula unit = 2) has edge lengths of the unit cell of 4 Å , 4 Å and 7 Å . Calculate the density of the substance if its formula weight is 80. Take $N_A = 6 \times 10^{23}$ and express your answer in mg/cc using four significant digits.
- 14.** Consider the arrangement of circles of equal radii with their centres arranged as per the 2-dimensional lattice defined by $a = b$, $\theta = 60^\circ$ such that each circle is touching all its nearest neighbours. If all the void areas present are additionally occupied by smaller circles of relevant size so that the void circles are just contacting their neighbours find the packing efficiency of the configuration in percent.
- 15.** 5.35 g of a salt ACl (of weak base AOH) is dissolved in 250 ml of solution. The pH of the resultant solution was found to be 4.827. Find the ionic radius of A^+ & Cl^- if ACl forms $CsCl$ type crystals having density 2.2 g/cc . Given $K_b(AOH) = 1.8 \times 10^{-5}$. $\frac{r^+}{r^-} = 0.731$ for this unit cell.
- 16.** How many units cells are there in a
- (a) 1.0 g cubic crystal of $NaCl$? and (b) along each edge of the crystal ?
- The cell content of $NaCl$ is 4 and formula mass of $NaCl$ is 58.5 g/mole .
- 17.** Calculate the Avogadro constant from the following data :
- Density of solid $NaCl = 2.165\text{ g/cc}$
- Distance between centres of adjacent Na^+ and $Cl^- = 0.2819\text{ nm}$
- Also, calculate the edge length of a cube containing 1 mole of $NaCl$ and the number of ions ($Na^+ + Cl^-$) along one edge of the cube. (6.02×10^{23} , 3.0 cm , 1.064×10^8).
- 18.** A solid cube of edge length = 25.32 mm of an ionic compound which has $NaCl$ type lattice is added to 1 kg of water. The boiling point of this solution is found to be 100.52°C (assume 100% ionisation of ionic compound). If radius of anion of ionic solid is 200 pm then calculate radius of cations of solid in pm (picometer).
- (k_b of water = $0.52\text{ K kg mole}^{-1}$, Avogadro number, $N_A = 6 \times 10^{23}$, $\left(\sqrt[3]{75}\right) = 4.22$)

ANSWER KEY

SINGLE CHOICE CORRECT

- 1.(A) 2.(B) 3.(B) 4.(D) 5. (B) 6. (B) 7. (C) 8. (B) 9.(C) 10.(A)
 11.(C) 12.(D) 13.(C) 14.(A) 15.(C) 16.(B) 17.(C) 18.(B) 19.(D) 20.(C)
 21.(D) 22.(B) 23.(B) 24.(A) 25.(B) 26.(A) 27.(C) 28.(C) 29.(B) 30.(C)
 31.(D) 32.(B) 33.(B) 34.(D) 35.(C) 36.(A) 37.(D) 38.(A) 39.(A) 40.(C)
 41.(C) 42.(B) 43.(A) 44.(A)

ONE OR MORE THAN CORRECT

45. (ALL) 46.(B,C) 47.(C) 48.(B,C)

MATCH THE COLUMN

49. (A → S); (B → R) ; (C → P); (D → Q)
 50. (A → P, Q, R, S); (B → P, R, S) ; (C → P, S); (D → P, R, S)
 51. (A → Q); (B → S) ; (C → R); (D → P)
 52. (A → P,Q,S); (B → P,Q,R,S); (C → Q,R); (D → Q,S)

LEVEL-1 (SUBJECTIVE TYPE QUESTIONS)

1. 41.67 g cm⁻³ 2. 3.5 g cm⁻³
 3. 59.95%, +4 4. 267 pm, 534 pm, 378 pm
 5. 3.472×10^{24} atoms 6. 0.24%
 7. $a = 5.94 \times 10^{-8}$ cm, $V = 2.096 \times 10^{-22}$ cm⁻³ 8. 1.81 Å
 9. 7.887 g/cc, 8.59 gm/cm³ 10. 15.053
 11. 4(r + 2R) 12. 4 Å
 13. 220 pm

LEVEL-2 (SUBJECTIVE TYPE QUESTIONS)

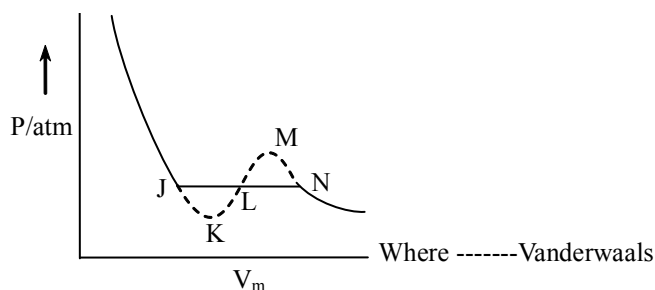
1. 103.4 pm, No 2. 4, 6, 8
 3. (a) 1.143, (b) 1.172 4. 0.414 r, 79.3%
 5. (a) MnF₃, (b) 6, (c) 4.02 Å 6. 4 molecules of H₂O
 7. (a) 4.5 Å, (b) 5.2 Å, (c) 8, (d) 6, (e) 0.92 g/cm³
 8. 6.02×10^{18} mol⁻¹ 9. (i) 3.90, (ii) 0.120 g/cc
 10. 2.376% 11. (i) $\frac{a}{2}$ (ii) $\frac{a}{\sqrt{2}}$
 12. (i) $r^2 \sqrt{3}$ (ii) $\frac{2r\sqrt{2}}{\sqrt{3}}$ (iii) Yes 13. 2381 mg/cc. 14. 95%
 15. $r_+ = 1.26 \times 10^{-8}$ cm and $r_- = 1.71 \times 10^{-8}$ cm 16. (a) 2.57×10^{21} (b) 1.37×10^7
 17. 3 cm, 1.064×10^8 18. 100 pm

SURFACE CHEMISTRY

Single Answer correct (3 Marks each, -1 for wrong answer)

1. (i) At 298 K the volume of NH_3 adsorbed by 1g of charcoal is higher than that of H_2 under similar conditions.
 (ii) The movement of colloidal particles towards the oppositely charged electrodes on passing electric current is known as Brownian movement.
 (A) T, T (B) T, F (C) F, T (D) F, F
2. **STATEMENT-1** : A gas with higher critical temperature is adsorbed more than a gas with lower critical temperature on the same absorbent.
STATEMENT-2 : Higher critical temperature implies that the gas is difficult to be liquified.
 (A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
 (C) Statement-1 is True, Statement-2 is False
 (D) Statement-1 is False, Statement-2 is True
3. When 9.0 ml of arsenious sulphide sol and 1.0 ml of $1.0 \times 10^{-4} \text{ M BaCl}_2$ are mixed, turbidity due to precipitation just appears after 2 hours. The effective ion and its coagulating value are respectively
 (A) Cl^- , 10 m mol/L (B) Cl^- , 20 m mol/L (C) Ba^{2+} , 10 m mol/L (D) Ba^{2+} , 20 m mol/L
4. 100 ml each of two sols of AgI, one obtained by adding AgNO_3 to slight excess of KI and another obtained by adding KI to slight excess of AgNO_3 , are mixed together. Then
 (A) The two sols will stabilize each other (B) The sol particles will acquire more electric charge
 (C) The sols will coagulate each other mutually (D) A true solution will be obtained
5. Select the correct statements for classical adsorption isotherm :
 (A) Each surface site can adsorb multiple molecules.
 (B) Multilayer formation will take place.
 (C) During each adsorption, constant amount of heat is involved.
 (D) adsorption process is non-spontaneous.
6. Which of the following phenomenon is not related to the adsorption on the surface of a substance?
 (A) Occlusion (B) Heterogeneous catalysis
 (C) Peptization (D) Flocculation
7. Which is universally correct for catalyst –
 (A) A catalyst remain unchanged chemically at the end of chemical reaction.
 (B) A catalyst takes part in a chemical reaction
 (C) All kinds of catalysts undergo catalyst poisoning
 (D) A catalyst physically changes at the end of the reaction
8. Which of the following constitute irreversible colloidal system in water as dispersion medium –
 (A) Clay (B) Platinum (C) $\text{Fe}(\text{OH})_3$ (D) All of these
9. Which one among the following sols is hydrophobic –
 (A) Gum (B) Gelatin (C) Starch (D) Sulphur
10. A gel is converted into a sol by shaking it with a dispersion medium. After some time it again becomes gel. This property is called –
 (A) Weeping (B) Thixotropy (C) Syneresis (D) None
11. In which of following system the dispersed phase and dispersion medium are both solid –
 (A) Foam (B) Dust storm (C) Coloured glass (D) Paints

12. The process of separation of colloids by passing through semi permeable membrane is called –
 (A) Filtration (B) Electrophoresis (C) Dialysis (D) Ultra filtration
13. The redispersal of a freshly precipitated substance into a sol by the addition of an electrolyte in common is known as –
 (A) Aggregation (B) Condensation (C) Coagulation (D) Peptization
14. Which of the following chemical equation represents the formation of colloidal solution –
 (A) $\text{Cu} + \text{CuCl}_2 \rightarrow \text{Cu}_2\text{Cl}_2$ (B) $2\text{Mg} + \text{CO}_2 \rightarrow 2\text{MgO} + \text{C}$
 (C) $2\text{HNO}_3 + 3\text{H}_2\text{S} \rightarrow 3\text{S} + 4\text{H}_2\text{O} + 2\text{NO}$ (D) Both (B) and (C)
15. In which of the following Tyndall effect is not observed –
 (A) Suspensions (B) Emulsions (C) Sugar solution (D) Gold sol.
16. The average molecular mass of colloids can be determined by –
 (A) Tyndall effect (B) Boiling of colloidal
 (C) Osmotic pressure measurement (D) Flocculation value
17. Which type of property is the Brownian movement of colloidal sol. –
 (A) Electrical (B) Optical (C) Mechanical (D) Colligative
18. The stability of the dispersed phase in a lyophobic colloids is due to –
 (A) High viscosity of the medium
 (B) The formation of electrical layer between two phases
 (C) High surface tension of sol
 (D) None of these
19. The ability of ion to bring about coagulation of a given colloidal solution depends upon –
 (A) Its size of ion (B) The magnitude of charge
 (C) The sign of charge (D) Both magnitude and sign of charge
20. Among the following, the surfactant that will form micelles in aqueous solution at the lowest molar concentration at ambient condition is –
 (A) $\text{CH}_3(\text{CH}_2)_{15}\text{N}^+(\text{CH}_3)_3\text{Br}^-$ (B) $\text{CH}_3(\text{CH}_2)_{11}\text{OSO}_3^-\text{Na}^+$
 (C) $\text{CH}_3(\text{CH}_2)_6\text{COO}^-\text{Na}^+$ (D) $\text{CH}_3(\text{CH}_2)_{11}\text{N}^+(\text{CH}_3)_3\text{Br}^-$
21. Among the electrolytes Na_2SO_4 , CaCl_2 , $\text{Al}_2(\text{SO}_4)_3$ and NH_4Cl , the most effective coagulating agent for Sb_2S_3 sol is –
 (A) Na_2SO_4 (B) CaCl_2 (C) $\text{Al}_2(\text{SO}_4)_3$ (D) NH_4Cl
22. The arsenious sulphide sol has negative charge the maximum coagulating power for precipitating it is of –
 (A) 0.1(N) $\text{Zn}(\text{NO}_3)_2$ (B) 0.1(N) Na_3PO_4 (C) 0.1(N) ZnSO_4 (D) 0.1(N) AlCl_3
23. Which of the following is/are correct ?



- (A) At points between J and K we obtained metastable superheated liquid
 (B) At points between N and M we obtained supercooled vapour
 (C) At the isotherm portion KLM, $\left(\frac{\partial P}{\partial V_m}\right)_T > 0$. Hence K L M part does not have any physical significance
 (D) All of the above are correct
24. The name aquadag is given to the colloidal sol of –
 (A) Copper in water (B) Platinum in water (C) Graphite in water (D) None of the above

- 25.** Fog is an example of -
 (A) Solid dispersed in gas (B) Liquid dispersed in gas
 (C) Liquid dispersed in solid (D) Solid dispersed in liquid
- 26.** Cod liver oil is-
 (A) Fat dispersed in water (B) Water dispersed in fat
 (C) Water dispersed in oil (D) Fat dispersed in fat
- 27.** Alcosol is colloidal system -
 (A) Of a solid dispersed in a gas (B) Of a gas dispersed in water
 (C) In which dispersion medium is benzene (D) In which dispersion medium is alcohol
- 28.** The dispersion medium in Gel is -
 (A) Solid (B) Liquid (C) Water (D) Gas
- 29.** Liquid-Liquid sols. are known as -
 (A) Aerosols (B) Emulsions (C) Foam (D) Gels
- 30.** Purple of cassius is -
 (A) Colloidal solution of gold (B) Colloidal solution of silver
 (C) Colloidal solution of platinum (D) Oxyacids of gold
- 31.** Cheese is a colloidal system of -
 (A) Liquid dispersed in solid (B) Solid dispersed in liquid
 (C) Gas dispersed in solid (D) Solid dispersed in gas
- 32.** A dispersion of AgCl in water is -
 (A) Hydrophilic colloid (B) An emulsion
 (C) An alcosol (D) Hydrophobic sol.
- 33.** Which of the following metal sols. cannot be prepared by Bredig's arc method -
 (A) Copper (B) Potassium (C) Gold (D) Platinum
- 34.** Flocculation value is expressed in terms of -
 (A) Millimole/litre (B) Mole/litre (C) Grams/litre (D) Mole/millilitre
- 35.** Which of the following has largest protecting power -
 (A) Gelating (Gold no. = 0.01) (B) Dextrin (Gold no. = 15)
 (C) Potato starch (Gold no. = 25) (D) Albumin (Gold no = 0.25)
- 36.** The coagulation of 10cm^3 of Gold sol is completely prevented by addition of 0.025g of starch to it. The gold number of starch is-
 (A) 0.025 (B) 0.25 (C) 2.5 (D) 25
- 37.** Gelatin protects -
 (A) Gold sol. (B) As_2S_3 sol. (C) $\text{Fe}(\text{OH})_3$ Sol. (D) All
- 38.** In Faraday-Tyndall effect the colloiddally suspended particles -
 (A) Scatter the strong beam of light (B) Co-agulation
 (C) Show electrophoresis (D) Show brownian movement

39. Which of the following indicates the charge on colloidal particles -
 (A) Brownian movement (B) Electrophoresis
 (C) Filtration (D) Tyndall effect
40. Adding alum to turbid water, impurities are removed due to -
 (A) Coagulation (B) Brownian movement
 (C) Electrophoresis (D) Tyndall effect
41. A soap ($C_{17}H_{35}COONa$) solution becomes a colloidal sol at a concentration of 1.2×10^{-3} M. On the average, 2.4×10^{13} colloidal particles are present in 1 mm^3 . What is the average number of stearate ions in one colloidal particle (micelle) : **Take : Avogadro's number = 6×10^{23}**
 (A) 30 (B) 40 (C) 50 (D) 60
42. Graph between $\log\left(\frac{x}{m}\right)$ and $\log P$ is a straight line inclined at an angle of 45° . When pressure of 0.5 atm and $\log K = 0.699$, amount of solute adsorbed for gram of adsorbent will be :
 (A) 1 gm / g adsorbent (B) 1.5 gm / g adsorbent
 (C) 2.5 gm / g adsorbent (D) 0.25 gm / g adsorbent
43. Which of the following has minimum flocculation value for positive colloids-
 (A) Cl^- (B) SO_4^{2-} (C) PO_4^{3-} (D) $[Fe(CN)_6]^{4-}$
44. Lyophilic sols are more stable than lyophobic sols because :
 (A) The colloidal particles have positive charge
 (B) The colloidal particles have no charge
 (C) The colloidal particles are solvated
 (D) There are strong electrostatic repulsions between the negatively charged colloidal particles
45. Which ion has least coagulation value for +ve sol ?
 (A) $[Fe(CN)_6]^{4-}$ (B) Cl^- (C) SO_4^{2-} (D) PO_4^{3-}
46. According to Freundlich on increasing the pressure change in extent of adsorption with respect to pressure (at low pressure range) -
 (A) Constant (B) Increase linearly
 (C) Increase but not linearly (D) Decreases
47. Which of the following statements is incorrect regarding physisorptions ?
 (A) It may form multi layers on adsorbent surface
 (B) Enthalpy of adsorption ($\Delta H_{\text{adsorption}}$) is low and positive
 (C) It occurs because of Van der Waal's forces
 (D) More easily liquefiable gases are adsorbed readily
48. Which of the following is a correct statement -
 (A) High temperature is favourable for physical adsorption
 (B) Flocculation value is measure of protective power
 (C) $Fe(OH)_3$ solution can be used for cogulation of As_2S_3 solution
 (D) Aqueous gold solution is an example of positively charged solution

Multiple Correct Answer Type (4 Marks each, -1 for wrong answer)

49. Select the incorrect options :
- (A) Lyophobic are self stabilized.
 - (B) For a protective colloid, higher the gold number greater its protecting power.
 - (C) 0.025 gm of a protective colloid is needed to prevent coagulation of 10 mL standard gold sol on addition of 1mL of 10% NaCl solution, then its gold no. is "0.025".
 - (D) Lyophilic sols are more stable than lyophobic sols due to solvation of colloidal particles.
50. For the following statement for micelles, which are correct ?
- (A) At critical micelle concentration, several properties of solution of surfactant such as molar conductivity, surface tension and osmotic pressure change
 - (B) Micelles from ionic surfactants can be formed only above a certain temperature called Kraft temperature
 - (C) Micelles formation is exothermic
 - (D) Micelles are associated colloids
51. Which of the following is characteristic of chemisorption
- (A) It is irreversible
 - (B) It is specific
 - (C) It is multilayer phenomenon
 - (D) Heat of adsorption is 80-240 kJ
52. Which of the following statement(s) is/are true for a colloidal solution of gold in water (Gold sol.) :
- (A) The path of light beam is invisible when passed through solution.
 - (B) The colloidal solution can be stabilised by addition of small amount of NaCl.
 - (C) The colloidal solution can be preserved for long time if small amount of starch is added to sol.
 - (D) The colloidal solution can be destroyed by prolonged electrophoresis of solution.
53. Select the correct options -
- (A) Chemisorption is highly specific in nature.
 - (B) Physisorption is reversible in nature.
 - (C) A gas that can show physisorption at low temperature can also show chemisorption at high temperature at same metal surface.
 - (D) In physisorption on increasing temperature extent of adsorption decreases.
54. Select the correct statement for lyophobic sol -
- (A) Can be coagulated by dialysis
 - (B) Can be coagulated by adding electrolyte
 - (C) Viscosity is much lesser than dispersion medium
 - (D) Can be prepared by peptisation
55. Select the **INCORRECT** statement for the $\text{Fe}(\text{OH})_3$ sol :
- (A) Homogeneous mixture
 - (B) Dispersed phase move towards anode during electro-osmosis
 - (C) Dispersed phase move towards cathode during electro-osmosis
 - (D) Can be formed by adding excess of FeCl_3 in NaOH solution

Paragraph Type (3 Marks each, -1 for wrong answer)

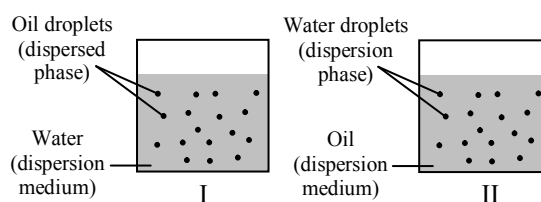
Passage # 1 (Q. 56 to 58)

If a pair of immiscible liquids such as olive oil and water are mixed and shaken vigorously, a dispersion of very fine droplets of one liquid in the other will result. Such a dispersion is termed emulsion.

- One of the components in emulsion is water and the other one an oily substance immiscible in water.
- Emulsion droplets are bigger than sol particles (10^{-6} m) and can be seen under an ordinary microscope or sometimes even with a magnifying glass.
- Emulsions resemble lyophobic sols in some of the properties.

Emulsion can be identified by dye-test.

56. In the following emulsion –



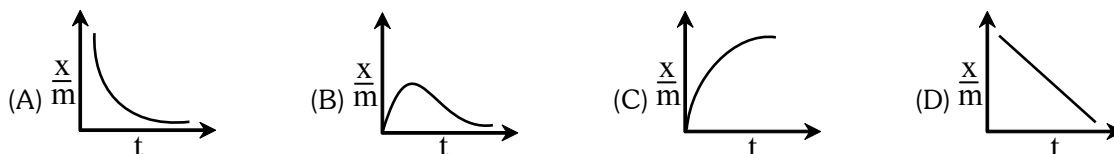
- (A) I is of oil-in-water type and II is of water-in-oil type
 (B) I is of water-in-oil type and II is of oil-in-water type
 (C) both are of oil-in-water type
 (D) both are of water-in-oil type
57. An oil-soluble dye is shaken with the given emulsion under study. We observe that whole background appears coloured. this indicates that emulsion is–
 (A) water-in-oil type (B) oil-in-water type
 (C) liquid under study is pure oil (D) liquid under study is pure water
58. An oil-soluble dye is shaken with the given emulsion under study. we observe coloured drops when seen under microscope. Thus emulsion is–
 (A) water-in-oil type (B) oil-in-water type
 (C) liquid under study is pure oil (D) liquid under study is pure water

Passag # 2 (Q. 59 to 62)

Adsorption is spontaneous process and involves unequal distribution of the molecules of the gaseous substance on the surface solid or liquid. Adsorption is exothermic process. The attractive forces between adsorbate and adsorbent are either van der Waals forces or chemical bonds. Adsorption of gases on solids is generally controlled by the factors like temperature, pressure and nature of adsorbate and adsorbent.

59. In physisorption process the attractive forces between adsorbate and adsorbent are -
 (A) Covalent bonds (B) ionic bonds (C) Van der Waals forces (D) H-bonds
60. What is not true about chemisorption ?
 (A) It forms monomolecular layer
 (B) The extent of adsorption is generally related to ease of liquefaction of gases
 (C) It is irreversible
 (D) It is highly specific
61. Which of the following is known as Langmuir adsorption isotherm -
 (A) $x/m = kP^{1/n}$ (B) $\log x/m = \log k + \frac{1}{n} \log P$
 (C) $x/m = \frac{k'P}{1 + kP}$ (D) $x/m = kP$

62. Which of the following graph represents the variation of physical adsorption with temperature



Paragraph for Q.63 to Q.64

Colloidal particles always carry a charge. The nature of this charge is same on all the particles in a given colloidal solution. Preferential adsorption of ions is the most accepted reason for charge when excess of an electrolyte is added the colloidal particles are precipitated. The reason is that colloids interact with ions carrying charge opposite to that present on themselves. The electrolyte which has maximum precipitating power has minimum coagulating value.

63. In which of the following case the colloidal particle is negatively charged

- (A) FeCl_3 is added to excess of hot water (B) KI solution is added to AgNO_3 solution
 (C) FeCl_3 is added to NaOH solution (D) KCl solution is added to AgNO_3 solution

64. Which of the following electrolyte has minimum coagulation value for a colloidal solution obtained by adding silver nitrate solution to potassium iodide solution -

- (A) Ba^{2+} (B) PO_4^{3-} (C) Al^{3+} (D) $[\text{Fe}(\text{CN})_6]^{4-}$

Paragraph for Q. 65 to Q. 66

Adsorption is the tendency of accumulation of molecular species at the surface of solid or liquid. Depending upon the nature of bonds or forces of attraction between adsorbate and adsorbent, it is classified between physisorption and chemisorption.

65. Which of the following statements are incorrect :

- (A) Usually adsorption leads to a decrease in enthalpy and entropy of the system
 (B) Adsorption is a surface phenomenon. In it, the concentration of substance is greater on the interface than in the bulk.
 (C) Adsorption increases with rise in temperature in high temperature range.
 (D) In physisorption, multimolecular layer of adsorption takes place.

66. Which of the following characteristics is not correct for physical adsorption ?

- (A) The extent of adsorption increases with increase of surface area of the adsorbent
 (B) Enthalpy of adsorption is low
 (C) It is reversible in nature
 (D) Degree of adsorption increases with increase in temperature

Match the column Type (8 Marks each, 0 for wrong answer)

67 . Match the column

Column-I

- (P) Starch
(Q) $\text{Fe}(\text{OH})_3$ sol
(R) Gold sol
(S) Soap in H_2O

Column-II

- (1) Multimolecular collide
(2) +ve collide
(3) Associated collide
(4) Lyophilic collide

Code :

	P	Q	R	S
(A)	3	2	1	4
(B)	2	1	4	3
(C)	4	2	1	3
(D)	1	3	4	2

68. Match the column-

Column-I

- (A) Dispersion of $\text{Al}(\text{OH})_3$ by small quantity of AlCl_3
(B) Stabilisation of $\text{Al}(\text{OH})_3$ by AlCl_3
(C) Solution of haemoglobin in water
(D) Mixing of $\text{Al}(\text{OH})_3$ with Al_2S_3

Column-II

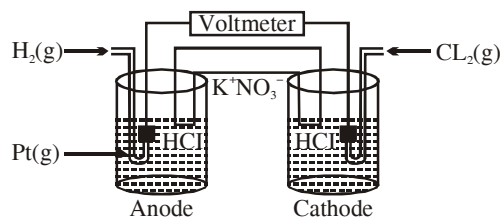
- (P) Macromolecular colloid
(Q) Selective adsorption
(R) Flocculation or coagulation
(S) Peptisation

ANSWER-KEY

1. (B) 2. (C) 3. (C) 4. (C) 5. (C) 6. (D) 7. (A) 8. (D) 9. (D) 10. (B)
11. (C) 12. (C) 13. (D) 14. (C) 15. (C) 16. (C) 17. (C) 18. (B) 19. (D) 20. (A)
21. (C) 22. (D) 23. (D) 24. (C) 25. (B) 26. (C) 27. (D) 28. (A) 29. (B) 30. (A)
31. (A) 32. (D) 33. (B) 34. (A) 35. (A) 36. (D) 37. (D) 38. (A) 39. (B) 40. (A)
41. (A) 42. (C) 43. (D) 44. (C) 45. (A) 46. (A) 47. (B) 48. (C) 49. (A,B,C)
50. (A, B, D) 51. (A,B, D) 52. (C,D) 53. (A, B,C,D) 54. (A,B,C,D)
55. (A, B, C) 56. (A) 57. (A) 58. (B) 59. (C) 60. (B) 61. (C) 62. (A) 63. (C)
64. (C) 65. (C) 66. (D) 67. (C) 68. (A) $\rightarrow$ S ; (B) $\rightarrow$ Q ; (C) $\rightarrow$ P ; (D) $\rightarrow$ R

Single Option Correct (+3, -1)

1. Consider the following Galvanic cell.



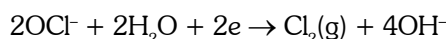
By what value the cell voltage change when concentration of ions in anodic and cathodic compartments both increased by factor of 10 at 298 K : -

- (A) + 0.0591 (B) -0.0591 (C) -0.1182 (D) 0
2. When a solution of AgNO_3 (1M) is electrolysed using platinum anode and copper cathode. What are the products obtained at two electrons : -

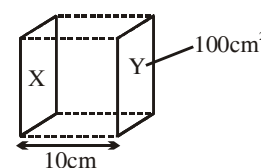
Given $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = + 0.34 \text{ volt}$; $E^\circ_{\text{O}_2\text{H}^+/\text{H}_2\text{O}} = + 1.23 \text{ volt}$

$E^\circ_{\text{H}^+/\text{H}_2} = + 0.00 \text{ volt}$; $E^\circ_{\text{Ag}^+/\text{Ag}} = + 0.8 \text{ volt}$

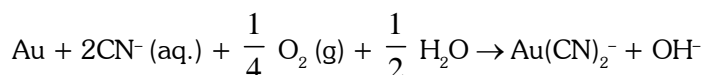
- (A) $\text{Cu} \rightarrow \text{Cu}^{2+}$ at Anode ; $\text{Ag}^+ \rightarrow \text{Ag}$ at Cathode
 (B) $\text{H}_2\text{O} \rightarrow \text{O}_2$ at Anode ; $\text{Cu}^{2+} \rightarrow \text{Cu}$ at Cathode
 (C) $\text{H}_2\text{O} \rightarrow \text{O}_2$ at Anode ; $\text{Ag}^+ \rightarrow \text{Ag}$ at Cathode
 (D) $\text{NO}_3^- \rightarrow \text{N}_2\text{O}_5$ at Anode ; $\text{Ag}^+ \rightarrow \text{Ag}$ at Cathode
3. A current of 9.65 A is placed for 3 hr between nickel cathode and Pt anode in 0.5L of a 2 M solution of $\text{Ni}(\text{NO}_3)_2$. The molarity of Ni^{2+} after electrolysis would be : -
- (A) 0.46 M (B) 0.92 M (C) 1.08 M (D) None
4. The standard reduction potential's of half cell OCl^-/Cl^- , OH^- and Cl_2/Cl^- are 0.94 volt and + 1.36 volt respectively. What is the reduction potential of half cell whose cell reaction is represented as: -



- (A) 0.21 V (B) 0.52 V (C) 1.04 V (D) 2.1 V
5. The conductance of a salt solution (AB) measured by two parallel electrodes of area 100 cm^2 separated by 10 cm was found to be $0.0001 \Omega^{-1}$. If volume enclosed between two electrode contain 0.5 mole of salt, what is the molar conductivity ($\text{Scm}^2 \text{ mol}^{-1}$) of salt at same conc.

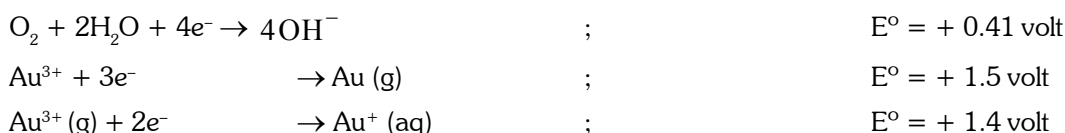


- (A) 0.01 (B) 0.02
 (C) 2×10^{-5} (D) None of these
6. How much time is required for complete decomposition of two moles of water using 4 ampere
- (A) $3.86 \times 10^5 \text{ sec}$ (B) $1.93 \times 10^5 \text{ sec}$ (C) 96500 sec (D) 48250 sec
7. Consider the reaction of extraction of gold from its ore



Use the following data to calculate ΔG° for the reaction

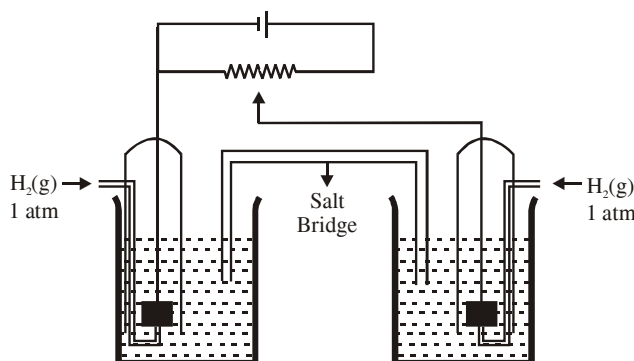
$$K_f \{ \text{Au}(\text{CN})_2^- \} = X$$



(A) $-RT \ln X + 1.29 F$ (B) $-RT \ln X - 2.11 F$

(C) $-RT \ln \frac{1}{X} + 2.11 F$ (D) $-RT \ln X - 1.29 F$

8. Equivalent conductance of 0.1 M HA (weak acid) solution is $10 \text{ Scm}^2 \text{ equivalent}^{-1}$ and that at infinite dilution is $200 \text{ Scm}^2 \text{ equivalent}^{-1}$. Hence pH of HA solution is : -
 (A) 1.3 (B) 1.7 (C) 2.3 (D) 3.7
9. Which of the following arrangement will produce oxygen at anode during electrolysis?
 (A) Dilute H_2SO_4 with Pt electrodes (B) Fused NaOH with inert electrodes
 (C) Dilute H_2SO_4 with Cu electrodes (D) Concentrate aq. NaCl with Pt electrodes
10. Carefully observe the given figure and using data provided find the EMF of shown Galvanic cell in volt :



Solution A is 0.1 M each in NH_4OH and NH_4Cl and solution B is 0.1 M each in CH_3COOH and $\text{CH}_3\text{COO}^-\text{Na}^+$.

[Given : $K_a(\text{CH}_3\text{COOH}) = 10^{-5}$, $K_b(\text{NH}_4\text{OH}) = 10^{-5}$ and $\frac{2.303RT}{F} = 0.06 \text{ volt}$]

- (A) 0.24 volt (B) 0.12 volt (C) 0.06 volt (D) -0.24 volt
11. The molar conductivity of 0.04 M solution of MgCl_2 is $200 \text{ Scm}^2 \text{ mol}^{-1}$ at 298 K. A cell with electrodes that are 4 cm^2 in surface area & 8 cm apart is filled with MgCl_2 solution.
 How much current will flow when the potential difference between the two electrodes is 10 V?
 (A) 0.32 A (B) $4 \times 10^{-2} \text{ A}$ (C) $4 \times 10^{-4} \text{ A}$ (D) None of these
12. $\text{K}_2\text{Cr}_2\text{O}_7$ is a powerful oxidising agent in acidic medium. By how many volt's oxidising power of $\text{K}_2\text{Cr}_2\text{O}_7$ decreases when pH changed from 0 to 14 at 298 K. (Take : $\frac{2.303RT}{F} = 0.06 \text{ volt}$)
 (A) 1.96 volt (B) 11.76 volt (C) 1.17 volt (D) 0.98 volt

Multiple choice questions (+4, -1)

13. The EMF of the following is 0.22 volt.
 $\text{Ag(s)} \mid \text{AgCl(s)} \mid \text{KCl (1M)} \mid \text{H}^+ (1\text{M}) \mid \text{H}_2(\text{g}) (1\text{atm}) ; \text{Pt(s)}$.
 Which of the following will decrease the EMF of cell.
 (A) increasing pressure of $\text{H}_2(\text{g})$ from 1 atm to 2 atm
 (B) increasing Cl^- concentration in Anodic compartment
 (C) increasing H^+ concentration in cathodic compartment
 (D) decreasing KCl concentration in Anodic compartment

Integer/Subjective type (+3, 0)

14. Find the log of equilibrium constant for the reaction at 298 K ($\log_{10} K$)
 $2\text{H}^+(\text{aq}) + \text{D}_2(\text{g}) \rightleftharpoons \text{H}_2(\text{g}) + 2\text{D}^+(\text{aq})$

Given $E^\circ_{\text{D}^+/\text{D}_2} = -2.955 \text{ mV}$; Use $\left(\frac{RT2.303}{F} \right) = 0.0591$

15. Calculate the magnitude of emf of a cell ($|E|$) in volt having reaction: $\text{Ag}_2\text{S(s)} + 2^- \rightleftharpoons 2\text{Ag(s)} + \text{S}^{2-}(\text{aq.})$ in a solution buffered at $\text{pH} = 3$ and which is also saturated with $0.1 \text{ M H}_2\text{S}(\text{aq.})$.

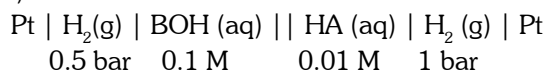
Given For H_2S : $K_{a_1} = 10^{-7}$ and $K_{a_2} = 10^{-14}$, $K_{sp}(\text{Ag}_2\text{S}) = 10^{-49}$, $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.8$.

Use $2.303 \times \frac{RT}{F} = 0.06$:-

16. The specific conductivity of a 0.5 M aq. solution of monobasic acid HA at 27°C is 0.006 Scm^{-1} . Its molar conductivity at infinite dilution is $200 \text{ Scm}^2 \text{ mol}^{-1}$. Calculate osmotic pressure (in atm) of $0.5 \text{ M HA}(\text{aq})$ solution at 27°C .

Given $R = 0.08 \frac{\text{atmL}}{\text{molK}}$

17. Calculate the EMF (in V) of the cell



Given : $K_a(\text{HA}) = 4 \times 10^{-6}$, $K_b(\text{BOH}) = 10^{-5}$, $2.303 \frac{RT}{F} = 0.06$, $\log 2 = 0.3$

18. The chlorate ion can disproportionate in basic solution according to reaction



What is the equilibrium concentration of per chlorate ions (in mole/litre) from a solution initially at 0.1 M in chlorate ions at 298 K .

Given : $E^\circ_{\text{ClO}_4^-/\text{ClO}_3^-} = 0.36 \text{ V}$ and $E^\circ_{\text{ClO}_3^-/\text{ClO}_2^-} = 0.33 \text{ V}$

Take : $2.303 \frac{RT}{F} = 0.06$, $\sqrt{10} = 3.16$

All data at 298 K

ANSWER KEY (QUIZ # 1)

- | | | | | | | | | |
|------------|------------------------------|---------|-----------|-----------|-------------|--------|-------------|--------|
| 1. (C) | 2. (C) | 3. (B) | 4. (B) | 5. (B) | 6. (C) | 7. (D) | 8. (C) | 9. (B) |
| 10. (A) | 11. (B) | 12. (A) | 13. (A,D) | 14. (0.1) | 15. (-0.19) | | 16. (12.72) | |
| 17. (0.43) | 18. (1.9×10^{-2}) | | | | | | | |

QUIZ # 2

Single Option Correct (+3, -1)

1. What is the EMF of represented cell at 298
 $\text{Ag} | \text{Ag}^+(\text{aq}, 0.1 \text{ M}) || \text{H}^+(\text{aq}, 0.1 \text{ M}) | \text{H}_2(\text{g}, 0.1 \text{ bar}) | \text{Pr}$
 Given : $E^\circ_{\text{Ag}/\text{Ag}^+} = -0.8 \text{ V}$
 (A) 0.77 V (B) -0.829 V (C) -0.77 V (D) None of these
2. 96.5 gm of A oxidized as $\text{A} \rightarrow \text{A}^{3+} + 3\text{e}^-$, where 2F charge is passed through solution with current efficiency of 90% . What is electrochemical equivalent of A ?
 (A) 53.61 (B) 48.25 (C) 5.55×10^{-4} (D) 5.55×10^{-3}
3. During discharging of lead storage battery, which of the following is/are true?
 (A) H_2SO_4 is produced (B) H_2O is consumed
 (C) PbSO_4 is formed at both electrodes (D) Density of electrolytic solution decreases

Given : $\left[\frac{2.303RT}{F} = 0.06 \right]$

- (A) 0.05 (B) 0.069 (C) 1.18 (D) 0.059

4. What is the value of $E_{\text{Cd}^{2+}|\text{Cd}}$ at pH = 10

Given : $E_{\text{Cd}|\text{Cd}^{2+}}^0 = 0.40 \text{ V}$; $K_{\text{SP}(\text{Cd}(\text{OH})_2)} = 10^{-14}$

- (A) -0.5773 V (B) -0.2227 V (C) +0.2227 V (D) -0.1045 V

5. Calculate acid dissociation constant for the reaction $\text{H}_2\text{O}(\text{aq}) \rightleftharpoons \text{H}^+(\text{aq}) + \text{HO}_2^-(\text{aq})$

Given : $\text{H}_2\text{O}_2(\text{aq}) + 2\text{H}^+(\text{aq}) + 2e^- \rightarrow 2\text{H}_2\text{O}(\text{l})$; $E^0 = 1.77$

$\text{HO}_2^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) + 2e^- \rightarrow 3\text{OH}^-(\text{aq})$; $E^0 = 0.87$

$K_w = 10^{-14}$; $2.303 \times R \times T = 5790 \text{ J/mol}$

- (A) 10^{-30} (B) 10^{-16} (C) 10^{-12} (D) 10^{-6}

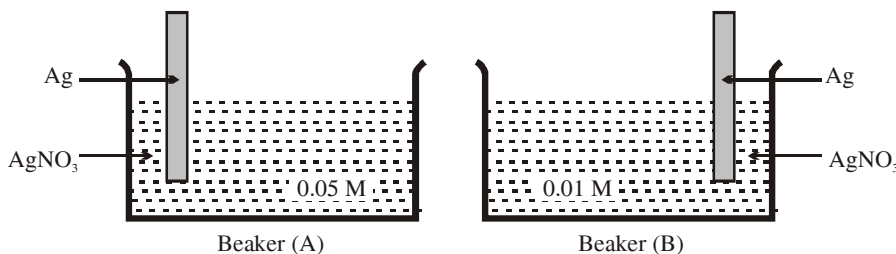
6. The pH of 10 L of concentrate KCl (aq) solution after the electrolysis for 100 sec. using 9.65 A current at 298 K is : -

- (A) 10 (B) 11 (C) 11.5 (D) 12

7. Two beakers A and B contains AgNO_3 solutions of concentrations 0.05 M and 0.01 M respectively as shown in figures. Pure silver electrodes are now lowered in each beaker. Now NH_3 is added in each beaker till conc. of NH_3 in beaker A and B become 0.1 M and 0.2 M respectively. If two beakers are now connected in Galvanic cell arrangement, find the EMF (in Volts) of the resulting cell at 298 K.

Use : $\log_{10} 2 = 0.3$, $\frac{RT}{F} (2.303) = 0.06$ at 298 K

Given : $E_{\text{Ag}^+/\text{Ag}}^0 = 0.8 \text{ Volt}$, $K_f [\text{Ag}(\text{NH}_3)_2^+] = 10^8$



- (A) 9.6×10^{-2} (B) 7.8×10^{-2} (C) 5.8×10^{-3} (D) None of these

Passage type (+3, -1)

Paragraph for question nos. 8 to 11

A half cell is prepared by $\text{K}_2\text{Cr}_2\text{O}_7$ in a buffer solution of pH = 1. Concentration of $\text{K}_2\text{Cr}_2\text{O}_7$ is 1 M. To 3 litre of this solution 570 gm of SnCl_2 is added which is oxidised completely to SnCl_4 .

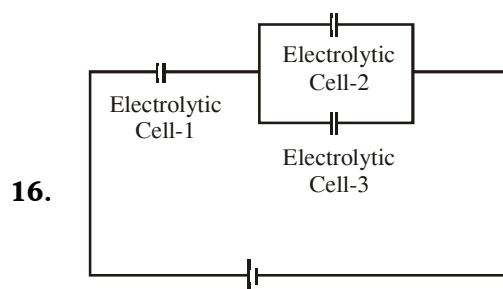
Given : $E_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{+3}, \text{H}^+}^0 = 1.33 \text{ V}$; $\frac{2.303}{F} RT = 0.06$;

Atomic mass of Sn = 119 ; $E_{\text{Sn}^{+4}/\text{Sn}^{+2}}^0 = 0.15$

8. Number of moles of Cr^{+3} formed are

- (A) 2 (B) 6 (C) 4 (D) 3

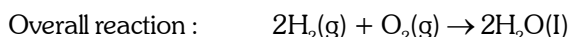
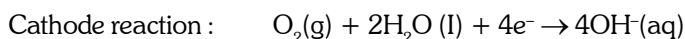
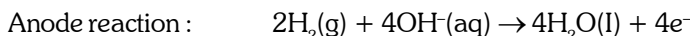
9. Half cell potential $E_{\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{+3}}$ after the reaction of SnCl_2 is : -
 (A) 1.187 (B) 1.191 (C) 1.285 (D) None of these
10. Emf of the cell $\text{Pt} | \text{Sn}^{+2}, \text{Sn}^{+4}, \text{H}^+ || \text{Cr}_2\text{O}_7^{2-}, \text{Cr}^{+3}, \text{H}^+ | \text{Pt}$
 (0.1 M) (0.2 M) (1M) (0.2 M) (1 M) (1 M)
 (A) -1.18 V (B) 1.164 V (C) 1.18 V (D) None of these
11. Equal volume of 0.015 M CH_3COOH & 0.015 M NaOH are mixed together. What would be molar conductivity of mixture if conductivity of CH_3COONa is $6.3 \times 10^{-4} \text{ S cm}^{-1}$.
 (A) 8.4 S cm^2 (B) $84 \text{ S cm}^2 \text{ mol}^{-1}$ (C) $4.2 \text{ S cm}^2 \text{ mol}^{-1}$ (D) $42 \text{ S cm}^2 \text{ mol}^{-1}$
12. An aqueous solution of Na_2SO_4 was electrolysed for 10 min. 82 ml of a gas was produced at anode over water at 27°C at a total pressure of 580 torr. Determine the current that was used.
 (Vapour pressure of H_2O at $27^\circ\text{C} = 10 \text{ torr}$) ($R = 0.082 \text{ atm lit. / mol / K}$)
 (A) 0.1 Amp (B) 1.25 Amp (C) 2.23 Amp (D) 1.61 Amp
13. Na-Amalgam is prepared by electrolysis of aq. NaCl solution using 10 gm Hg cathode. How many Faraday's of electricity is required to prepare 18.7% Na-Amalgam. (current Eff. = 50%)
 (A) 0.081 (B) 0.16 F (C) 0.2 F (D) 0.1 F
14. Select the correct option if it is known that $K_{\text{sp}}(\text{AgCl}) > K_{\text{sp}}(\text{AgBr}) > K_{\text{sp}}(\text{AgI})$
 (A) $E^\circ_{\text{I}^-/\text{AgI}/\text{Ag}} > E^\circ_{\text{Br}^-/\text{AgBr}/\text{Ag}} > E^\circ_{\text{Cl}^-/\text{AgCl}/\text{Ag}}$ (B) $E^\circ_{\text{I}^-/\text{AgI}/\text{Ag}} < E^\circ_{\text{Br}^-/\text{AgBr}/\text{Ag}} < E^\circ_{\text{Cl}^-/\text{AgCl}/\text{Ag}}$
 (C) $E^\circ_{\text{Ag}/\text{AgI}/\text{I}^-} < E^\circ_{\text{Ag}/\text{AgBr}/\text{Br}^-} < E^\circ_{\text{Ag}/\text{AgCl}/\text{Cl}^-}$ (D) None of these
15. If $E_{\text{I}_2/\text{I}^-} = +0.50 \text{ V}$ and $E_{\text{I}_2/\text{I}_3^-} = +0.20 \text{ V}$, then $E_{\text{I}_3^-/\text{I}^-}$ is : -
 (A) 0.30 V (B) 0.65 V (C) 1.30 V (D) -0.067 V



	Content	Resistance	Volume of gas produced (at same T & P)
Cell-1	$\text{CuSO}_4(\text{aq})$	10Ω	at anode is V_1
Cell-2	$\text{NaCl}(\text{aq})$	2Ω	at node is V_2
Cell-3	$\text{ZnSO}_4(\text{aq})$	6Ω	at cathode is V_3
(A) 2 : 3 : 1	(B) 3 : 2 : 1	(C) 3 : 1 : 2	(D) 2 : 1 : 3

Paragraph for question nos. 17 to 19.

A fuel cell is a device to convert the energy of a fuel into electrical energy without the use of a heat engine, where the fuel is burnt directly. Such conversions are possible because the overall reaction is highly exothermic. Electrical energy can be obtained indefinitely from a fuel cell as long as the outside supply of fuel is maintained. In a hydrogen-oxygen fuel cell, the following reaction takes place : -



$$\Delta H_f^\circ [\text{H}_2\text{O}(\text{l})] = -285.8 \text{ kJ mol}^{-1} \text{ and } \Delta G_f^\circ [\text{H}_2\text{O}(\text{l})] = -237.39 \text{ kJ mol}^{-1} \text{ at } 25^\circ\text{C}.$$

17. What is the standard EMF of the cell?
 (A) 0.615 V (B) 1.23 V (C) 2.46 V (D) 0.74 V
18. How many litres of gaseous H_2 , when combined with excess O_2 in the fuel cell at 25°C and 1.00 bar, are needed to produce 23.739 KJ of useful work under standard conditions?
 (A) 2.47 L (B) 4.94 L (C) 1.23 L (D) 2.26 L
19. The theoretical efficiency of the fuel cell given by : -
 (A) 83.06% (B) 100% (C) 67.53% (D) 97.88%
20. A Daniel Cell is made at 25°C by connecting Zn/Zn^{+2} (0.1 M, 1 litre) & Cu/Cu^{+2} (0.9 M, 1 litre) electrode. The cell is discharged till its emf reaches 1.10 volt, then it is charged (reversing the discharge process) by passing 0.6 F charge.

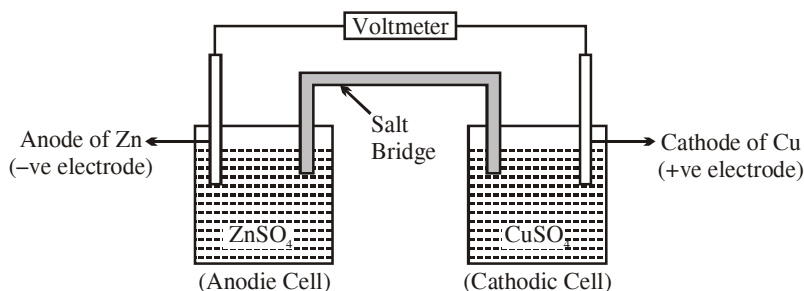
[Given : $E^\circ_{\text{Zn}/\text{Zn}^{+2}} = 0.76 \text{ V}$, $E^\circ_{\text{Cu}/\text{Cu}^{+2}} = -0.34 \text{ V}$, $\frac{2.303RT}{F} = 0.06$, $\log 2 = 0.30$]

Select the correct option(s)

- (A) Concentration of Zn^{+2} becomes 1 M when EMF reach 1.10 volt
 (B) Concentration of Cu^{+2} becomes 0.5 M when EMF reach 1.10 volt
 (C) EMF of the cell just after charging is 1.118 volt
 (D) After charging concentration of Cu^{+2} becomes 0.8 M

Paragraph for question nos. 21 to 23.

The most popular electrochemical cell, Daniel cell was originally developed by the English chemist John F. Daniel. The general assembly of the cell is as given below



An undergraduate student set up a Daniel cell using 100 cm^3 of 0.100 M CuSO_4 and 0.100 M ZnSO_4 solution respectively. The two compartments are connected by suitable salt bridge.

Given : $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34 \text{ V}$, $E^\circ_{\text{Zn}^{2+}/\text{Zn}} = -0.76 \text{ V}$, $\frac{2.303RT}{F} = 0.06$, $\log 2 = 0.3$.

21. Calculate the emf (in Volt) of the above cell. Assume the temperature of the laboratory to be 25°C and the solutions are dilute enough so that molar concentrations can be used instead of activities of the solutions : -
 (A) 0 (B) 1.1 (C) 1.16 (D) 1.04
22. A lab mate of the student asked her for some solid CuCl_2 . While she was lifting the bottle from a shelf, the lid of the bottle slipped and some amount of CuCl_2 fell in the CuSO_4 compartment at constant volume. She measured the emf of the cell again and found that it had increased by 9 mV. She used this data to calculate the amount of CuCl_2 that had split in the compartment.
 Calculate the mass of CuCl_2 in grams that had split into the Daniel cell?
 (Molecular mass of $\text{CuCl}_2 = 135 \text{ g}$)
 (A) 1.35 (B) 13.5 (C) 27 (D) 2.7

23. After making this calculation, the student discarded the solutions from both the compartments and washed the cell thoroughly. She refilled the compartments with 100 cm³ of the original CuSO₄ and ZnSO₄ solutions. A current of 0.965 A was drawn from the cell for a period of 20 min. What will be the cell potential after 20 min.?
- (A) 1.091 V (B) 1.109 V (C) 1.118 V (D) 1.082 V

Paragraph for question nos. 24 to 25

An element (X) having same chemical properties as that of hydrogen atom. The diatomic molecule X₂ is gaseous in nature. The energy required to remove an electron from the outmost shell of (X) atom is 13.5 eV. It is also found that the time taken for diffusion of equal volume of gaseous X₂ & O₂ at the same pressure is in the ratio $\sqrt{3} : 4$. The normal freezing point of pure X₂O is 0°C. However on adding 0.02 moles of a non-electrolyte solute to 0.8 moles of X₂O, the freezing point of solution is found to be -1.25°C. X-atom also obey Bohr's model.

24. When 3 gm of a weak monoprotic acid (M_w = 60) is added to 0.8 moles of X₂O, the resulting solution freeze at -3.75°C. What is degree of dissociation for weak acid.
- (A) 0.10 (B) 0.20 (C) 0.30 (D) 0.40
25. Energy required to promote electrons in the sample from ground state to 3rd state
- (A) 50 eV (B) 60 eV (C) 40 eV (D) 90 eV

Multiple Option Correct (+4, -2, +1 for each correct answer)

26. The following data are obtained in an experiment

Solution	Conductivity
0.01 M HA	$3.8 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$
100 ml of 0.01 M HA + 1 mmol NaOH	$80.0 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$

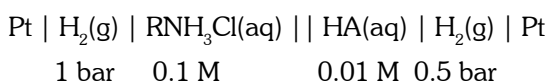
Calculate

Given : $\lambda_m^\infty (\text{Na}^+) = 50 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, $\lambda_m^\infty (\text{H}^+) = 350 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$

- (A) $\Lambda_m (\text{NaA})$ is $80 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ (B) $\Lambda_m (\text{A}^-)$ is $30 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$
- (C) $\Lambda_m (\text{HA})$ is $3.8 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ (D) degree of dissociation of 0.01 M HA is 0.01

Integer type (+3, 0)

27. Calculate the EMF (in milliVolt) of the cell



[Given : $K_{a(\text{HA})} = 4 \times 10^{-6}$, $K_{b(\text{RNH}_2)} = 10^{-5}$, $2.303 \frac{RT}{F} = 0.06$; $\log 2 = 0.3$]

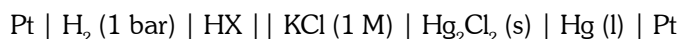
28. Half cells Zn | Zn²⁺ (1 litre, 0.1 M) and Cu | Cu²⁺ (1 litre, 1M) are connected to form as cell. Calculate the concentration of Cu²⁺ in solution when cell potential becomes 0.8 volt.

Given : $E^\circ_{\text{Cu}^{2+}/\text{Cu}} = 0.34$; $E^\circ_{\text{Zn}/\text{Zn}^{2+}} = 0.76$; $\frac{2.303RT}{F} = 0.06$

[Fill your answer by multiplying it with 10¹²].

29. On passing electricity through nitrobenzene solution, it is converted into azobenzene. Calculate the mass of azobenzene (in mg) if same quantity of electricity produces oxygen just sufficient to burn 96 mg of fullerene (C₆₀):
30. The electrolysis of cold sodium chloride solution produces NaOH and Cl₂. The Cl₂ produced, the disproportionates into NaOH solution to give sodium hypochlorite (NaClO) and sodium chloride. How long will a cell operate to produce 1.00 × 10³ L of 7.45% (w/w) solution of NaClO if the cell current is 9.65 ampere? Assume that the density of solution is 1.00 gm/ml : - [Fill your answer by multiplying it with 10⁻⁵]

31. Consider the following galvanic cell :-

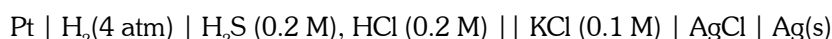


At 25°C, emf of the cell is 600 mV. When HX is replaced by a buffer solution at pH 6.80, the emf of cell becomes 750 mV. Calculate the pH of HX solution originally used.

$$\text{Given : } \frac{2.303RT}{F} = 0.06 \text{ V.}$$

(Answer by multiplying the pH by 1000)

32. Calculate E_{cell} of the given cell



$$K_{a_1} (\text{H}_2\text{S}) = 10^{-8} \text{ M}; \quad K_{a_2} (\text{HS}^-) = 10^{-13} \text{ M}; \quad K_{\text{sp}} (\text{AgCl}) = 10^{-10} \text{ M}^2$$

$$E_{\text{Ag}^+/\text{Ag}}^\circ = 0.8 \text{ V}; \quad \frac{2.303RT}{F} = 0.06$$

[Fill your answer by multiplying it with 100.]

ANSWERS (QUIZ # 2)

1. (C) 2. (C) 3. (C) 4. (A) 5. (C) 6. (B) 7. (B) 8. (A) 9. (B)

10. (B) 11. (B) 12. (D) 13. (C) 14. Sol. $E_{\text{I}^-/\text{AgI}/\text{Ag}}^\circ = E_{\text{Ag}^+/\text{Ag}}^\circ + \frac{0.059}{1} \log K_{\text{sp}}$

15. (B) 16. (A) 17. (B) 18. (A) 19. (A) 20. (C) 21. (B) 22. (B)

$$23. \text{ Sol. } n_e = \frac{0.965 \times 20 \times 60}{96500} = 0.012, \quad E = 1.1 - \frac{0.06}{2} \log \frac{0.01 + \frac{n_e}{2}}{0.01 - \frac{n_e}{2}}$$

24. (B) 25. (B) 26. (A,B,C,D) 27. (87) 28. (1.1×10^{-10})

29. (0.728) 30. (2×10^7) 31. (4.3) 32. (0.32)

QUIZ # 3

Single Option Correct (+3, -1)

1. In the electrolysis of CuCl_2 solution using Cu electrodes, the weight of Cu increased by 2 gram at cathode. At the anode -

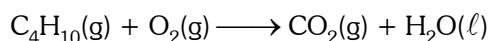
(A) 0.2 mole of Cu^{2+} will go into solution

(B) Cl_2 is produced

(C) No loss in weight

(D) 2 gram of copper goes into solution as Cu^{2+}

2. During combustion of butane at 1 atm and 298 K in a fuel cell as per reaction



$$\Delta G^\circ = -3900 \text{ kJ/mole}$$

E_{cell}° will be

(A) +6.735 V

(B) +0.775 V

(C) +3.36 V

(D) +1.55 V

3. The resistance of 0.05 M solution of oxalic acid is 200 ohm and cell constant is 2.0 cm^{-1} , the equivalent conductance (in $\text{S cm}^2 \text{ eq}^{-1}$) of 0.05 M oxalic acid is :-
 (A) 100 (B) 10 (C) 1 (D) None of these
4. ΔG° of the cell reaction
 $\text{AgCl(s)} + \frac{1}{2} \text{H}_2(\text{g}) \rightleftharpoons \text{Ag(s)} + \text{H}^+(\text{aq.}) + \text{Cl}^-(\text{aq.})$
 is -21.52 kJ . Calculate the EMF for the cell reaction
 $2\text{AgCl(s)} + \text{H}_2(\text{g}) \rightleftharpoons 2\text{Ag(s)} + 2\text{H}^+(\text{aq.}) + 2\text{Cl}^-(\text{aq.})$
 (A) 0.223 V (B) 0.446 V (C) 0.112 V (D) 0.337 V
5. The aq. solutions of the following substances were electrolysed using inert electrodes. In which case, the pH of the solution does not change -
 (A) $\text{AgNO}_3(\text{aq.})$ (B) $\text{CuSO}_4(\text{aq.})$ (C) $\text{dil. NaCl}(\text{aq.})$ (D) $\text{K}_2\text{SO}_4(\text{aq.})$
6. During electrolysis of NaCl, if 3 mole of H_2O are electrolysed then how much charge is required if current efficiency is 75% -
 (A) 1 F (B) 2 F (C) 4 F (D) 8 F
7. If E° for the reaction
 $\text{MnO}_4^-(\text{aq.}) + 8\text{H}^+(\text{aq.}) + 5e^- \longrightarrow \text{Mn}^{2+}(\text{aq.}) + 4\text{H}_2\text{O(l)}$
 is 1.51 volt and E° for the reaction
 $\text{MnO}_2(\text{s}) + 4\text{H}^+(\text{aq.}) + 2e^- \longrightarrow \text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O(l)}$
 is 1.21 volt then the E° for the reaction -
 $\text{MnO}_4^-(\text{aq.}) + 4\text{H}^+(\text{aq.}) + 3e^- \longrightarrow \text{MnO}_2(\text{s}) + 3\text{H}_2\text{O(l)}$ is -
 (A) 1 volt (B) 2.42 volt (C) 2.71 volt (D) 1.71 volt
8. The emf of an electrolytic cell does not depends upon :
 (A) Valency of reacting material used in cell
 (B) concentration of metal ion present in the electrolytic solution
 (C) quantity of electrolytic solution used in the cell
 (D) temperature of electrolytic cell
9. Which of the following reaction occurs at anode when the electrolysis of aqueous CuSO_4 solution is carried out using copper electrodes?
 (A) $\text{Cu} \rightarrow \text{Cu}^{2+} + 2e^-$ (B) $2\text{SO}_4^{2-} + 2\text{H}_2\text{O} \rightarrow 2\text{H}_2\text{SO}_4 + \text{O}_2 + 4e^-$
 (C) $2\text{H}_2\text{O} \rightarrow \text{O}_2 + 4\text{H}^+ + 4e^-$ (D) $2\text{Cu} \rightarrow 2\text{Cu}_2^{2+} + 2e^-$
10. $\text{Pt}(\text{H}_2) \mid \text{pH} = 2 \mid \text{pH} = 3 \mid \text{Pt}(\text{H}_2)$
 1 atm 1 atm
 The cell reaction for the given cell is
 (A) spontaneous (B) non-spontaneous (C) equilibrium (D) None of these
11. The change in potential of mercuric to mercurous ion at 298K if mercuric ion concentration is reduced ten times while mercurous ion concentration is kept same, will be -
 (A) increased by 0.059 V (B) decreased by 0.059 V
 (C) increased by 0.0295 V (D) decreased by 0.0295 V
12. Which among the following metals form a protective layer of oxide to prevent corrosion
 (A) Cu (B) Ag (C) Au (D) Al

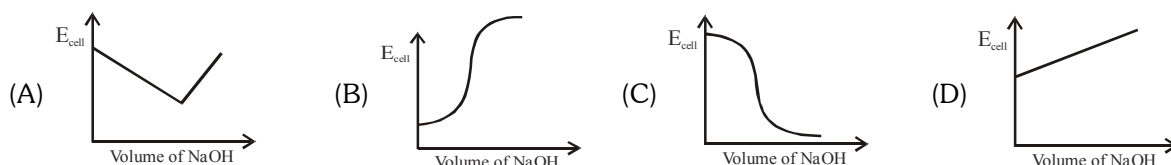
Passage type (+3, -1)**Paragraph for Q. 13 & Q.14**

For the titration of a solution of HCl against NaOH solution, hydrogen electrode is dipped into HCl solution. Then this electrode is connected to normal calomel electrode (NCE).

In such experiment 25 ml of HCl (0.1M) is titrated against a standard 0.1 M NaOH solution

$$E_{\text{NCE}^+ (\text{reduction})} = 0.24\text{V} ; \frac{2.303RT}{F} = 0.06$$

13. Select the correct graph for titration



14. E_{cell} after adding 24.95 ml of NaOH

(A) 0.3 (B) 0 V (C) 0.48 V (D) 0.64 V

Paragraph for Q.15 & 16

A solution of 0.1 M CH_3COOH is placed between parallel electrodes of cross-section area 4cm^2 , separated by 2cm. For this solution resistance measured is 100Ω .

$$K_b = 0.5 \text{ K kg/mol} ; \Lambda_m^\infty(\text{H}^+) = 300 \text{ Scm}^2 \text{ mol}^{-1} \quad \Lambda_m^\infty(\text{CH}_3\text{COO}^-) = 100 \text{ Scm}^2 \text{ mole}^{-1}$$

15. Molar conductance of 0.1M CH_3COOH solution filled is (in $\text{Scm}^2\text{mol}^{-1}$)

(A) 50 (B) 100 (C) 400 (D) 800

16. Elevation in boiling point for the given 0.1M CH_3COOH solution is -

(A) $\frac{1}{8} \times 10^{-2}$ (B) $\frac{45}{8} \times 10^{-2}$ (C) $\frac{9}{8} \times 10^{-2}$ (D) 9×10^{-2}

Paragraph for Q. 17 & Q.18

At 298 K following observation was made for KCl solution

Concentration (Mol/L) 0.01 0.04

Λ_m ($\text{S cm}^2 \text{ mol}^{-1}$) 123 112

17. What is limiting molar conductance of KCl at 298 K (in $\text{S cm}^2 \text{ mol}^{-1}$)

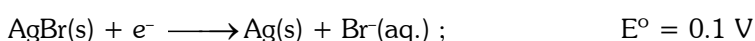
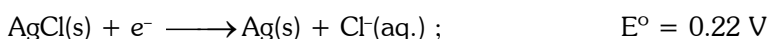
(A) 134 (B) 145 (C) 150 (D) 166

18. Specific conductance of 0.25 M KCl solution at 298 K is (in S m^{-1})

(A) 0.017 (B) 1.7 (C) 0.01975 (D) 1.975

Paragraph for Q.19 & Q.20

A galvanic cell is made by using the following electrodes.



$$\text{Use : } 2.303 \frac{RT}{F} = 0.06$$

$$K_{\text{sp}}(\text{AgCl}) = 10^{-10}$$

$$K_{\text{sp}}(\text{AgBr}) = 5 \times 10^{-13}$$

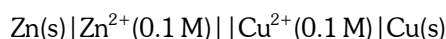
19. What is the emf of this cell in standard condition
 (A) 0.32 V (B) 0.12 V (C) 0 V (D) 0.21 V
20. For what ratio of $\frac{[Cl^-]}{[Br^-]}$ the emf of the cell is 0.06 volt -
 (A) $\frac{1}{20}$ (B) $\frac{20}{1}$ (C) $\frac{1}{50}$ (D) $\frac{50}{1}$

Multiple Option Correct (+4, -2, +1 for each correct answer)

21. For electrolysis of aq. Na_2SO_4 using Hg cathode & Pt anode. Select correct statement -
 (A) Na^+ will be reduced at anode
 (B) H_2O will be reduced at cathode
 (C) SO_4^{2-} will be oxidised at higher concentration of Na_2SO_4
 (D) H_2O will be oxidised at lower concentration of Na_2SO_4

Integer type (+3, 0)

22. Consider a Galvanic cell,



by what factor, the electrolyte in anodic half cell should be diluted to increase the emf by 9 milli volt at 298 K.

$$\left(\frac{2.303RT}{F} = 0.06 \right)$$

23. E°_{cell} of $Cu | Cu^{+2} || Ag^+ | Ag$
 Given : $E^\circ_{Cu^{2+}/Cu} = 0.34$ (with respect to standard hydrogen electrode)
 $E^\circ_{Ag^+/Ag} = 1.05$ (with respect to calomel electrode)
 $E^\circ_{\text{Calomel(reduction)}} = -0.25$ (with respect to standard hydrogen electrode)

Write your answer by multiplying with 100.

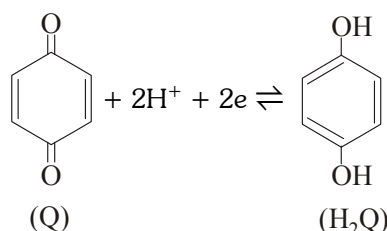
24. A 250 ml of aqueous NaCl solution is electrolysed for 1.0 min using a current of 16 mA at 25°C. What is the pH of the final solution (**F = 96000 C**).
25. A conductivity cell filled with 0.01M $Na_2C_2O_4$ solution. If resistance of solution is 200 ohm then find cell constant in cm^{-1} . If degree of dissociation of salt is 80%.

$$\text{Given : } \Lambda^\circ_{eq}(Na^+) = 75 \text{ ohm}^{-1} \text{cm}^2 \text{ gm eq}^{-1}$$

$$\Lambda^\circ_{eq}(C_2O_4^{2-}) = 50$$

Fill your answer by multiplying it with 10

26. Quinhydrone is a sparingly soluble one-to-one addition compound formed from hydroquinone and quinone. When solid quinhydrone is dissolved in an aqueous medium, equal concentrations of hydroquinone and quinone result. If a platinum wire is dipped into the solution, the potential of the electrode is governed by the reversible reaction:



Where Q denotes quinone and H_2Q denotes hydroquinone. The E° value of this half cell reaction is +0.46 V with respect to the saturated calomel reference electrode (SCE). Because the potential of the quinone-hydroquinone

half reaction depends on the concentration of hydrogen ion, it is possible to use this system to measure pH. A sample solution of unknown pH was saturated with quinhydrone and a platinum electrode was dipped into it. If the potential of the such electrode was found to be +0.22 V with respect to SCE, what was the pH of the sample solution?

[Given : $\frac{2.303}{F} RT = 0.06$]

ANSWER KEY (QUIZ # 3)

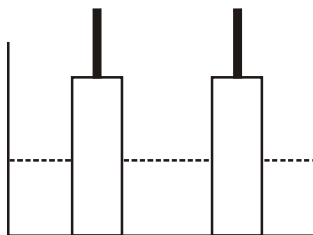
- | | | | | | | | | |
|----------------|----------------|------------------|----------------|------------------|------------------|----------------|----------------|----------------|
| 1. (D) | 2. (D) | 3. (A) | 4. (A) | 5. (C) | 6. (C) | 7. (D) | 8. (C) | 9. (A) |
| 10. (B) | 11. (B) | 12. (D) | 13. (B) | 14. (C) | 15. (A) | 16. (B) | 17. (A) | 18. (D) |
| 19. (B) | 20. (B) | 21. (C,D) | 22. (2) | 23. (046) | 24. (009) | 25. (4) | 26. (4) | |

ELECTROCHEMISTRY

SINGLE CHOICE CORRECT

- What is cell entropy change of the following cell ?
 $\text{Pt(s)} \mid \text{H}_2(\text{g}) \mid \text{CH}_3\text{COOH}, \text{HCl} \parallel \text{KCl(aq)} \mid \text{Hg}_2\text{Cl}_2(\text{s}) \mid \text{Hg}$
 $P = 1 \text{ atm} \quad 0.1 \text{ M} \quad 0.1 \text{ M}$
 Emf of the cell is found to be 0.045 V at 298 K and temperature coefficient is $3.4 \times 10^{-4} \text{ VK}^{-1}$.
 (A) 60 J/K/mole (B) 65.2 J/K/mole (C) 69.2 J/K/mole (D) 63.5 J/K/mole
- Na-amalgam is prepared by electrolysis of NaCl solution using liquid Hg as cathode. How long should the current of 10 amp. is passed to produce 10% Na-Hg on a cathode of 10 gm Hg. (atomic mass of Na=23)
 (A) 7.77 min (B) 9.44 min. (C) 5.24 min. (D) 11.39 min.
- Following cell has EMF 0.7995 V.
 $\text{Pt} \mid \text{H}_2(1 \text{ atm}) \mid \text{HNO}_3(1\text{M}) \parallel \text{AgNO}_3(1\text{M}) \mid \text{Ag}$
 If we add enough KCl to the Ag cell so that the final Cl^- is 1M. Now the measured emf of the cell is 0.222 V. The K_{sp} of AgCl would be :
 (A) $1 \times 10^{-9.8}$ (B) $1 \times 10^{-19.6}$ (C) 2×10^{-10} (D) 2.64×10^{-14}
- At 298 K the standard free energy of formation of $\text{H}_2\text{O}(\ell)$ is -237.20 kJ/mole while that of its ionisation into H^+ ion and hydroxyl ions is 80 kJ/mole , then the emf of the following cell at 298 K will be (take $F = 96500 \text{ C}$)
 $\text{Pt} \mid \text{H}_2(\text{g}, 1 \text{ bar}) \mid \text{H}^+(1\text{M}) \parallel \text{OH}^-(1\text{M}) \mid \text{O}_2(\text{g}, 1 \text{ bar}) \mid \text{Pt}$
 (A) 0.40 V (B) 0.81 V (C) 1.23 V (D) -0.40 V
- A hydrogen electrode X was placed in a buffer solution of sodium acetate and acetic acid in the ratio a : b and another hydrogen electrode Y was placed in a buffer solution of sodium acetate and acetic acid in the ratio b : a. If reduction potential values for two cells are found to be E_1 and E_2 respectively w.r.t. standard hydrogen electrode, the pK_a value of the acid can be given as
 (A) $\frac{E_1 - E_2}{0.118}$ (B) $-\frac{E_1 + E_2}{0.118}$ (C) $\frac{E_1}{E_2} \times 0.118$ (D) $\frac{E_2 - E_1}{0.118}$
- Which of following cell can produce more electrical work.
 $(K_a(\text{CH}_3\text{COOH}) = 1.8 \times 10^{-5}, K_b(\text{NH}_3) = 1.8 \times 10^{-5})$
 (A) $\text{Pt} \mid \text{H}_2 \mid 0.1 \text{ M NH}_4\text{Cl} \parallel 0.1 \text{ M CH}_3\text{COOH} \mid \text{H}_2 \mid \text{Pt}$
 (B) $\text{Pt} \mid \text{H}_2 \mid 0.1 \text{ M HCl} \parallel 0.1 \text{ M NaOH} \mid \text{H}_2 \mid \text{Pt}$
 (C) $\text{Pt} \mid \text{H}_2 \mid 0.1 \text{ M HCl} \parallel 0.1 \text{ M CH}_3\text{COOK} \mid \text{H}_2 \mid \text{Pt}$
 (D) $\text{Pt} \mid \text{H}_2 \mid 0.1 \text{ M CH}_3\text{COOK} \parallel 0.1 \text{ M HCl} \mid \text{H}_2 \mid \text{Pt}$
- A hydrogen electrode is immersed in a solution with $\text{pH} = 0$ (HCl). By how much will the potential (reduction) change if an equivalent amount of NaOH is added to the solution. (Take $P_{\text{H}_2} = 1 \text{ atm}$)
 $T = 298 \text{ K}$.
 (A) increase by 0.41 V (B) increase by 59 mV
 (C) decrease by 0.41 V (D) decrease by 59 mV
- At what $\frac{[\text{Br}^-]}{\sqrt{[\text{CO}_3^{2-}]}}$ does the following cell have its reaction at equilibrium ?
 $\text{Ag(s)} \mid \text{Ag}_2\text{CO}_3(\text{s}), \text{Na}_2\text{CO}_3(\text{aq}) \parallel \text{KBr(aq)}, \text{AgBr(s)} \mid \text{Ag(s)}$
 $K_{\text{sp}} = 8 \times 10^{-12}$ for Ag_2CO_3 and $K_{\text{sp}} = 4 \times 10^{-13}$ for AgBr
 (A) $\sqrt{1} \times 10^{-7}$ (B) $\sqrt{2} \times 10^{-7}$ (C) $\sqrt{3} \times 10^{-7}$ (D) $\sqrt{4} \times 10^{-7}$

9. Calculate the EMF of the cell at 298 K
 $\text{Pt} | \text{H}_2(1\text{atm}) | \text{NaOH} (x\text{M}) || \text{NaCl} (x\text{M}) | \text{AgCl} (s) | \text{Ag}$
 If $E_{\text{Cl}^- / \text{AgCl} / \text{Ag}}^0 = +0.222 \text{ V}$
 (A) 1.048 V
 (B) -0.04 V
 (C) -0.604 V
 (D) emf depends on x and cannot be determined unless value of x is given
10. A current of 0.1 A was passed for 2hr through a solution of cuprocyanide and 0.3745 g of copper was deposited on the cathode. Calculate the current efficiency for the copper deposition. (Cu – 63.5)
 (A) 79% (B) 39.5% (C) 63.25% (D) 63.5%
11. With 't' taken in seconds and 'i' taken in Amp, the variation of 'i' follows the equation
 $t^2 + i^2 = 25$
 what amount of Ag will be electrodeposited with this current flowing in the interval 0 to 5 second ?
 (Atomic mass of Ag = 108)
 (A) 22 mg (B) 66 mg (C) 77 mg (D) 88 mg
12. A flashlight cell has the cathodic reaction
 $2\text{MnO}_2 (s) + \text{Zn}^{+2} + 2e^- \longrightarrow \text{Zn Mn}_2\text{O}_4 (s)$
 If the flashlight cell is to give out 4.825 mA, how long could it run if initially 8.7 g of the limiting reagent MnO_2 is present? [Mn = 55, O = 16]
 (A) $2 \times 10^6 \text{ sec}$ (B) $4 \times 10^6 \text{ sec}$ (C) $6 \times 10^6 \text{ sec}$ (D) $8 \times 10^6 \text{ sec}$
13. The following cell has a potential of 0.55 V at 25°C
 $\text{Pt} (s) | \text{H}_2(1\text{atm}) | \text{H}^+ (? \text{ M}) || \text{Cl}^- (1\text{M}) | \text{Hg}_2\text{Cl}_2 (s) | \text{Hg} (l)$
 What is the pH of the solution in the anodic chamber ? $E_{\text{Cl}^- / \text{Hg}_2\text{Cl}_2 / \text{Hg}}^0 = 0.28 \text{ V}$
 (A) 4.57 (B) 14.07 (C) 9.15 (D) 7.15
14. A resistance of 50Ω is registered when two electrodes are suspended into a beaker containing a dilute solution of a strong electrolyte such that exactly half of them are submerged into solution. If the solution is diluted by adding pure water (negligible conductivity) so as to just completely submerge the electrodes, the new resistance offered by the solution would be



- (A) 50Ω (B) 100Ω (C) 25Ω (D) 200Ω
15. The standard reduction potential of a silver chloride electrode is 0.2 V and that of a silver electrode is 0.79V. The maximum amount of AgCl that can dissolve in 10^6 L of a 0.1 M AgNO_3 solution is
 (A) 0.5 m mol (B) 1.0 m mol (C) 2.0 m mol (D) 2.5 m mol
16. Calculate the cell EMF in mV for
 $\text{Pt} | \text{H}_2(1\text{atm}) | \text{HCl} (0.01 \text{ M}) | \text{AgCl} (s) | \text{Ag}(s)$ at 298 K
 if ΔG_f° values are at 25°C
 $-109.56 \frac{\text{kJ}}{\text{mol}}$ for $\text{AgCl}(s)$ and $-130.79 \frac{\text{kJ}}{\text{mol}}$ for $(\text{H}^+ + \text{Cl}^-)(\text{aq})$
 (A) 456 m V (B) 654 m V (C) 546 m V (D) None of these

- 17.** Adiponitrile is manufactured electrolytically from acrylonitrile
 $2 \text{CH}_2 = \text{CHCN} \longrightarrow \text{CN} - (\text{CH}_2)_4 - \text{CN}$
 How many kg of adiponitrile (molecular mass = 108) is produced in 9.65 hr using a current of 3750 A with 80% efficiency ?
 (A) 30 kg (B) 58 kg (C) 60 kg (D) 80 kg
- 18.** It is observed that the voltage of a galvanic cell using the reaction $\text{M(s)} + x\text{H}^+ \longrightarrow \text{M}^{x+} + \frac{x}{2} \text{H}_2$ varies linearly with the log of the square root of the hydrogen pressure and the cube root of the M^{x+} concentration. The value of x is
 (A) 2 (B) 3 (C) 4 (D) 5
- 19.** During the electrolysis of 0.1 M CuSO_4 solution using copper electrodes, a depletion of $[\text{Cu}^{++}]$ occurs near the cathode with a corresponding excess near the anode, owing to inefficient stirring of the solution. If the local concentrations of $[\text{Cu}^{++}]$ near the anode & cathode are respectively 0.12 M & 0.08 M, calculate the back e.m.f. developed. Temperature = 298 K.
 (A) 22 mV (B) 5.2 mV (C) 29 mV (D) 59 mV
- 20.** Electrolysis of a solution of HSO_4^- ions produces $\text{S}_2\text{O}_8^{2-}$. Assuming 75% current efficiency, what current should be employed to achieve a production rate of 1 mole of $\text{S}_2\text{O}_8^{2-}$ per hour ?
 (A) 71.5 amp (B) 35.7 amp (C) 142.96 amp (D) 285.93 amp
- 21.** Consider the cell $\text{Ag(s)} | \text{AgBr(s)} | \text{Br}^-(\text{aq}) || \text{AgCl(s)} | \text{Cl}^-(\text{aq}) | \text{Ag(s)}$ at 25°C . The solubility product constants of AgBr & AgCl are respectively 5×10^{-13} & 1×10^{-10} . For what ratio of the concentrations of Br^- & Cl^- ions would the emf of the cell be zero ?
 (A) 1 : 200 (B) 1 : 100 (C) 1 : 500 (D) 200 : 1
- 22.** Value of Λ_m^∞ for SrCl_2 in water at 25°C from the following data :

Conc. (mol/l)	0.25	1
$\Lambda_m (\Omega^{-1} \text{cm}^2 \text{mol}^{-1})$	260	250

 (A) $270 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ (B) $260 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ (C) $250 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$ (D) $255 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$
- 23.** Select the correct statement if -
 $E^\circ_{\text{Mg}^{2+}/\text{Mg}} = -2.4 \text{ V}$, $E^\circ_{\text{Sn}^{4+}/\text{Sn}^{2+}} = 0.1 \text{ V}$, $E^\circ_{\text{MnO}_4^-, \text{H}^+/\text{Mn}^{2+}} = 1.5 \text{ V}$, $E^\circ_{\text{I}_2/\text{I}^-} = 0.5 \text{ V}$
 Here,
 (A) MnO_4^- is the strongest Oxidizing Agent and Mg is the strongest Reducing Agent.
 (B) $\text{Sn}^{4+} + 2\text{I}^- \longrightarrow \text{Sn}^{2+} + \text{I}_2$ is a spontaneous reaction.
 (C) $\text{Mg}^{2+} + \text{Sn}^{2+} \longrightarrow \text{Mg} + \text{Sn}^{4+}$ is a spontaneous reaction.
 (D) Here, Weakest oxidizing agent is Sn^{4+} and weakest reducing agent is Mn^{2+}
- 24.** What is the value of pK_b (CH_3COO^-) if $\lambda_m^\infty = 390$ & $\lambda_m = 7.8$ for 0.04 M of CH_3COOH at 25°C
 (A) 9.3 (B) 9.2 (C) 4.7 (D) 4.8
- 25.** A cell $\text{Ag} | \text{Ag}^+ || \text{Cu}^{++} | \text{Cu}$ initially contains 2M Ag^+ and 2M Cu^{++} ions. The change in cell potential after the passage of 10 amp current for 4825 sec is (Both cathodic and anodic compartments contains 1 litre solution).
 (A) -0.00738 V (B) -1.00738 V (C) -0.0038 V (D) none

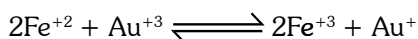
26. The standard oxidation potentials for Mn^{3+} ion acid solution are

$\text{Mn}^{2+} \xrightarrow{-1.5\text{ V}} \text{Mn}^{3+} \xrightarrow{-1.0\text{ V}} \text{MnO}_2$. Is the reaction $2\text{Mn}^{3+} + 2\text{H}_2\text{O} \longrightarrow \text{Mn}^{2+} + \text{MnO}_2 + 4\text{H}^+$ spontaneous under conditions of unit activity? What is the change in free energy?

- (A) spontaneous, -48250 J (B) non-spontaneous, $+48250\text{ J}$
 (C) no change in free energy (D) spontaneous, -96500 J

ONE OR MORE THAN ONE CORRECT OPTION

27. Find the equilibrium constant of the reaction



and tell whether it is spontaneous under standard conditions.

Given: $E^\circ_{\text{Au}^+/\text{Au}^{+3}} = -1.50\text{ V}$, $E^\circ_{\text{Fe}^{+2}/\text{Fe}^{+3}} = -0.77$ will be:

- (A) 10^{11} (B) 10^{22} (C) spontaneous (D) Non-spontaneous

28. In which of the following cell (s): $E_{\text{cell}} = E^\circ_{\text{cell}}$?

- (A) $\text{Cu(s)} \mid \text{Cu}^{2+}(0.01\text{ M}) \parallel \text{Ag}^+(0.1\text{ M}) \mid \text{Ag(s)}$ (B) $\text{Pt(H}_2\text{)} \mid \text{pH} = 1 \mid \text{Zn}^{2+}(0.01\text{ m}) \mid \text{Zn(s)}$
 (C) $\text{Pt(H}_2\text{)} \mid \text{pH} = 1 \mid \text{Zn}^{2+}(1\text{ M}) \mid \text{Zn(s)}$ (D) $\text{Pt(H}_2\text{)} \mid \text{H}^+ = 0.01\text{ M} \mid \text{Zn}^{2+}(0.01\text{ M}) \mid \text{Zn(s)}$

29. Which one is/are correct among the followings?

Given, the half cell emfs $E^\circ_{\text{Cu}^{+2}/\text{Cu}} = 0.337$, $E^\circ_{\text{Cu}^{+1}/\text{Cu}} = 0.521$

- (A) Cu^{+1} disproportionates
 (B) Cu and Cu^{2+} comproportionates (reverse of disproportionation into Cu^+).
 (C) $E^\circ_{\text{Cu} \mid \text{Cu}^{+2}} + E^\circ_{\text{Cu}^{+1} \mid \text{Cu}}$ is positive
 (D) All of these

30. Indicate the correct statements:

- (A) conductivity cells have cell constant values independent of the solution filled into the cell
 (B) DC (direct current) is not used for measuring the resistance of a solution.
 (C) Kohlrausch law is valid both for strong and weak electrolytes.
 (D) The k decreases but Λ_m and Λ_{eq} increase on dilution.

31. The standard redox potentials E° of the following systems are

System	E° (volts)
(i) $\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51
(ii) $\text{Sn}^{2+} \longrightarrow \text{Sn}^{4+} + 2\text{e}^-$	-0.15
(iii) $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.33
(iv) $\text{Ce}^{3+} \longrightarrow \text{Ce}^{4+} + \text{e}^-$	-1.61

The oxidising power of the various species are related as

- (A) $\text{Cr}_2\text{O}_7^{2-} > \text{MnO}_4^-$ (B) $\text{Ce}^{4+} > \text{Sn}^{4+}$ (C) $\text{Ce}^{4+} > \text{MnO}_4^-$ (D) $\text{MnO}_4^- > \text{Sn}^{4+}$

32. How can the oxidation potential of a hydrogen electrode be decreased?

- (A) increasing the $[\text{H}^+]$ (B) decreasing the p_{H_2}
 (C) decreasing the p_{OH} of the solution (D) increasing the p_{H_2}

- 33.** Acetic acid is titrated with NaOH solution. Which of the following statement is correct for this titration?
- (A) conductance increases upto equivalence point, then it decreases
 (B) conductance decreases upto equivalence point, then it increases
 (C) first conductance increases slowly upto equivalence point and then increases rapidly
 (D) first conductance increases slowly upto equivalence point and then drops rapidly
- 34.** During an electrolysis of conc. H_2SO_4 , perdisulphuric acid ($\text{H}_2\text{S}_2\text{O}_8$) and O_2 form in equimolar amount. The amount of H_2 that will form simultaneously will be ($2\text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_8 + 2\text{H}^+ + 2\text{e}^-$)
- (A) thrice that of O_2 in moles
 (B) twice that of O_2 in moles
 (C) equal to that of O_2 in moles
 (D) thrice that of $\text{H}_2\text{S}_2\text{O}_8$ in moles
- 35.** Consider the sequence $\text{A}^{a+} \xrightarrow{\text{E}_1^0} \text{A}^{b+} \xrightarrow{\text{E}_2^0} \text{A}^{c+}$
 $\text{A}^{a+} \xrightarrow{\text{E}^0} \text{A}^{c+}$
- Given that $\text{E}^0 = \frac{\text{E}_1^0 + \text{E}_2^0}{2}$, which of the following might represent correct set of values for a, b and c?
- (A) a = 3, b = 2, c = 1
 (B) a = 5, b = 3, c = 1
 (C) a = 6, b = 4, c = 2
 (D) a = 6, b = 4, c = 1
- 36.** In a standard solution of X^- and Y^- ions, gas Z_2 is passed. X_2 is liberated and Z^- is formed while Y^- remain unaffected. E_x^0 , E_y^0 and E_z^0 represent standard reduction potentials of X_2 , Y_2 and Z_2 respectively. Which of the following can be inferred from this equation?
- (A) $\text{E}_z^0 > \text{E}_x^0$
 (B) $\text{E}_z^0 > \text{E}_y^0$
 (C) $\text{E}_y^0 > \text{E}_x^0$
 (D) $\text{E}_z^0 > 1/2(\text{E}_x^0 + \text{E}_y^0)$
- 37.** Which is/are true about the salt bridge?
- (A) Any salt can be employed to construct the bridge
 (B) Only salts containing cation and ion having nearly equal ionic mobilities are employed
 (C) During the operation of cell, anions of the salt rush into the anodic half cell and cations into the cathodic half cell to maintain the electrical neutrality of the two solutions
 (D) Salt bridge gives rise to liquid-liquid junction potential.
- 38.** When an aqueous solution of Lithium chloride is electrolyzed using graphite electrodes
- (A) pH of the resulting solution increases
 (B) pH of the resulting solution decreases
 (C) as the current flows, pH of the solution around the cathode increases
 (D) as the current flows, pH of the solution around the anode increases.
- 39.** Which of the following mixtures, when dissolved in water, will give significant quantity of hydrogen gas at cathode when electrolysed using inert electrodes under standard conditions?
- (A) K_2CO_3 and Na_2SO_4
 (B) NaCl and CaCl_2
 (C) $\text{Ba}(\text{NO}_3)_2$ and $\text{Pb}(\text{NO}_3)_2$
 (D) CuCl_2 and potash alum
- 40.** On passing 0.5 mole of electrons through CuSO_4 and $\text{Hg}_2(\text{NO}_3)_2$ solutions in series using inert electrodes:
- (A) 0.5 mol of Cu deposited
 (B) 0.5 mol of Hg deposited
 (C) 0.125 mol of O_2 produced in each electrolytic cell
 (D) 0.5 mol of O_2 produced
- 41.** Which of the following statements are correct?
- (A) The electrolysis of aqueous NaCl produces hydrogen gas at the cathode and chlorine gas at the anode
 (B) The electrolysis of a dilute solution of sodium fluoride produces oxygen gas at the anode and hydrogen gas at the cathode
 (C) The electrolysis of concentrated sulphuric acid produces SO_2 gas at the anode and O_2 gas at the cathode
 (D) After the electrolysis of aqueous sodium sulphate, the solution becomes acidic

42. H_2 and O_2 gases will evolve at the cathode and anode respectively during the electrolysis of aqueous solutions of :
- (A) KNO_3 , Pt electrodes (B) $AgNO_3$, Ag electrodes
 (C) $CuSO_4$, Cu electrodes (D) Na_2SO_4 inert electrodes
43. If 90g of water is electrolysed completely with 50% current efficiency
- (A) 10 Faraday of electricity will be consumed (B) 20 Faraday of electricity will be consumed
 (C) 168 L (STP) of gases will be produced (D) 84 L (STP) of gases will be produced
44. During conductance measurement of an electrolyte (based upon wheatstone bridge principle), alternating current is used because direct current leads to the complications.
- (A) Polymerization
 (B) Ionization
 (C) Electrolysis resulting in the change of concentration and in the resistance of solution.
 (D) Polarisation at the electrodes resulting in the change of resistance of solution.

COMPREHENSION BASED QUESTIONS

Comprehension # 1

The potential associated with each electrode is known as electrode potential. If the concentration of each species taking part in the electrode reaction is unit (if any gas appears in the electrode reaction, it is confined to 1 atmospheric pressure) and further the reaction is carried out at 298 K, then the potential of each electrode is said to be the standard electrode potential. By convention, the standard electrode potential of hydrogen electrode is 0.0 volt. The electrode potential value for each electrode process is a measure of relative tendency of the active species in the process to remain in the oxidised/reduced form. A negative E° means that the redox couple is a stronger reducing agent than the H^+/H_2 couple. A positive E° means that the redox couple is a weaker reducing agent than the H^+/H_2 couple. The metal with greater positive value of standard reduction potential forms the oxide of greater thermal stability.

45. The emf of the cell, $Ag | AgI \text{ in } KI | NH_4NO_3 | AgNO_3 | Ag$
0.05M Sol. 0.05M
- is 0.788 volt at 25 °C. The degree of dissociation of KI and silver nitrate in the above solution is 0.90 each. Calculate the solubility product of AgI (Use : $\frac{2.303RT}{F} = 0.059$)
- (A) 8.93×10^{-17} (B) 4×10^{-17} (C) 11×10^{-17} (D) 2.105×10^{-17}
46. The standard reduction potential at 25°C of the reaction $2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$ is - 0.8277 volt. Calculate the equilibrium constant for the reaction in which water is forming from its ions at 25°C.
- (A) 10^{-14} (B) 10^{-7} (C) 10^{14} (D) 10^7

Comprehension # 2

An aqueous solution of NaCl on electrolysis gives $H_2(g)$, $Cl_2(g)$ and NaOH according to reaction



A direct current of 25 ampere with a current efficiency of 62% is passed through 20 litre of NaCl solution (20% by weight)

47. Reaction taking place at cathode and at anode respectively are:
- (A) $2e^- + 2H_2O \longrightarrow 2OH^- + H_2$; $2Cl^- \longrightarrow Cl_2 + 2e^-$
 (B) $Na^+ + e^- \longrightarrow Na$; $2OH^- \longrightarrow H_2O + \frac{1}{2} O_2 + 2e^-$
 (C) $2e^- + H_2 \longrightarrow 2H^-$; $H_2 \longrightarrow 2H^+ + 2e^-$
 (D) $4e^- + O_2 \longrightarrow 2O^{2-}$; $H_2 + 2OH^- \longrightarrow 2H_2O + 2e^-$
48. Time required to produce 1 kg of Cl_2 is -
- (A) 30.2 hr (B) 15.1 hr (C) 24.35 hr (D) 48.71 hr
49. Molarity of solution with respect to OH^- ions is - (Use data of q. no. 57)
- (A) 2.8 (B) 0.35 (C) 1.4 (D) 0.7

LEVEL-1 (SUBJECTIVE TYPE QUESTIONS)

- Write cell reaction of the following cells :
 (A) $\text{Ag} | \text{Ag}^+ (\text{aq}) || \text{Cu}^{2+} (\text{aq}) | \text{Cu}$ (B) $\text{Pt} | \text{Fe}^{2+}, \text{Fe}^{3+} || \text{MnO}_4^-, \text{Mn}^{2+}, \text{H}^+ | \text{Pt}$
 (C) $\text{Pt}, \text{Cl}_2 | \text{Cl}^- (\text{aq}) || \text{Ag}^+ (\text{aq}) | \text{Ag}$ (D) $\text{Pt}, \text{H}_2 | \text{H}^+ (\text{aq}) || \text{Cd}^{2+} (\text{aq}) | \text{Cd}$
- Write cell representation for following cells.
 (A) $\text{Cd}^{2+} (\text{aq}) + \text{Zn} (\text{s}) \longrightarrow \text{Zn}^{2+} (\text{aq}) + \text{Cd} (\text{s})$
 (B) $2\text{Ag}^+ (\text{aq}) + \text{H}_2 (\text{g}) \longrightarrow 2\text{H}^+ (\text{aq}) + 2\text{Ag} (\text{s})$
 (C) $\text{Cr}_2\text{O}_7^{2-} (\text{aq}) + 14\text{H}^+ (\text{aq}) + 6\text{Fe}^{2+} (\text{aq}) \longrightarrow 6\text{Fe}^{3+} (\text{aq}) + 2\text{Cr}^{3+} (\text{aq}) + 7\text{H}_2\text{O} (\text{l})$
- Write half cells of each cell with following cell reactions :
 (A) $\text{Zn} (\text{s}) + 2\text{H}^+ (\text{aq}) \rightleftharpoons \text{Zn}^{2+} (\text{aq}) + \text{H}_2 (\text{g})$
 (B) $2\text{Fe}^{3+} (\text{aq}) + \text{Sn}^{2+} (\text{aq}) \rightleftharpoons 2\text{Fe}^{2+} (\text{aq}) + \text{Sn}^{4+} (\text{aq})$
 (C) $\text{MnO}_4^- (\text{aq}) + 8\text{H}^+ (\text{aq}) + 5\text{Fe}^{2+} (\text{aq}) \rightleftharpoons 2\text{Fe}^{3+} (\text{aq}) + \text{Mn}^{2+} (\text{aq}) + 4\text{H}_2\text{O} (\text{l})$
 (D) $\text{Pb} (\text{s}) + \text{Br}_2 (\text{l}) \rightleftharpoons \text{Pb}^{2+} (\text{aq}) + 2\text{Br}^- (\text{aq})$
- Is 1.0 M H^+ solution under H_2SO_4 at 1.0 atm capable of oxidising silver metal in the presence of 1.0 M Ag^+ ion?
 $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.80 \text{ V}, E^\circ_{\text{H}^+/\text{H}_2(\text{Pt})} = 0.0 \text{ V}$
- For a cell $\text{Mg}(\text{s}) | \text{Mg}^{2+}(\text{aq}) || \text{Ag}^+ (\text{aq}) | \text{Ag}$,
 (i) Calculate the equilibrium constant at 25°C .
 (ii) Also find the maximum work per mole Mg that can be obtained by operating the cell.
 $E^\circ (\text{Mg}^{2+}/\text{Mg}) = -2.37\text{V}, E^\circ (\text{Ag}^+/\text{Ag}) = 0.8 \text{ V}$.
- The EMF of the cell $\text{M} | \text{M}^{n+} (0.02\text{M}) || \text{H}^+ (1\text{M}) | \text{H}_2(\text{g}) (1 \text{ atm}), \text{Pt}$ at 25°C is 0.81V. Calculate the valency of the metal if the standard oxidation of the metal is 0.76V.
- Equinormal Solutions of two weak acids, HA ($\text{pK}_a = 3$) and HB ($\text{pK}_a = 5$) are each placed in contact with equal pressure of hydrogen electrode at 25°C . When a cell is constructed by interconnecting them through a salt bridge, find the emf of the cell.
- In two vessels each containing 500ml water, 0.5m mol of aniline ($K_b = 10^{-9}$) and 25mmol of HCl are added separately. Two hydrogen electrodes are constructed using these solutions. Calculate the emf of cell made by connecting them appropriately.
- Calculate E° and E for the cell $\text{Sn} | \text{Sn}^{2+} (1\text{M}) || \text{Pb}^{2+} (10^{-3}\text{M}) | \text{Pb}$, $E^\circ (\text{Sn}^{2+} | \text{Sn}) = -0.14\text{V}$, $E^\circ (\text{Pb}^{2+} | \text{Pb}) = -0.13\text{V}$. Is cell representation is correct?
- Calculate the equilibrium constant for the reaction $\text{Fe} + \text{CuSO}_4 \rightleftharpoons \text{FeSO}_4 + \text{Cu}$ at 25°C .
 Given $E^\circ (\text{Fe}/\text{Fe}^{2+}) = 0.44\text{V}$,
 $E^\circ (\text{Cu}/\text{Cu}^{2+}) = -0.337\text{V}$.
- For the reaction, $4\text{Al}(\text{s}) + 3\text{O}_2(\text{g}) + 6\text{H}_2\text{O} + 4\text{OH}^- \rightleftharpoons 4 [\text{Al}(\text{OH})_4^-]$; $E^\circ_{\text{cell}} = 2.73 \text{ V}$.
 If $\Delta G_f^\circ (\text{OH}^-) = -157 \text{ kJ mol}^{-1}$ and $\Delta G_f^\circ (\text{H}_2\text{O}) = -237.2 \text{ kJ mol}^{-1}$, determine $\Delta G_f^\circ [\text{Al}(\text{OH})_4^-]$.
- Calculate the EMF of the cell,
 $\text{Zn} - \text{Hg}(\text{c}_1\text{M}) | \text{Zn}^{2+} (\text{aq}) | \text{Hg} - \text{Zn}(\text{c}_2\text{M})$
 at 25°C , if the concentrations of the zinc amalgam are: $\text{c}_1 = 10\text{g per } 100\text{g of mercury}$ and $\text{c}_2 = 1\text{g per } 100\text{g of mercury}$.

- 13.** Calculate pH using the following cell :
 $\text{Pt} (\text{H}_2) \mid \text{H}^+ (x \text{ M}) \parallel \text{H}^+ (1 \text{ M}) \mid \text{Pt} (\text{H}_2)$ if $E_{\text{cell}} = 0.2364 \text{ V}$.
 1 atm 1 atm
- 14.** The electrosynthesis of MnO_2 is carried out from a solution of MnSO_4 in H_2SO_4 (aq). If a current of 25.5 ampere is used with a current efficiency of 85%, how long would it take to produce 1 kg of MnO_2 ?
- 15.** Calculate the quantity of electricity that would be required to reduce 12.3 g of nitrobenzene to aniline, if the current efficiency for the process is 50 percent. If the potential drop across the cell is 3.0 volts, how much energy will be consumed?
- 16.** A metal is known to form fluoride MF_2 . When 10A of electricity is passed through a molten salt for 330 sec. 1.95g of metal is deposited. Find the atomic weight of M. What will be the quantity of electricity required to deposit the same mass of Cu from CuSO_4 ?
- 17.** 10g solution of CuSO_4 is electrolyzed using 0.01F of electricity. Calculate:
 (a)The weight of resulting solution
 (b)Equivalents of acid or alkali in the solution.
- 18.** After electrolysis of NaCl solution with inert electrodes for a certain period of time. 600 mL of the solution was left. Which was found to be 1N in NaOH. During the same time, 31.75 g of Cu was deposited in the copper voltameter in series with the electrolytic cell. Calculate the percentage yield of NaOH obtained.
- 19.** A current of 3.7A is passed for 6hrs. between Ni electrodes in 0.5L of 2M solution of $\text{Ni}(\text{NO}_3)_2$. What will be the molarity of solution at the end of electrolysis?
- 20.** The conductivity of pure water in a conductivity cell with electrodes of cross sectional area 4 cm^2 and 2 cm apart is $8 \times 10^{-7} \text{ S cm}^{-1}$.
 (i) What is resistance of conductivity cell ?
 (ii) What current would flow through the cell under an applied potential difference of 1 volt?
- 21.** Given the following molar conductivities at 25°C : HCl, $426 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ NaCl, $125 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, NaC (sodium crotonate), $83 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$. Calculate the degree of dissociation and dissociation constant of crotonic acid. The conductivity of a 0.001 ml dm^{-3} acid solution is $3.83 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$. Neglect the variation of conductivity with concentration.
- 22.** The conductivity of $0.001 \text{ M Na}_2\text{SO}_4$ solution is $2.6 \times 10^{-4} \Omega^{-1} \text{ cm}^{-1}$ and increases to $7.0 \times 10^{-4} \Omega^{-1} \text{ cm}^{-1}$ when the solution is saturated with CaSO_4 . The molar conductivities of Na^+ and Ca^{2+} are $50 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and $120 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$, respectively. Calculate (A) the conductivity of only CaSO_4 in the solution, (B) solubility of CaSO_4 , and (C) solubility product of CaSO_4 . The conductivity of water used is $0.50 \times 10^{-6} \Omega^{-1} \text{ cm}^{-1}$
- 23.** The following data are obtained in an experiment to determine the dissociation constant of the weak acid HA.

Solution	Conductivity
0.01 M HA	$3.8 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$
100 cm^3 of 0.01 of M HA	$80.0 \times 10^{-5} \Omega^{-1} \text{ cm}^{-1}$
+ 1 cm^3 of 1.0 M NaOH	(neglecting dilution)

Calculate $\wedge_m(\text{NaA})$, $\lambda_m(\text{A}^-)$ and $\wedge_m(\text{H}^+ \text{A}^-)$ and the degree of dissociation of 0.01 M HA.
 Given : $\lambda_m(\text{Na}^+) = 50 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and $\lambda_m(\text{H}^+) = 350 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$.
- 24.** If the electrode potential for the half cell ,
 $\text{HA} \mid \text{Pt}(\text{H}_2)$ is -0.236 V , then calculate the dissociation constant of HA.
 0.1M 2 atm
- 25.** The standard reduction potential of $E^\circ_{\text{Bi}^{3+}/\text{Bi}}$ and $E^\circ_{\text{Cu}^{2+}/\text{Cu}}$ are 0.226 V and 0.344 V respectively. A mixture of salts of Bi^{3+} and Cu^{2+} at unit concentration each is electrolysed at 25°C . To what value can $[\text{Cu}^{2+}]$ be brought down before bismuth starts to deposit during electrolysis.
- 26.** The cell $\text{Pt}(\text{H}_2) (1\text{atm}) \mid \text{H}^+ (\text{pH}= X) \parallel$ Normal calomel electrode has e.m.f. of 0.67 V at 25°C . Calculate pH of solution. The oxidation potential of calomel electrode on H scale is -0.28 V .

27. A cell designed below contains one litre of buffer mixture of CH_3COOH and CH_3COONa each 1 M in two compartments using platinum electrodes. pK_a for $\text{CH}_3\text{COOH} = 4.74$.



Calculate :

- (i) the pH in the each compartment
 (ii) the e.m.f. of cell
 (iii) the pH and e.m.f. of the cell each compartment after passage of 1.25 A current for 241.25 minute. (Both anodic and cathodic compartment contains 1 litre solution.)
28. Calculate the e.m.f. of the cell at 18°C for : ($K_w(\text{H}_2\text{O}) = 10^{-16}$ at 18°C).

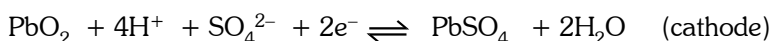
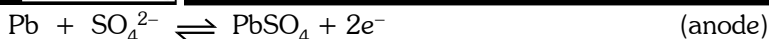


HCl is 90 % dissociated and KOH is 75 % ionised .

29. Electrolysis of NaCl (aq) gives NaOH at cathode. Assuming 100 % current efficiency, determine the quantity of electricity required to convert 10 g NaCl into NaOH .
30. Find out the volume of gases evolved by passing 0.965 A current for 1 hr through an aqueous solution of CH_3COONa at 25°C and 1 atm.
31. Calculate the quantity of current required to liberate :
 (i) H_2 gas at STP at the rate of 1 cc per second
 (ii) O_2 gas at STP at the rate of 1 mm^3 per minute
 (iii) Cl_2 gas at STP at the rate of 1 dm^3 per hour
32. The density of copper is 8.94 g ml^{-1} . Find out the number of coulomb needed to plate an area of $10 \times 10 \text{ cm}^2$ to a thickness of 10^{-2} cm using CuSO_4 solution as electrolyte.
33. A current of 40 microampere is passed through a solution of AgNO_3 for 32 minutes using Pt electrodes. A uniformly single atom thick layer of Ag is deposited covering 43% cathode surface. What is the total surface area of cathode if each Ag atom covers $5.4 \times 10^{-16} \text{ cm}^2$?
34. A cell contains two hydrogen electrodes. The negative electrode is in contact with a solution of 10^{-6} M hydrogen ions. The emf of the cell is 0.118 volt at 25°C . Calculate the concentration of hydrogen ions at the positive electrode.
35. For the galvanic cell, $\text{Ag} | \text{AgCl(s)} | \text{KCl}(0.2\text{M}) || \text{KBr}(0.001 \text{ M}) | \text{AgBr(s)} | \text{Ag}$. Calculate the emf generated and assign correct polarity to each electrode for a spontaneous process after taking into account the cell reaction at 25°C . [Given : $K_{\text{spAgCl}} = 2.8 \times 10^{-10}$; $K_{\text{spAgBr}} = 3.3 \times 10^{-13}$]
36. Electrolysis of a solution of MnSO_4 in aqueous sulphuric acid is a method for the preparation of MnO_2 as per reaction, $\text{Mn}^{2+}(\text{aq}) + 2\text{H}_2\text{O} \longrightarrow \text{MnO}_2(\text{s}) + 2\text{H}^+(\text{aq.}) + \text{H}_2(\text{g})$
 Passing a current of 27A for 24 hour gives one kg of MnO_2 . What is the value of current efficiency ? Write the reaction taking place at the cathode and at the anode.
37. Calculate the equilibrium constant for the reaction, $2\text{Fe}^{3+} + 3\text{I}^- \rightleftharpoons 2\text{Fe}^{2+} + \text{I}_3^-$
 The standard reduction potential in acidic condition are 0.77 and 0.54 V respectively for $\text{Fe}^{3+}/\text{Fe}^{2+}$ and I_3^-/I^- couples.
38. Two students make Daniel cells in laboratory. They take ZnSO_4 from common stock with Cu as positive electrode. The emf of one cell is 0.03 V more than the other. The concentration of CuSO_4 in cell of higher emf is 0.5 M. Find the concentration of CuSO_4 in second cell.

[Given : $\frac{2.303RT}{F} = 0.06 \text{ V}$]

39. At 25°C , $\lambda_\infty(\text{H}^+) = 3.4982 \times 10^{-2} \text{ S m}^2 \text{ mol}^{-1}$ and $\lambda_\infty(\text{OH}^-) = 1.98 \times 10^{-2} \text{ S m}^2 \text{ mol}^{-1}$.
 Given: Sp. conductance = $5.1 \times 10^{-6} \text{ S m}^{-1}$ for H_2O , determine pH and K_w .
40. During the discharge of a lead storage battery the density of sulphuric acid fell from 1.294 to 1.139 g ml^{-1} . H_2SO_4 of density 1.294 g mL^{-1} is 39% and that of density 1.39 g mL^{-1} is 20% by weight. The battery holds 3.5L of acid and the volume practically remains constant during the discharge. Calculate the number of ampere hours for which the battery must have been used. The discharging reactions are:



41. K_d for complete dissociation of $[\text{Ag}(\text{NH}_3)_2]^+$ into Ag^+ and 2NH_3 is 6×10^{-8} . Calculate E^0 for the following half reaction; $\text{Ag}(\text{NH}_3)_2^+ + e^- \rightleftharpoons \text{Ag} + 2\text{NH}_3$



42. Determine at 298 for cell



(A) it's emf when $\text{pH} = 5.0$

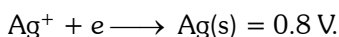
(B) the pH when $E_{\text{cell}} = 0$

(C) the positive electrode when $\text{pH} = 7.5$

(Given $E^0_{\text{RP(RHS)}} = 0.28$, $E^0_{\text{RP(LHS)}} = 0.699$)

LEVEL-2 (SUBJECTIVE TYPE QUESTIONS)

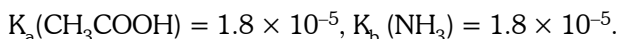
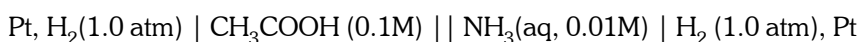
1. Voltage of the cell $\text{Pt}, \text{H}_2(1 \text{ atm}) | \text{HOCN}(1.3 \times 10^{-3} \text{ M}) || \text{Ag}^+(0.8 \text{ M}) | \text{Ag}(\text{s})$ is 0.982 V . Calculate the K_a for HOCN . Neglect $[\text{H}^+]$ because of oxidation of $\text{H}_2(\text{g})$.



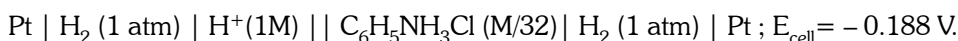
2. Calculate the potential of an indicator electrode versus the standard hydrogen electrode, which originally contained 0.1 M MnO_4^- and 0.8 M H^+ and which was treated with 90% of the Fe^{2+} necessary to reduce all the MnO_4^- to Mn^{2+} .



3. Calculate the emf of the cell



4. Determine the degree of hydrolysis and hydrolysis constant of aniline hydrochloride in $\text{M}/32$ solution of salt at 298 K from the following cell data at 298 K .



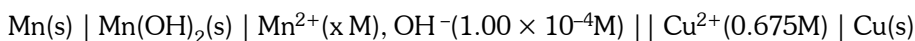
5. The emf of the cell, $\text{Pt} | \text{H}_2(1 \text{ atm}), | \text{H}^+(0.1 \text{ M}, 30 \text{ ml}) || \text{Ag}^+(0.8 \text{ M}) | \text{Ag}$ is 0.9 V . Calculate the emf when 40 ml of 0.05 M NaOH is added.

6. The emf of the cell $\text{Ag} | \text{AgI} | \text{KI}(0.05 \text{ M}) || \text{AgNO}_3(0.05 \text{ M}) | \text{Ag}$ is 0.788 V . Calculate the solubility product of AgI .

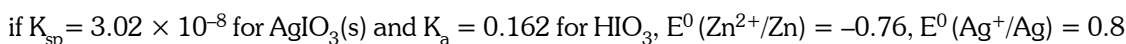
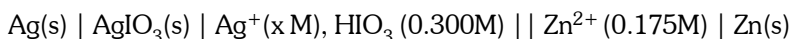
7. The $\text{p}K_{\text{sp}}$ of AgI is 16.07 . If the E^0 value for $\text{Ag}^+ | \text{Ag}$ is 0.7991 V . Find the E^0 for the half cell reaction $\text{AgI}(\text{s}) + e^- \longrightarrow \text{Ag} + \text{I}^-$.

8. Given, $E^0 = -0.268 \text{ V}$ for the $\text{Cl}^- | \text{PbCl}_2 | \text{Pb}$ couple and -0.126 V for the $\text{Pb}^{2+} | \text{Pb}$ couple, determine K_{sp} for PbCl_2 at 25°C ?

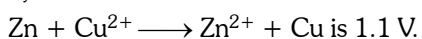
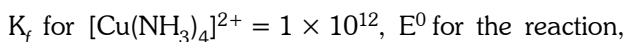
9. Calculate the voltage, E , of the cell at 25°C



10. Calculate the voltage, E , of the cell

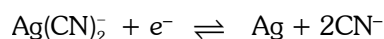
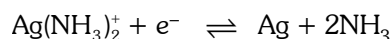


11. Estimate the cell potential of a Daniel cell having 1 M Zn^{++} & originally having 1 M Cu^{++} after sufficient NH_3 has been added to the cathode compartment to make NH_3 concentration 2 M .



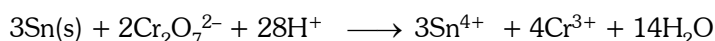
12. The overall formation constant for the reaction of 6 mol of CN^- with cobalt (II) is 1×10^{19} . The standard reduction potential for the reaction $[\text{Co}(\text{CN})_6]^{3-} + e^- \rightarrow \text{Co}(\text{CN})_6^{4-}$ is -0.83 V . Calculate the formation constant of $[\text{Co}(\text{CN})_6]^{3-}$.
 Given $\text{Co}^{3+} + e^- \rightarrow \text{Co}^{2+}$; $E^\circ = 1.82 \text{ V}$

13. Calculate E° for the following reactions at 298 K,



Given: $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.7991 \text{ V}$, $K_{\text{ins}}[\text{Ag}(\text{NH}_3)_2^+] = 6.02 \times 10^{-8}$ and $K_{\text{ins}}[\text{Ag}(\text{CN})_2^-] = 1.995 \times 10^{-19}$

14. Calculate the equilibrium constant for the reaction:



E° for $\text{Sn}/\text{Sn}^{2+} = 0.136 \text{ V}$ E° for $\text{Sn}^{2+}/\text{Sn}^{4+} = -0.154 \text{ V}$

E° for $\text{Cr}_2\text{O}_7^{2-}/\text{Cr}^{3+} = 1.33 \text{ V}$

15. An acidic solution of Cu^{2+} salt containing 0.4 g of Cu^{2+} is electrolyzed until all the copper is deposited. The electrolysis is continued for seven more minutes with the volume of solution kept at 100 ml and the current at 1.2 amp. Calculate the volume of gases evolved at NTP during the entire electrolysis.

16. At 25°C , $\Delta H_f(\text{H}_2\text{O}, l) = -56700 \text{ cal/mol}$ and energy of ionization of $\text{H}_2\text{O}(l) = 19050 \text{ cal/mol}$. What will be the reversible EMF at 25°C of the cell,

$\text{Pt} | \text{H}_2(\text{g}) (1 \text{ atm}) | \text{H}^+ || \text{OH}^- | \text{O}_2(\text{g}) (1 \text{ atm}) | \text{Pt}$, if at 26°C the emf increases by 0.0001158 V .

17. Calculate the cell potential of a cell having reaction: $\text{Ag}_2\text{S} + 2e^- \rightleftharpoons 2\text{Ag} + \text{S}^{2-}$ in a solution buffered at $\text{pH} = 3$ and which is also saturated with $0.1 \text{ M H}_2\text{S}$.

For H_2S : $K_1 = 10^{-8}$ and $K_2 = 1.1 \times 10^{-13}$, $K_{\text{sp}}(\text{Ag}_2\text{S}) = 2 \times 10^{-49}$, $E^\circ_{\text{Ag}^+/\text{Ag}} = 0.8$.

18. The equivalent conductance of 0.10 N solution of MgCl_2 is $97.1 \text{ mho cm}^2 \text{ eq}^{-1}$ at 25°C . a cell with electrode that are 1.5 cm^2 in surface area and 0.5 cm apart is filled with 0.1 N MgCl_2 solution. How much current will flow when potential difference between the electrodes is 5 volt .

19. The specific conductance at 25°C of a saturated solution of SrSO_4 is $1.482 \times 10^{-4} \text{ ohm}^{-1} \text{ cm}^{-1}$ while that of water used is $1.5 \times 10^{-6} \text{ mho cm}^{-1}$. Determine at 25°C the solubility in gm per litre of SrSO_4 in water. Molar ionic conductance of Sr^{2+} and SO_4^{2-} ions at infinite dilution are 59.46 and $79.8 \text{ ohm}^{-1} \text{ cm}^2 \text{ mole}^{-1}$ respectively. [$\text{Sr} = 87.6$, $\text{S} = 32$, $\text{O} = 16$]

20. Calculate the solubility and solubility product of $\text{Co}_2[\text{Fe}(\text{CN})_6]$ in water at 25°C from the following data:

Conductivity of a saturated solution of $\text{Co}_2[\text{Fe}(\text{CN})_6]$ is $2.06 \times 10^{-6} \Omega^{-1} \text{ cm}^{-1}$ and that of water used $4.1 \times 10^{-7} \Omega^{-1} \text{ cm}^{-1}$. The ionic molar conductivities of Co^{2+} and $\text{Fe}(\text{CN})_6^{4-}$ are $86.0 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and $444.0 \Omega^{-1} \text{ cm}^2 \text{ mol}^{-1}$.

21. The conductivity of 0.1 M NaOH solution is 0.022 S cm^{-1} . When an equal volume of 0.1 M HCl solution is added, the conductivity decreases to 0.0056 S cm^{-1} . A further addition of HCl solution, the volume of which is equal to that of the first portion added, the conductivity increases to 0.017 S cm^{-1} . Calculate $\Lambda_m(\text{NaCl})$ in units of $\text{S cm}^2 \text{ mol}^{-1}$ and also calculate $\Lambda_m(\text{HCl})$.

ANSWER KEY

SINGLE CHOICE CORRECT

- | | | | |
|--------|--------|--------|---------|
| 1.(B) | 2.(A) | 3.(A) | 4. (A) |
| 5.(B) | 6.(D) | 7.(C) | 8.(B) |
| 9.(D) | 10.(A) | 11.(A) | 12.(A) |
| 13.(A) | 14.(A) | 15.(B) | 16.(A) |
| 17.(B) | 18.(B) | 19.(B) | 20. (A) |
| 21.(A) | 22.(A) | 23.(A) | 24.(B) |
| 25.(A) | 26.(A) | | |

ONE OR MORE THAN CORRECT

- | | | | |
|-------------|------------|-----------|-----------|
| 27. (C) | 28.(A,B,D) | 29.(A,C) | 30.(ALL) |
| 31. (B,C,D) | 32.(A,B) | 33. (C) | 34.(A,D) |
| 35. (A,B,C) | 36. (A,C) | 37. (B,C) | 38. (A,C) |
| 39.(A,B) | 40.(B,C) | 41.(A,B) | 42.(A,D) |
| 43.(B,C) | 44.(C,D) | | |

COMPREHENSION BASED QUESTIONS

- | | | |
|---------|---------|---------|
| 45. (A) | 46. (C) | 47. (A) |
| 48. (D) | 49. (C) | |

LEVEL-1 (SUBJECTIVE TYPE QUESTIONS)

- (A) $2\text{Ag} + \text{Cu}^{2+} \longrightarrow 2\text{Ag}^+ + \text{Cu}$, (B) $\text{MnO}_4^- + 5\text{Fe}^{2+} + 8\text{H}^+ \longrightarrow \text{Mn}^{2+} + 5\text{Fe}^{3+} + 4\text{H}_2\text{O}$
(C) $2\text{Cl}^- + 2\text{Ag}^+ \longrightarrow 2\text{Ag} + \text{Cl}_2$, (D) $\text{H}_2 + \text{Cd}^{2+} \longrightarrow \text{Cd} + 2\text{H}^+$
- (A) $\text{Zn} | \text{Zn}^{2+} || \text{Cd}^{2+} | \text{Cd}$, (B) $\text{Pt}, \text{H}_2 | \text{H}^+ || \text{Ag}^+ | \text{Ag}$,
(C) $\text{Pt} | \text{Fe}^{2+}, \text{Fe}^{3+} || \text{Cr}_2\text{O}_7^{2-}, \text{H}^+ | \text{Cr}^{3+} | \text{Pt}$
- | Anode | Cathode |
|--|--|
| (A) $\text{Zn} \text{Zn}^{2+}$ | $\text{H}^+, \text{H}_2 \text{Pt}$ |
| (B) $\text{Pt} \text{Sn}^{2+}, \text{Sn}^{4+}$ | $\text{Fe}^{3+}, \text{Fe}^{2+} \text{Pt}$ |
| (C) $\text{Pt} \text{Fe}^{2+}, \text{Fe}^{3+}$ | $\text{MnO}_4^-, \text{Mn}^{2+} \text{Pt}$ |
| (D) $\text{Pb} \text{Pb}^{2+}$ | $\text{Br}_2, \text{Br}^- \text{Pt}$ |
- $-0.80 \text{ V}, \text{NO}$
- $K_c = 2.868 \times 10^{107}, \Delta G^0 = -611.8 \text{ kJ}$
- $n = 2$
- $E = 0.059$
- $E = 0.395 \text{ V}$
- $E_{\text{cell}}^0 = +0.01 \text{ V}, E_{\text{cell}} = -0.0785 \text{ V}$, correct representation is $\text{Pb} | \text{Pb}^{2+} (10^{-3} \text{ M}) || \text{Sn}^{2+} (1 \text{ M}) | \text{Sn}$
- $K_c = 1.96 \times 10^{26}$
- $-1.30 \times 10^3 \text{ kJ mol}^{-1}$
- 0.0295 V

13. pH = 4
14. $1.023 \times 10^5 \text{ sec}$
15. 115800C, 347.4 kJ
16. A = 114, Q = 5926.8C
17. Final weight = 9.6g, 0.01 Eq of acid
18. 60 %
19. 2M
20. (i) $6.25 \times 10^5 \text{ ohm}$, (ii) $1.6 \times 10^{-6} \text{ amp}$
21. [$\alpha = 0.1$, $K_a = 1.11 \times 10^{-5} \text{ M}$]
22. [(A) $4.4 \times 10^{-4} \Omega^{-1} \text{ cm}^{-1}$ (B) $1.567 \times 10^{-3} \text{ M}$ (C) $4.0386 \times 10^{-6} \text{ M}^2$]
23. [$80 \text{ S cm}^2 \text{ mol}^{-1}$, $30 \text{ S cm}^2 \text{ mol}^{-1}$, $380 \text{ S cm}^2 \text{ mol}^{-1}$, $\alpha = 0.01$]
24. [2×10^{-7}]
25. [10^{-4} M]
26. [6.6]
27. [(i) 4.74 (ii) 0.0V (iii) (Anode) $_{\text{pH}} = 4.5752$, (cathode) $_{\text{pH}} = 4.9048$]
28. [-0.856 V]
29. [16495.7 C]
30. [1.762 lire]
31. (i) 8.61 amp (ii) $2.87 \times 10^{-4} \text{ amp}$, (iii) 2.4 A
32. [27172 coulomb]
33. [602 cm^2]
34. 10^{-4} M
35. $E = -0.037 \text{ V}$, polarity of cells should be reversed
36. 94.8 %
37. $K_c = 6.26 \times 10^7$
38. 0.05 M
39. (i) 6.98 (ii) 1.08×10^{-14}
40. 265 Amp. hr.
41. 0.373V
42. (A) -0.124 V , (B) 7.1, (C) calomel electrode

LEVEL-2 (SUBJECTIVE TYPE QUESTIONS)

- | | |
|---|--|
| 1. $K_a = 6.74 \times 10^{-4}$ | 2. 1.39V |
| 3. -0.46 V | 4. $h = 2.12 \times 10^{-2}$, $K_h = 1.43 \times 10^{-5} \text{ M}$ |
| 5. 0.95 V | 6. $K_{sp} = 1.1 \times 10^{-16}$ |
| 7. $E^0 = -0.1511 \text{ V}$ | 8. $1.536 \times 10^{-5} \text{ M}^3$ |
| 9. 1.66V | 10. -1.188 V |
| 11. $E^0 = 0.71 \text{ V}$ | 12. $K_f = 8.227 \times 10^{63}$ |
| 13. 0.372 V , -0.307 V | 14. $K = 10^{268}$ |
| 15. $V(\text{O}_2) = 99.68 \text{ mL}$, $V(\text{H}_2) = 58.46 \text{ mL}$, Total vol. = 158.1 mL | |
| 16. 0.74985 V | 17. -0.167 V |
| 18. 0.1456 ampere | 19. 0.1934 gm/litre |
| 20. () | 21. (0112) |